MARK IV

SPEEDTRONIC *

GAS TURBINE CONTROL SYSTEM

FOR HEAVY DUTY GAS TURBINES

(GEK 83810)

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GEK 83810

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I. THE CONTROL CONCEPT

A. GENERAL

The control concept applies microcomputer technology to the gas turbine control system and provides improved reliability, quality and monitoring capability over traditional solid-state controls. Reliability is improved by having three (3) identical computers monitor and perform control and protection calculations simultaneously. This redundancy enables the turbine to keep running during a computer failure and enables maintenance personnel to repair a computer with out shutting down the machine. Quality is improved because the hardware is wired and packaged the same on every order with application variables contained in software. This repeatability facilitates automatic on-line and off-line diagnostics to assist testing the hardware in the field and in the factory. Monitoring capability includes access to any control parameter or logic signal in the control which can be displayed on the CRT in engineering units and grouped, if desired, with other parameters of the same type. Built in RS232 and RS422 serial links provide access to this data at the local printer or at a remote station on demand.

B. RELIABILITY

Three (3) computers, "Controllers [R], [S] and [T]", contain identical hardware and software, and are used for all calculations which are necessary to keep the gas turbine running after it has reached complete sequence and all calculations needed for a normal shutdown. The design of each computer contains circuitry to drive its outputs in a defined direction on loss of power or a computer stall condition but the two-thirds voting concept is the only reliable means of protecting against random component failures. A fourth computer, "The Communicator, [C]", monitors [R], [S] and [T], and initiates an audible alarm when there is a disagreement between any control parameter or logic signal in [R], [S] and [T]. The discrepancy is then displayed on the CRT for maintenance personnel to evaluate the problem. Meanwhile the turbine continues to run because the control is responding to the median value.

Figure I-B-1 illustrates redundancy for contact inputs and contact outputs. A field trip contact is wired to Contact Input Module #1, [CIM1], or #2, [CIM2], where the signal is paralleled to three (3) optical isolators and fed to separate digital input cards in [R], [S] and [T]. [C] monitors the inputs seen by [R], [S] and [T], performs a majority vote and feeds the voted and individual values to the CRT for display. Field contacts which are not needed to keep the turbine running are connected to Contact Input Modules #3 - 6 and are optically isolated before being connected directly to [C]. Power for each computer is supplied by an independent power supply which is separately fused from the 125VDC battery system. Voting of logic outputs is performed by the relay driver card, HRDB, in the Relay Driver Module, [RDM]. Power for the HRDB cards and the relays is supplied by two (2) redundant 28VDC power supplies which are monitored by [C]. Relays used for trip circuits, such as fuel solenoids, have additional protection which removes power from the coil when a trip condition occurs. This protects against the relay driver failing in the turned-on mode. Magnetic relays are still used instead of solid-state relays due to their predictable failure mode, interface flexibility, i.e., AC or DC, and form "C" contacts.

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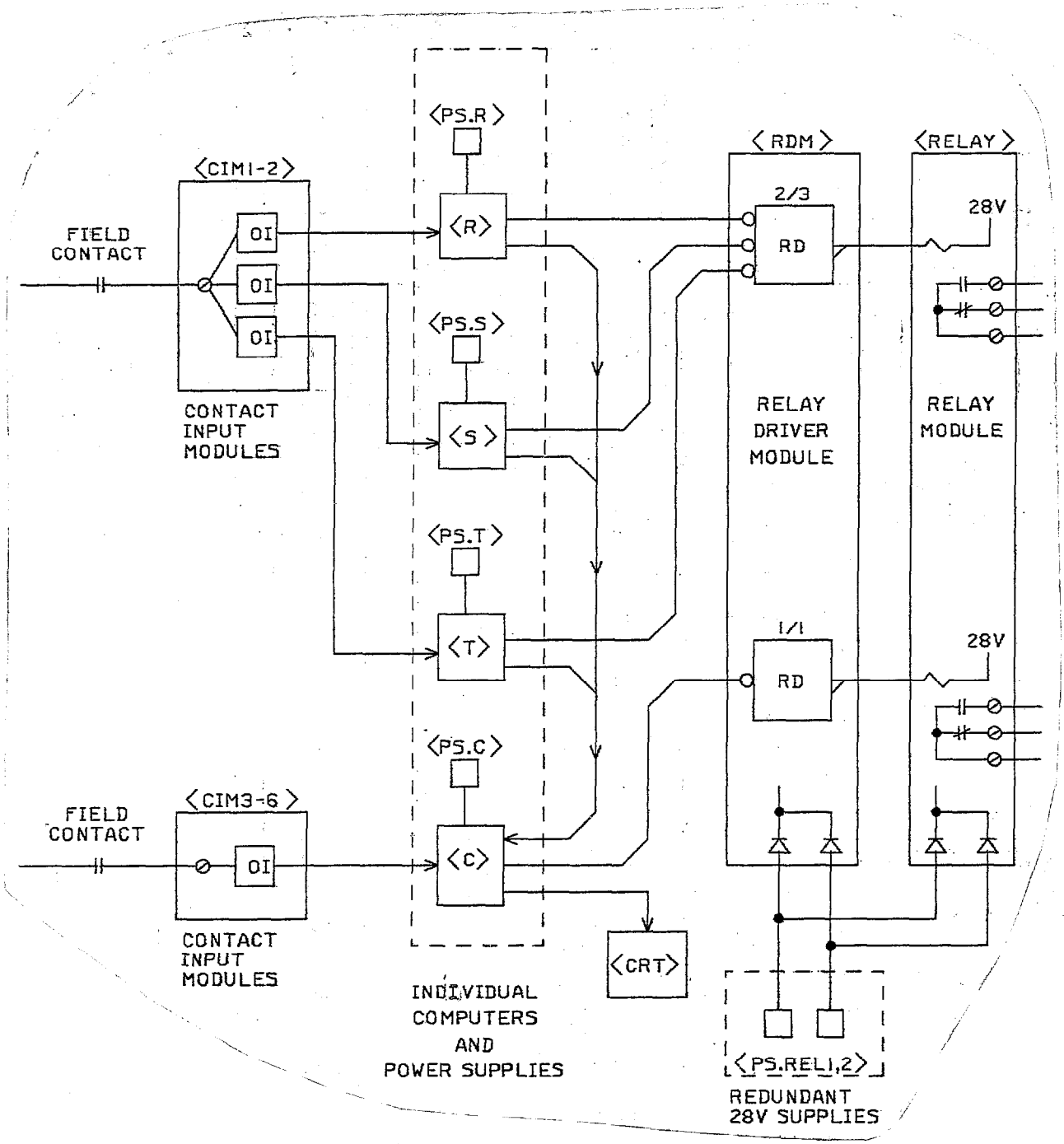


Figure I-B-1: LOGIC VOTING

B. RELIABILITY (continued)

Figure I-B-2 illustrates the analog control system. All analog input signals connect to screw terminals on the Analog I/O Modules, [AIO1-4], and the Thermocouple Modules, [TCM], where they are filtered prior to being fed to Controllers, [R], [S] and [T], or the Communicator, [C]. I/O modules are designed with no active components to reduce the likelihood of failure, and the connections to the computer cards are via card front connectors to confine I/O faults and noise to the cards rather than the computer back-planes. Incoming signals are multiplexed on computer cards dedicated to specific functions and then wired to an analog input card, HAIC, for a second stage of multiplexing and the final analog to digital conversion. Median selection of key parameters such as fuel stroke reference, "FSR", is calculated in [C] and fed back to [R], [S] and [T] to bias their individual calculations, and in case of a computer stall, to up-date the computer prior to reinitializing it for an automatic restart, i.e., recovery. All analog control loops are calculated in software except for the fuel servos which are regulated in hardware on the HSAA cards. In this case, [R], [S] and [T] calculate the required fuel flow which is converted from digital to analog on the HSAA card, compared with the LVDT position, and/or flow divider feedback, and used to drive the three (3) coil servos. The application of three (3) coil servos extends the voting principle to the gas turbine.

Thermocouple inputs are processed similar to other analog inputs. Trip and essential control thermocouples are connected to separate input modules, [TCM.R], [TCM.S] and [TCM.T], for filtering, cold junction reference and offset measurement prior to being fed to Controllers [R], [S] and [T] for multiplexing on the NTCF card. Monitoring thermocouples are wired to one (1) of three (3) input modules, [TCM.C1], [TCM.C2] and [TCM.C3], which feed the Communicator, [C]. A temperature sensing device is mounted on each module to measure the ambient temperature and to provide a reference for the cold junction calculation which is performed in software.

C. MONITORING

The advent of the CRT/Microprocessor monitoring system has made possible a quantum jump in the availability of information to the operator and maintenance personnel. Some of the features are:

1. The "Normal Display" shows all key gas turbine parameters relevant to a particular operating mode, i.e., start-up, running and shutdown, and automatically changes displays when the operating mode changes.

2. Any display can be copied on a panel mounted printer.

3. Two (2) field selectable "Demand Displays" can be provided with thirty-two (32) points each including any control or logic signal displayed in engineering units. It can be logged at specified time intervals or printed manually on demand.

4. Signals can be called by their signal names as shown in the control elementary diagram without referencing the computer memory map.

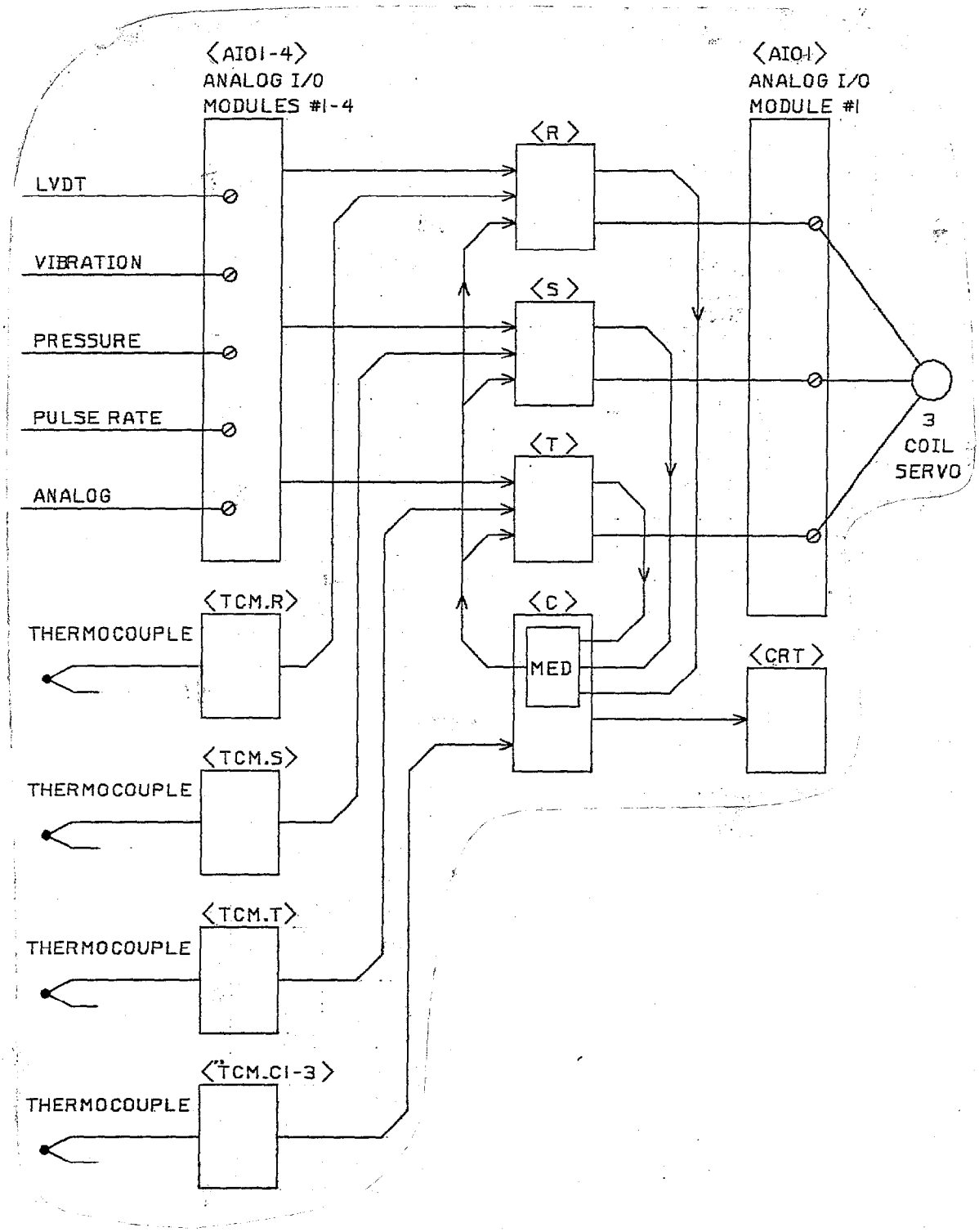


Figure I-B-2: ANALOG CONTROL

C. MONITORING (continued)

5. 256 alarm messages are available, and additional alarms are dedicated to internal diagnostics.

6. Gains, time constants and setpoints can be displayed and changed from the Operator Interface Module, [OPM], and the changes will be retained in BRAM, BATTERY-BACKED RAM, for up to ten (10) days at 25 degrees C if DC power is removed from the panel.

7. The status of all contact inputs can be displayed simultaneously to assist in troubleshooting.

8. Logic signals can be forced from the Operator Interface Module, [OPM], if the correct security code is entered in the computer by qualified maintenance personnel.

9. Two (2) levels of security codes prevent unauthorized adjustment of guarded signals.

10. An historical log monitors key parameters and records the time and date of alarms during a 72 hour period.

NOTES

II. DOCUMENTATION

A. DRAWINGS

1. The Panel "Max Case" Connection Diagram shows all wiring, cards and modules which are necessary to build a max case panel. The term Connection Diagram is used instead of elementary because it shows no functionality.

2. The Panel Assembly Diagram shows a panel layout, and it describes how to assemble the various components.

3. The Panel Outline shows the panel outside dimensions, cable entry and mounting arrangement.

4. The Control Elementary Diagram shows the gas turbine control system including external interface to the panel, hardware/software interface, sequencing in ladder diagram form, control algorithms and the motor control center (if supplied by Drive Systems Department).

B. INSTRUCTION MANUALS

1. GEK 83810 "The Mark IV SPEEDTRONIC* Gas Turbine Control System" manual describes the operation of the heavy duty gas turbine control system.

2. GEK 83821 "The Mark IV SPEEDTRONIC* Elementary and Connection Diagram Reading Aid" is used to illustrate the drawing symbols, cross referencing techniques, etc. used in these diagrams.

3. GEK 83811 "The Mark IV SPEEDTRONIC* Maintenance Manual" provides maintenance instructions for the panel.

4. GEK 83825 "The Mark IV SPEEDTRONIC* Operator's Manual" describes how the operator interfaces with the control panel.

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NOTES

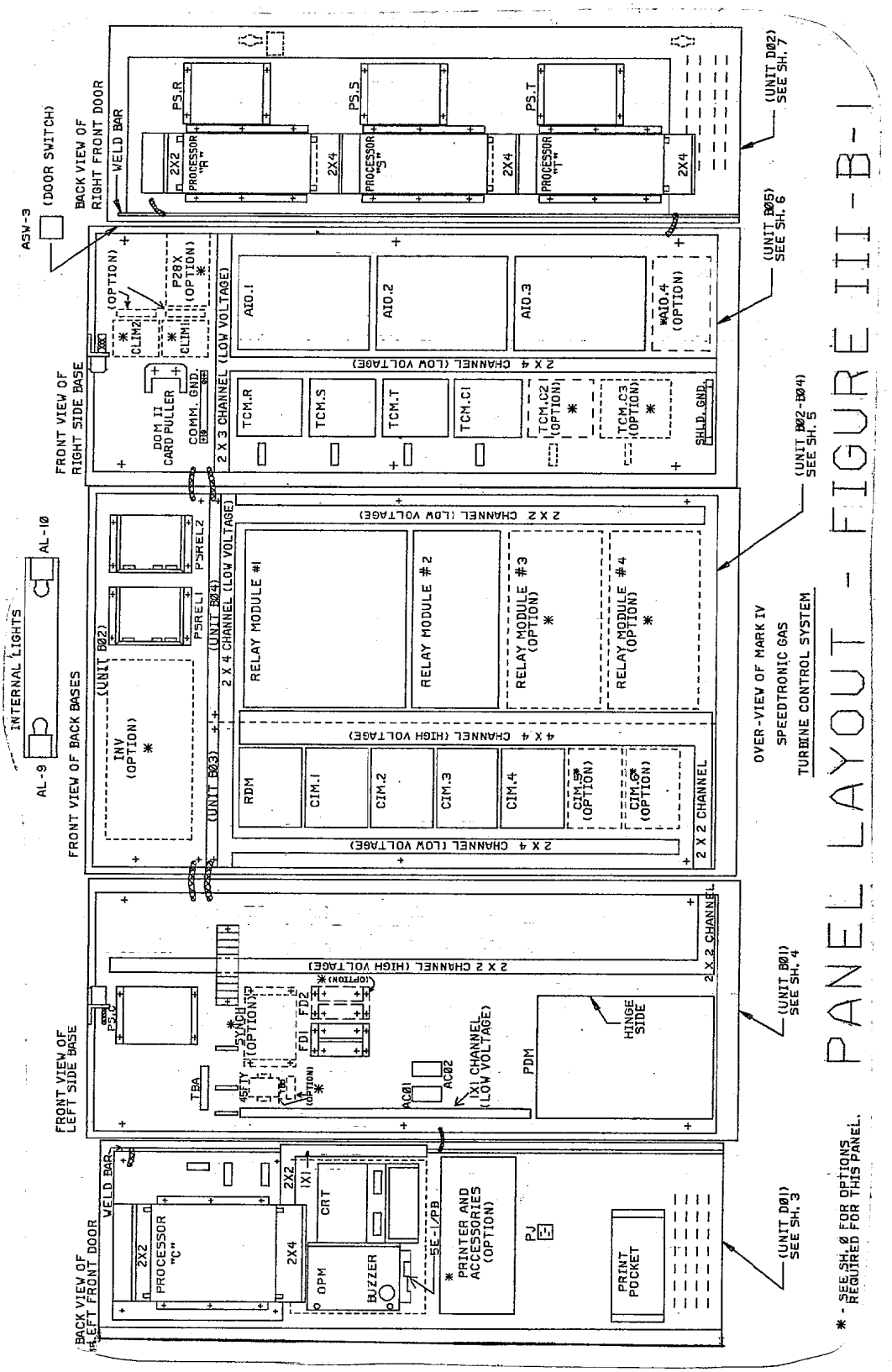


Figure III-B-1: PANEL LAYOUT

III. THE PANEL

A. DEVICE NOMENCLATURE

[AIO1]	Analog I/O Module #1
[AIO2]	Analog I/O Module #2
[AIO3]	Analog I/O Module #3
[AIO4]	Analog I/O Module #4
[C]	Communicator
[CIM1]	Contact Input Module #1
[CIM2]	Contact Input Module #2
[CIM3]	Contact Input Module #3
[CIM4]	Contact Input Module #4
[CIM5]	Contact Input Module #5
[CIM6]	Contact Input Module #6
[CLIM1]	Communication Line Isolation Module #1
[CLIM2]	Communication Line Isolation Module #2
[CRT]	Video Display
[FD1]	Flame Detector Module #1
[FD2]	Flame Detector Module #2
[INV]	Black Start Inverter
[OPM]	Operator Interface Module
[P28X]	28VDC Termination Module
[PDM]	Power Distribution Module
[PS.C]	Power Supply for the Communicator "C"
[PS.R]	Power Supply for Controller "R"
[PS.REL1]	Power Supply #1 for the Relay Modules
[PS.REL2]	Power Supply #2 for the Relay Modules
[PS.S]	Power Supply for Controller "S"
[PS.T]	Power Supply for Controller "T"
[PTR]	Printer
[R]	Controller "R"
[RDM]	Relay Driver Module
[RELAY1]	Relay Module #1
[RELAY2]	Relay Module #2
[RELAY3]	Relay Module #3
[RELAY4]	Relay Module #4
[RTD]	RTD Input Module
[S]	Controller "S"
[SYNCH]	Synchronizing Interface Module
[T]	Controller "T"
[TCM.C1]	Thermocouple Input Module #1 for the Communicator "C"
[TCM.C2]	Thermocouple Input Module #2 for the Communicator "C"
[TCM.C3]	Thermocouple Input Module #3 for the Communicator "C"
[TCM.R]	Thermocouple Input Module for Controller "R"
[TCM.S]	Thermocouple Input Module for Controller "S"
[TCM.T]	Thermocouple Input Module for Controller "T"

* Refer to the panel layout shown in Figure III-B-1.

B. PANEL LAYOUT

1. LEFT FRONT DOOR

COMMUNICATOR [C]

The Communicator monitors the health of the voting Controllers [R], [S] and [T] and drives the Video Display, [CRT] and the Printer, [PTR]. It contains diagnostic software, interfaces with logic and analog I/O and provides RS422 and RS232 serial links for remote interface. Refer to Figure III-B-2 for a typical block diagram. Cards contained in [C] are described briefly in the following table:

Card	Qty	Function	Application
HAFA	1	Analog/Digital Ins	16 Analog and 32 Digital Inputs
HAIC	1	Analog Input	16 Analog Ins. and [OPM] Interface
HCMA	2	RS232 Interface	RS232 Interface [R] [S] [T] [PTR]
HCVA	4	Analog Output	2 Analog Outputs per card
HIOD	3	Digital I/O	32 Digital Inputs and Outputs
HMHA	1	RS422 Interface	RS422 Remote Interface
HMPF	1	Microprocessor	CPU, 32K PROM, 8K RAM
HPRB	1	Pulse Rate Inputs	4 Pulse Rate Inputs
HRMB	1	Battery-backed RAM	Back up RAM on a power outage
HUMA	3	Universal Memory	RAM, EPROM or EEPROM
HVDB	1	Video Driver	Interface with [CRT]
HXPD	1	CPU Expander	PROM, 16 Digital I/O, 8 Interrupts
NTCF	3	Thermocouple Input	14 Thermocouple Inputs per Card

Many card slots are wired for several different types of cards. The table shows those cards which are most commonly used.

OPERATOR INTERFACE MODULE [OPM]

Reference Chapter V OPERATOR INTERFACE for a detailed description of the Operator Interface Module.

VIDEO DISPLAY [CRT]

The Video Display is a 12 inch, black and white CRT with the following characteristics:

Power:	60 Watts
Resolution:	800 lines center, 600 lines corner
Linearity:	2%
Horizontal Retrace:	11 microsec max at 15.72 KHZ
Scanning Frequency:	Horizontal = 15.72 KHZ
	Vertical = 50/60 HZ

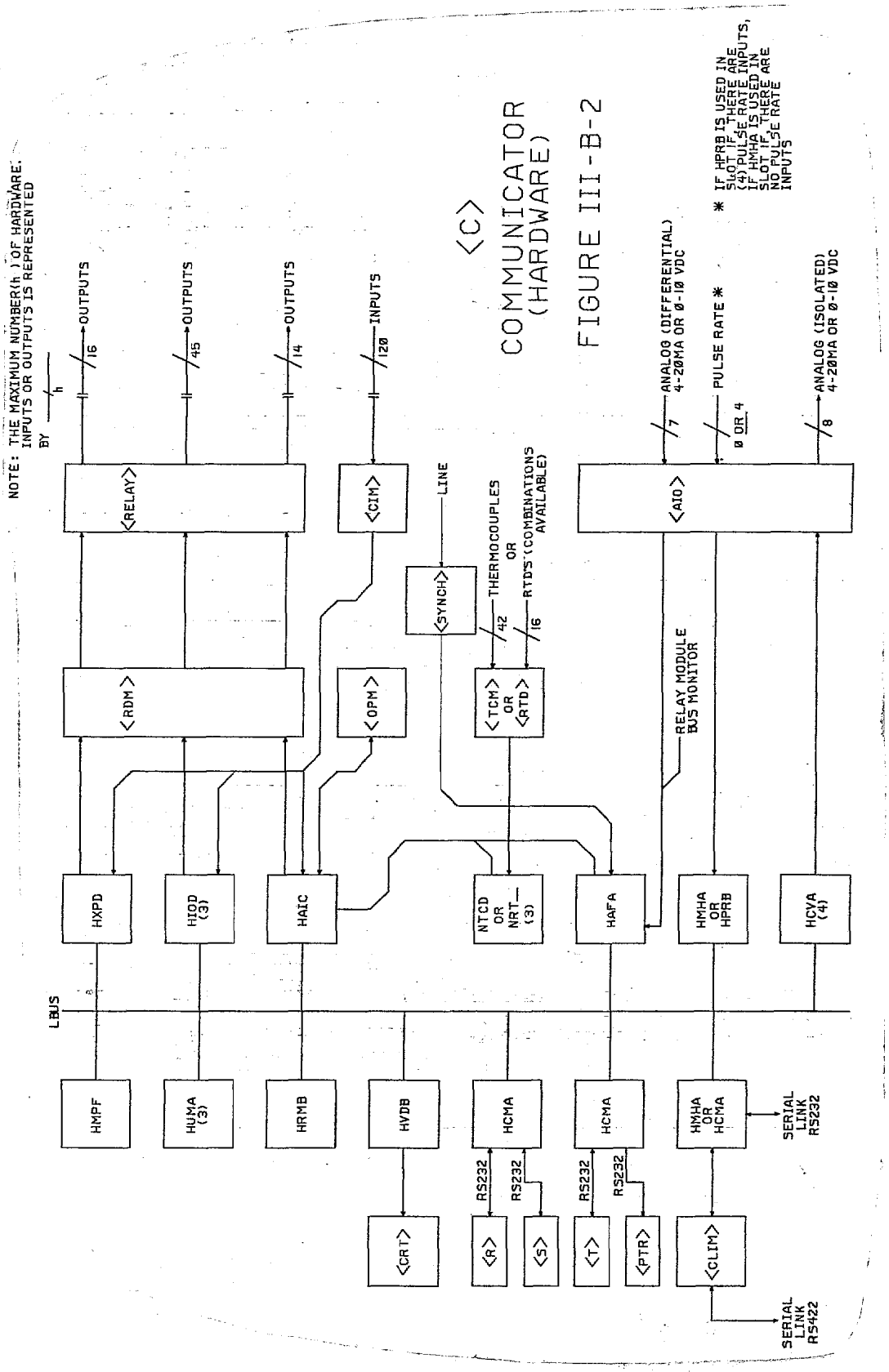


Figure III-B-2: COMMUNICATOR [C] BLOCK DIAGRAM

B. 1. LEFT FRONT DOOR (continued)

PRINTER [PTR]

The panel mounted printer is used for copying [CRT] displays and logging. Its characteristics are:

Power: 60 Watts
150 characters per second at 10 characters per inch
5, 8.18, 10, 16.36 characters per inch
Bi-directional
Take-up Reel
RS232-C Interface
768 Character Buffer
Cassette Ribbon

2. LEFT SIDE WALL

POWER DISTRIBUTION MODULE [PDM]

The power distribution module receives 125VDC from the battery bus and separately fuses all feeders to internal power supplies. Power to the Contact Input Modules, [CIM], is protected with current limiting resistors, an MOV and fuses. Each module can be separately fused or all modules can be wired in parallel. When 24VDC Contact Input Modules are required for intrinsically-safe applications, the 24VDC power input is wired to the Power Distribution Module where it is then internally distributed to the Contact Input Modules. 125VDC power to Relay Module #1, [RELAY1], is used for energizing field solenoids, and it is fused from the main bus, and it can be fuse isolated from the trip solenoids and non-trip solenoids. The module also contains:

Ignition Relay 2TVX
Space Heater (62.5 Watts)
Convenience Receptacle
Two (2) 2A dual element fuses for solenoids
Two (2) current limiting resistors for solenoids
Feeders for the Video Display, [CRT], the Printer, [PTR], panel lighting and the process panel.

FLAME DETECTORS [FD1] [FD2]

Flame detectors monitor ultraviolet scanners on the turbine and generate logic signals to controllers [R], [S] and [T]. The panel mounted detectors receive 28VDC from from the redundant 28VDC power supplies and produce 350VDC, 120HZ outputs to the scanners. Flame in the turbine's combustion chambers ionizes the gas in the scanners which causes a 120HZ current feedback to the panel detector. Voltage follower amplifiers in the detectors turn on switching transistors when a flame is present, thus generating a logic "0" output. The outputs are filtered in the Analog I/O Modules and fed to the HAFA cards in [R], [S] and [T] for monitoring.

B. 2. LEFT SIDE WALL (continued)

SYNCHRONIZING INTERFACE MODULE [SYNCH]

Figure VII-G-1 shows the Synchronizing Interface Module and its interface to the generator control panel and the turbine control panel's computers. The module receives two (2) 120VRMS 60/50HZ inputs from the generator and line potential transformers. These inputs are isolation transformers inside the module which step-down the voltage to 8VRMS prior to conditioning the inputs with zero volt crossover detector circuits and rectifiers. Every time the voltage crosses the zero volt threshold in a positive direction, a TTL square wave output is generated which is in phase with the AC inputs. These outputs, i.e., GR, GS, GT, LR, LS and LT, and the outputs of the voltage level detectors, i.e., DVR, DVS, DVT, SVLR, SVLS and SVLT, are fed to the HPRB and HAIC cards in Controllers [R], [S] and [T].

When the controllers calculate the appropriate time to close the breaker, the HPRB cards send logic outputs to the synchronizing interface module to energize the breaker closure relay "25". A 2 out of 3 vote is performed by the relay driver circuit which generates outputs POSK1 and NEGK1 to drive the relay coil. Three (3) diagnostic circuits monitor the status of the relay driver circuit, and the HAFA card in Communicator [C], monitors each signal for a permissive to synchronize. 1DIAG1 is true when the relay contacts are closed. 0DIAG2 is true when NEGK1 is greater than +6.2V. 0DIAG3 is true when POSK1 is less than +20V.

POWER SUPPLY [PS.C]

The Communicator has its own power supply which is fuse isolated from the other equipment on the 125VDC bus via 10A fuses in the Power Distribution Module, [PDM], and internal 5A fuses on both sides of the 125VDC bus. Power supply characteristics are:

Loading: P5V = 20A; P15V = 1A; N15V = 1A; or
P5V = 15A; P15V = 1.5A; N15V = 1.5A
Input Voltage Range = 80 to 140VDC
P5V Current Limit = 24A
Reverse input polarity protection
Switch SW1 enables output power
Yellow I'M OK light
Line-dip ride-through from $\geq$ 98VDC to 0VDC for 2 milliseconds

3. BACK BASE

POWER SUPPLIES [PS.REL1] [PS.REL2]

These are the redundant 28VDC, 6A power supplies used to energize trip relays and vital modules inside the panel. Their design is identical to [PS.C], described above.

B. 3. BACK BASE (continued)

RELAY MODULE #1 [RELAY1]

The relay modules are built with magnetic relays, instead of solid-state components, because of their predictable failure mode, application flexibility (AC or DC) and form "C" contacts. #24 AWG wire is wire wrapped on the pins of each relay socket, and metal jumpers are connected across screw terminals for summing the current of individual solenoid circuits. Each relay has an indicating light and is labelled with a unique identifier which matches the label mounted next to the socket. There are 15 voted solenoid circuits activated by application software in [R], [S] and [T], and 10 non-voted solenoid circuits with application software in [C]. Each circuit contains inductive suppression (150 ohms, 5 microfarads, diode) mounted on the SSFG cards.

SIXL cards provide an interface between the intermodule wiring and the relay module back-plane. An SIXJ uses diodes to select the higher of the two (2) 28VDC power supply buses and redistributes it to trip/control relays and modules. The bus is referred to as P28VR, the "R" designating the redundant 28VDC bus. An SIXK card reduces the 125VDC and 28VDC buses to 8V which is monitored in [C] for the proper level and possible grounds.

Each Controller, [R], [S] and [T], contains an output port for a master protective relay named 4R, 4S and 4T, respectively. Contacts from these relays vote to energize the 4 relays on a single master protective bus which is then used to supply 28VDC to the coils of trip relays. This ensures that a trip relay will de-energize even if its relay driver has failed in the turned-on mode. Contacts from the emergency stop pushbutton, "5E", and the fire protection relay, "45FTX", are wired in series with the "4" bus independent of the computers. A detailed description and diagram of this circuit are contained in Chapter VII, Section A.

RELAY MODULE #2 [RELAY2]

This module contains 16 voted and 14 non-voted form "C" contact outputs. All coils are on the redundant 28VDC bus, and selected coils can be placed on the "4" bus with jumpers.

RELAY MODULE #3 [RELAY3]

This module contains 30 relays which are activated by Communicator [C], and powered by a non-redundant 28VDC bus. Each relay has one (1) form "C" contact which is wired to pre-defined screw terminals. None of the relays are a function of the master protective "4" bus.

B. 3. BACK BASE (continued)

RELAY MODULE #4 [RELAY4]

This module contains 15 relays which are activated by Communicator [C], and powered by a non-redundant 28VDC bus. Each relay has one (1) form "C" contact which is wired to pre-defined screw terminals. None of the relays are a function of the master protective "4" bus.

CONTACT INPUT MODULES #1 [CIM1] & #2 [CIM2]

Contact inputs which are needed to keep the turbine running, to perform a normal shutdown or to trip the turbine are wired to Contact Input Modules #1 or #2. The signals are then optically isolated and reduced to 8 volts by the internal HSCG cards prior to being fed to the HAIC and HXPD cards in Controllers [R], [S] and [T]. Characteristics of the Contact Input Modules are:

16 contact inputs per module
125VDC and 24VDC versions
Range: 140VDC max to 62VDC min threshold voltage (125VDC version)
30VDC max to 11.7VDC min threshold voltage (24VDC version)
Load: 10.8 milliamps per contact input (125VDC version)
12.0 milliamps per contact input (28VDC version)
Flow-soldered back-plane
Fuse isolation is available in the Power Distribution Module [PDM]

CONTACT INPUT MODULES #3 [CIM3] - #6 [CIM6]

These modules contain 32 contact inputs per module which are connected to HAIC, HXPD and HIOD cards in Communicator [C]. Their characteristics are identical to those described for modules #1 and #2 except for their loading which is one-third (1/3) the load per contact input.

BLACK START INVERTER [INV]

The Black Start Inverter converts 125VDC to 120VAC for the turbine ignition transformers when 120VAC is not available, and it powers the Video Display, [CRT], for start-up monitoring. This function can only be supplied when a starting means, such as a diesel, is used to bring the turbine up to light-off speed without AC power. The module's features are:

Manual on/off switch

Input Voltage	=	105VDC to 250VDC
Output Voltage	=	112VAC
Output Current	=	3A continuous, 6A for 3 minutes
Power Rating	=	1000 Watts DC or 800 Watts AC
V-A Rating	=	1130 VA
Frequency	=	60HZ (type A282C) or 50HZ (type A282D)
Wave Shape	=	Square

B. 4. RIGHT SIDE WALL

ANALOG I/O MODULES #1 [AIO1] - #4 [AIO4]

The Analog I/O Modules provide screw terminal connections for remote analog interface, shield termination bars and passive filtering. All multiplexing and amplification is contained in cards mounted in Controllers [R], [S] and [T], and Communicator [C]. A typical I/O arrangement for heavy duty gas turbine applications is shown in the following table.

Module	Function	Chnl	Assignment	Interface
[AIO1]	LVDT/Servo	1	Modulated IGV	[R] [S] [T] HSAA 2D
		2	Liquid Fuel	[R] [S] [T] HSAA 2D
		3	Gas Control Valve	[R] [S] [T] HSAA 2E
		4	Gas Ratio Valve	[R] [S] [T] HSAA 2E
		5	Nozzle Control	[R] [S] [T] HSAA 2F
		6	Steam Turbine C V	[R] [S] [T] HSAA 2F
[AIO1]	Logic Inputs	1	Flame #1	[R] [S] [T] HAFA 1B
		2	Flame #2	[R] [S] [T] HAFA 1B
		3	Flame #3	[R] [S] [T] HAFA 1B
		4	Flame #4	[R] [S] [T] HAFA 1B
		5	Not Available	[R] [S] [T] HAFA 1B
		6	Not Available	[R] [S] [T] HAFA 1B
		7	Not Available	[R] [S] [T] HAFA 1B
		8	Not Available	[R] [S] [T] HAFA 1B
		9	Fire - 45FTX	[R] [S] [T] HAFA 1B
		10	"4" Permissive	[R] [S] [T] HAFA 1B
		11	"4's" Energized	[R] [S] [T] HAFA 1B
		12	4R, 4S, 4T Agree	[R] [S] [T] HAFA 1B
		13	Spare	[R] [S] [T] HAFA 1B
		14	Spare	[R] [S] [T] HAFA 1B
		15	Spare	[R] [S] [T] HAFA 1B
		16	Spare	[R] [S] [T] HAFA 1B
[AIO2]	Analog Outputs (Non-voted) 4-20MA or 0-10VDC	1	Spare	[C] HCVA 1B
		2	Spare	[C] HCVA 1B
		3	Spare	[C] HCVA 1C
		4	Spare	[C] HCVA 1C
		5	Spare	[C] HCVA 1D
		6	Spare	[C] HCVA 1D
		7	Spare	[C] HCVA 1E
		8	Spare	[C] HCVA 1E

B. 4. RIGHT SIDE WALL (continued)

ANALOG I/O MODULES #1 [AIO1] - #4 [AIO4]

Module	Function	Chnl	Assignment	Interface
[AIO2]	Vibration	1	Gas Turbine	[R] [S] [T] NVCB 2B
		2	Gas Turbine	[R] [S] [T] NVCB 2B
		3	Gas Turbine	[R] [S] [T] NVCB 2B
		4	Gas Turbine	[R] [S] [T] NVCB 2B
		5	Gas Turbine	[R] [S] [T] NVCB 2B
		6	Generator	[R] [S] [T] NVCB 2B
		7	Generator	[R] [S] [T] NVCB 2A
		8	Generator	[R] [S] [T] NVCB 2A
		9	Steam Turbine	[R] [S] [T] NVCB 2A
		10	Steam Turbine	[R] [S] [T] NVCB 2A
		11	Steam Turbine	[R] [S] [T] NVCB 2A
		12	Spare	[R] [S] [T] NVCB 2A
[AIO3]	Pressure and Analog Inputs	1	Compressor Disch.	[R] [S] [T] NVCB 2B
		2	Gas Fuel Pressure	[R] [S] [T] NVCB 2B
		3	Watts	[R] [S] [T] NVCB 2B
		4	VARS	[R] [S] [T] NVCB 2B
		5	Spare	[R] [S] [T] NVCB 2A
		6	Spare	[R] [S] [T] NVCB 2A
		7	Spare	[R] [S] [T] NVCB 2A
		8	Spare	[R] [S] [T] NVCB 2A
[AIO3]	Analog Inputs Differential 4-20MA or 0-10VDC	1	Spare	[R] [S] [T] HAFA 1B
		2	Spare	[R] [S] [T] HAFA 1B
		3	Spare	[R] [S] [T] HAFA 1B
		4	Spare	[R] [S] [T] HAFA 1B
		5	Spare	[R] [S] [T] HAFA 1B
		6	Spare	[R] [S] [T] HAFA 1B
		7	Spare	[R] [S] [T] HAFA 1B
		8	Spare	[R] [S] [T] HAFA 1B
		9	Spare	[R] [S] [T] HAFA 1B
		10	Spare	[R] [S] [T] HAFA 1B
		11	Spare	[R] [S] [T] HAFA 1B
		12	Spare	[R] [S] [T] HAFA 1B
		13	Spare	[R] [S] [T] HAFA 1B
		14	Spare	[R] [S] [T] HAFA 1B
		15	Spare	[C] HAFA 2F
		16	Spare	[C] HAFA 2F
		17	Spare	[C] HAFA 2F
		18	Spare	[C] HAFA 2F
		19	Spare	[C] HAFA 2F
		20	Spare	[C] HAFA 2F
		21	Spare	[C] HAFA 2F

B. 4. RIGHT SIDE WALL (continued)

ANALOG I/O MODULES #1 [AIO1] - #4 [AIO4]

Module	Function	Chnl	Assignment	Interface
[AIO3]	Pulse Rate	1	LP Speed/ST Spd	[R] [S] [T] HPRB 1A
		2	Starting Device Spd	[R] [S] [T] HPRB 1A
		3	Liquid Fuel Flow	[R] [S] [T] HPRB 1A
		4	HP Speed	[R] [S] [T] HPRB 1A
		5	Spare	[C] HPRB 1F
		6	Spare	[C] HPRB 1F
		7	Spare	[C] HPRB 1F
		8	Spare	[C] HPRB 1F
[AIO4]	Analog Outputs (Voted)	1	Spare	[R] [S] [T] HCVA 1E
		2	Spare	[R] [S] [T] HCVA 1E

THERMOCOUPLE MODULES [TCM.R], [TCM.S], [TCM.T], [TCM.C1], [TCM.C2] and [TCM.C3]

There are six (6) Thermocouple Modules which provide 14 thermocouple inputs, a cold junction reference and an offset measurement to each Controller, [R], [S] and [T], and 42 inputs to the Communicator, [C]. The card interface is:

[TCM.R] to [R] NTCF slot 1C
[TCM.S] to [S] NTCF slot 1C
[TCM.T] to [T] NTCF slot 1C
[TCM.C1] to [C] NTCF slot 2D
[TCM.C2] to [C] NTCF slot 2C
[TCM.C3] to [C] NTCF slot 2B

The first input provides the cold junction reference and can be isolated with a jumper if a remote copper junction reference is used. The last input is jumpered at the screw terminals for the offset measurement. Note that the cold junction and offset inputs are in addition to the 14 thermocouple inputs. A maximum incoming lead length of 1000 feet can be used with #16 or #18 AWG twisted pair, grounded or ungrounded. The modules flow-soldered back-plane contains high frequency decoupling to ground on each side of the thermocouple pairs. All modules are covered for a uniform temperature measurement.

RTD INPUT MODULE [RTD]

This module is available for ungrounded RTD's only, which excludes most generator applications. Sixteen (16) RTD's can be terminated on the module using either two (2) wire or three (3) wire RTD's. The module's flow-soldered back-plane contains high frequency decoupling to ground on all incoming signals. Monitoring of RTD's is performed by application software in Communicator [C] by NRT_ cards which have different forms corresponding to different types of RTD's. In addition, these cards share the same slots, i.e., 2B, and 2C, as some thermocouple inputs; therefore, the available combinations of thermocouples and RTD's in [C] are:

B. 4. RIGHT SIDE WALL (continued)

RTD INPUT MODULE [RTD] (continued)

42 thermocouples and 0 RTD's
28 thermocouples and 8 RTD's
14 thermocouples and 16 RTD's

The following card forms are available for RTD interface:

NRTA	10 ohm copper at 25 degrees C, range	=	-50 to 260 degrees C
NRTB	100 ohm platinum at 0 degrees C, range	=	-50 to 260 degrees C
NRTC	100 ohm platinum at 0 degrees C, range	=	-50 to 660 degrees C
NRTD	120 ohm nickel at 0 degrees C, range	=	-50 to 260 degrees C

COMMUNICATION LINE ISOLATION MODULE [CLIM]

This module provides an isolated, synchronous, RS422 serial link for remote interface to remote panels or data acquisition systems supplied by General Electric Company. Its characteristics are:

a. Optical isolation of 200V maximum common mode
b. 50K baud at 4,000 feet maximum
c. Field interface is via #18 AWG shielded, twisted pair
d. 8 drivers and 1 receiver or 1 driver and 8 receivers
e. The parity line is not disabled by powering down or plugging or unplugging the isolation card HIMA.

28VDC TERMINATION MODULE [P28X]

This module receives 28VDC from the redundant 28VDC power supplies and provides 0.5A for field use. The feeder from each power supply is used in a diode high select circuit which provides a redundant bus for the field in addition to separate buses from each supply. All buses are fused and contain LED indication.

CONTROL COMMON BAR, SHIELD BAR and GROUND BAR

All control commons are wired to a single termination bar at the top of the right side wall which is mounted on insulated stand-offs. Shields are wired to shield bars on the individual I/O modules, and then wired to a single termination bar at the bottom of the right side wall. The bar is insulated from the panel case with stand-offs. Ground straps connect the shield and control common termination bars to a single ground bar located directly below the shield termination bar at the bottom of the right side wall. This bar is bolted to the metal case. Its ground lug must be connected to the plant ground grid with a cable of up to #4 AWG.

B. 5. RIGHT FRONT DOOR

POWER SUPPLIES [PS.R], [PS.S] & [PS.T]

Each Controller has its own power supply which is fuse isolated from the other equipment on the 125VDC bus via 10A fuses in the Power Distribution Module, [PDM], and internal 5A fuses on both sides of the 125VDC bus. Power supply characteristics are:

Loading: P5V = 20A, P15V = 1A, N15V = 1A or
P5V = 15A, P15V = 1.5A, N15V = 1.5A
Input Voltage Range = 80 to 140VDC
P5V Current Limit = 24A
Reverse input polarity protection
Switch SW1 enables output power
Yellow I'M OK light
Line-dip ride-through from $\geq$ 98VDC to 0VDC for 2 milliseconds

CONTROLLERS [R], [S] & [T]

The Controllers contain all of the application software and computer hardware needed to keep the turbine running after it has completed its start-up sequence and all of the software and hardware necessary for a normal shutdown. All hardware and software are identical in each Controller. A typical assortment of cards is shown in the following table, and figure III-B-3 provides a block diagram of the Controllers.

Card	Qty	Function	Application
HAFA	1	Analog/Digital Ins	16 Analog and 32 Digital Inputs
HAIC	1	Analog Input	16 Analog Ins. and [OPM] Interface
HCVA	1	Analog Output	2 Analog Outputs
HMPF	1	Microprocessor	CPU, 32K PROM, 8K RAM
HPRB	1	Pulse Rate Inputs	4 Pulse Rate Inputs
HSAA	3	Servo Amplifier	2 LVDT/Servo Channels per card
HXPD	1	CPU Expander	PROM, 16 Digital I/O, 8 Interrupts
NTCF	1	Thermocouple Input	14 Thermocouple Inputs
NVCB	2	Vibration/Pressure	6 Vibration/4 Pressure per card

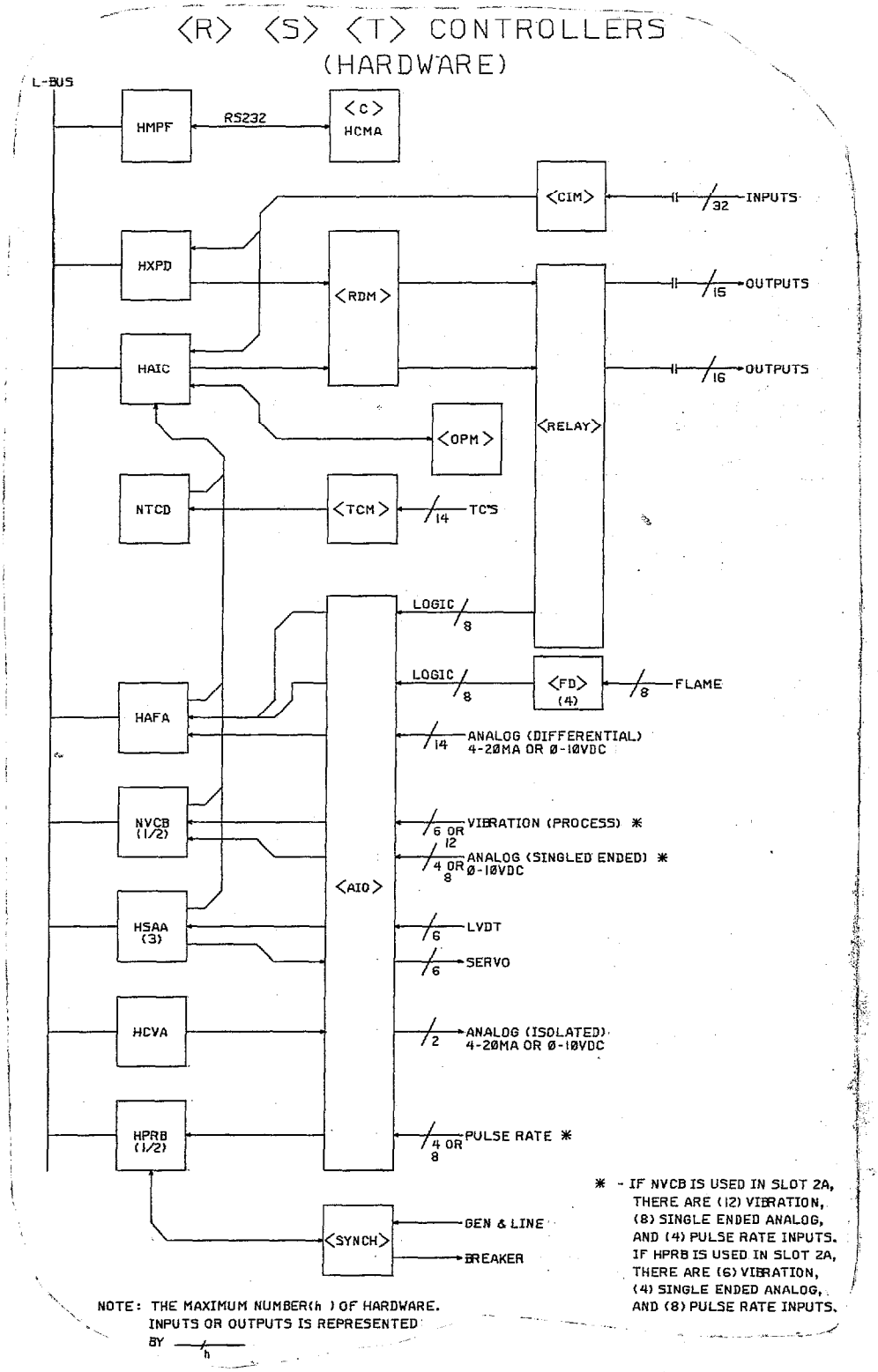


Figure III-B-3: CONTROLLERS [R], [S] & [T] BLOCK DIAGRAM

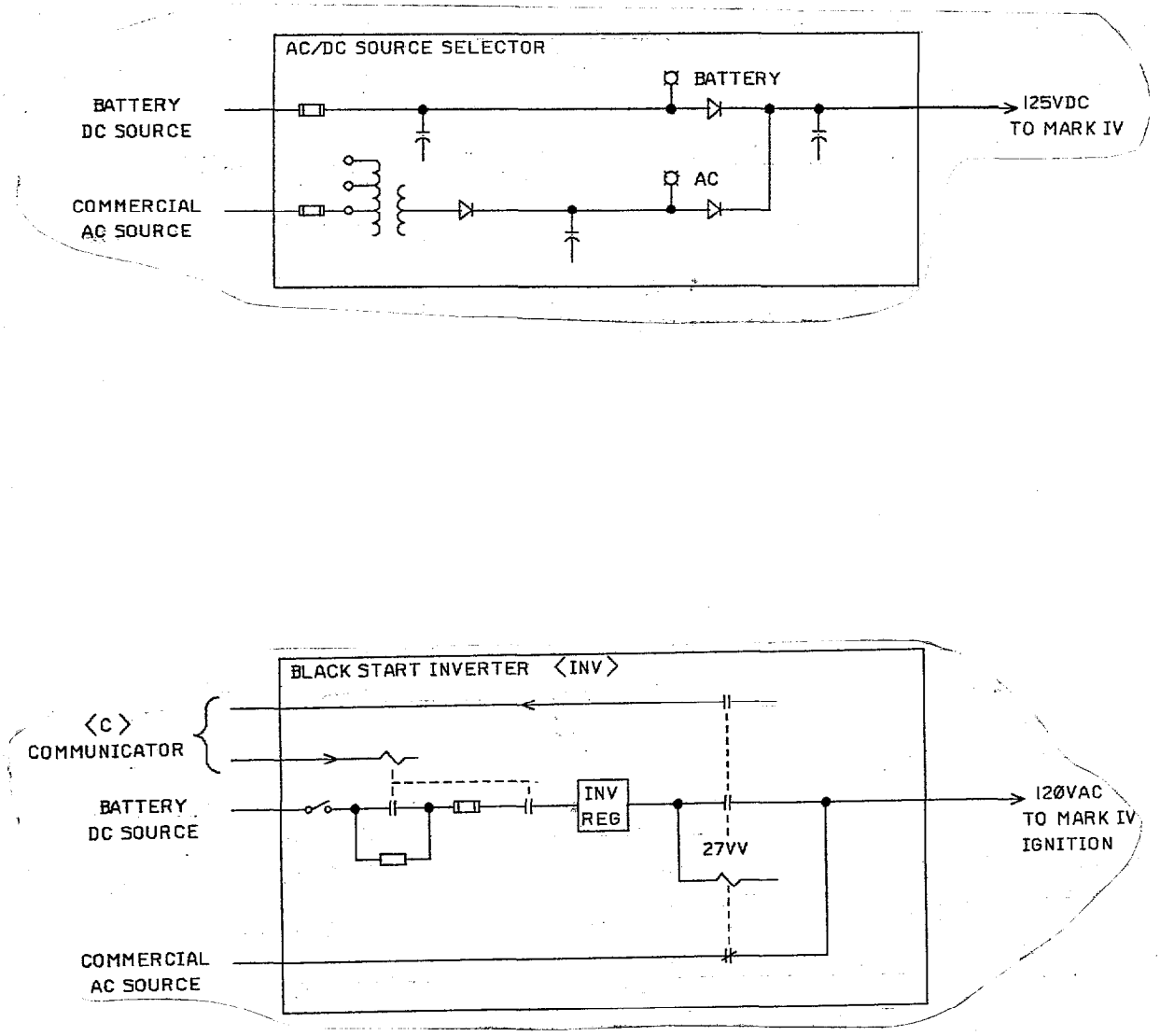


Figure IV-A-1: AUXILIARY POWER MODULES

IV. POWER SUPPLY AND GROUND SYSTEMS

A. POWER SUPPLY SYSTEM

1. GENERAL

125VDC is provided by the station batteries and connected to a 50A circuit breaker in the motor control center. A feeder is then run to the turbine control panel for its primary power source. The batteries usually consist of 56 calcium alloyed lead-acid cells floating at 2.35V per cell with a no load input of 2.0V per cell and a discharged input of 1.75V per cell. Nominal 125VDC load requirements of the panel are:

Power Supplies	=	7.5 A-DC	
External Solenoids	=	2.0 A-DC	(typical)
Contact Inputs	=	0.5 A-DC	
Total	=	10.0 A-DC	* not including Black Start Inverter

Range = 105VDC to 135VDC
Ripple = 5V Peak to Peak maximum

AC power to the panel is provided through an isolation transformer with ground on one of the center taps of the secondary winding. The secondary voltage is then connected to a circuit breaker in the motor control center which is connected to the turbine control panel. AC power is 115/230V, 50/60HZ with nominal 115V loads as follows:

CRT	=	0.5A	
Printer	=	0.5A	
Heater	=	0.5A	
Lights	=	1.0A	
Total	=	2.5A	* not including convenience receptacle

2. AUXILIARY POWER MODULES

Applications which use AC power sources or AC/DC combination sources require rectifier modules. Figure IV-A-1 shows a DC/AC Source Selector which compares the battery bus with the rectified AC commercial bus and selects the highest bus through a "diode high-select" circuit to generate the 125VDC output to the panel's power supplies. Each bus is protected with a fuse and an indicating light is provided.

The Black Start Inverter, [INV], is used to convert 125VDC to 120VAC for the turbine ignition transformers when 120VAC is not available. It is used for back start applications when a starting means, such as a diesel, is available to bring the turbine up to light-off speed without AC power. The module is mounted inside the turbine panel at

A. 2. AUXILIARY POWER MODULES (continued)

the top of the back base, and it is activated through software. Since the CRT is used to supervise the black start, the inverters square wave output feeds the CRT through a separately fused circuit in the Power Distribution Module, [PDM]. Figure IV-A-1 shows a one-line diagram of the Black Start Inverter. A disconnect switch and a fuse are in series with the battery bus. Relay 4VV is activated by Communicator [C], to initiate a black start, and the inverter regulator inverts the 125VDC to 120VAC. Undervoltage relay 27VV is used to switch the AC output between the AC commercial bus and the inverter bus. In addition, it provides a feedback to the Communicator to confirm that the inverter is working properly.

3. PANEL POWER DISTRIBUTION

The Power Distribution Module is located on the bottom of the left side wall. It is used to fuse isolate the 125VDC and 120VAC buses from the various modules in the panel and devices in the field. Figure IV-A-2 shows a diagram of the module. Each computer has its own power supply identified as [PS.R], [PS.S], [PS.T] and [PS.C]; all of the supplies are identical. They are rated (P5V = 20A, P15V = 1A, N15V = 1A) or (P5V = 15A, P15V = 1.5A, N15V = 1.5A). If a computer needs to be repaired while the control is on-line, the supply can be turned off by a switch on the supply, and if the supply needs to be replaced, the fuses in the Power Distribution Module can be removed to isolate the supply. Each supply contains an internal 5A fuse on each side of the 125VDC bus in addition to the 10A fuses in the Power Distribution Module. All connections to the Power Distribution Module from modules inside the panel are secured with 2-pin lockable connectors. The redundant 28V, 6A power supplies, [PS.REL1] and [PS.REL2], are used to power the relays. They are designed similar to the computer supplies. Fuse isolation between trip and non-trip solenoids is an available option provided by the two (2) 15A fuses feeding Relay Module #1, [RELAY1]. Feeders to the Contact Input Modules, [CIM], are conditioned by an MOV, filtering capacitor and current limit resistor. Fuse isolation between Contact Input Modules is an available option. It is monitored by the Communicator to alarm a blown fuse and specify which fuse has blown. The Communicator monitors all internal power supplies, and alarms when any supply is out of tolerance, or if there is an undervoltage or ground on the 125VDC battery bus. This ground calculation is performed by algorithm "64D".

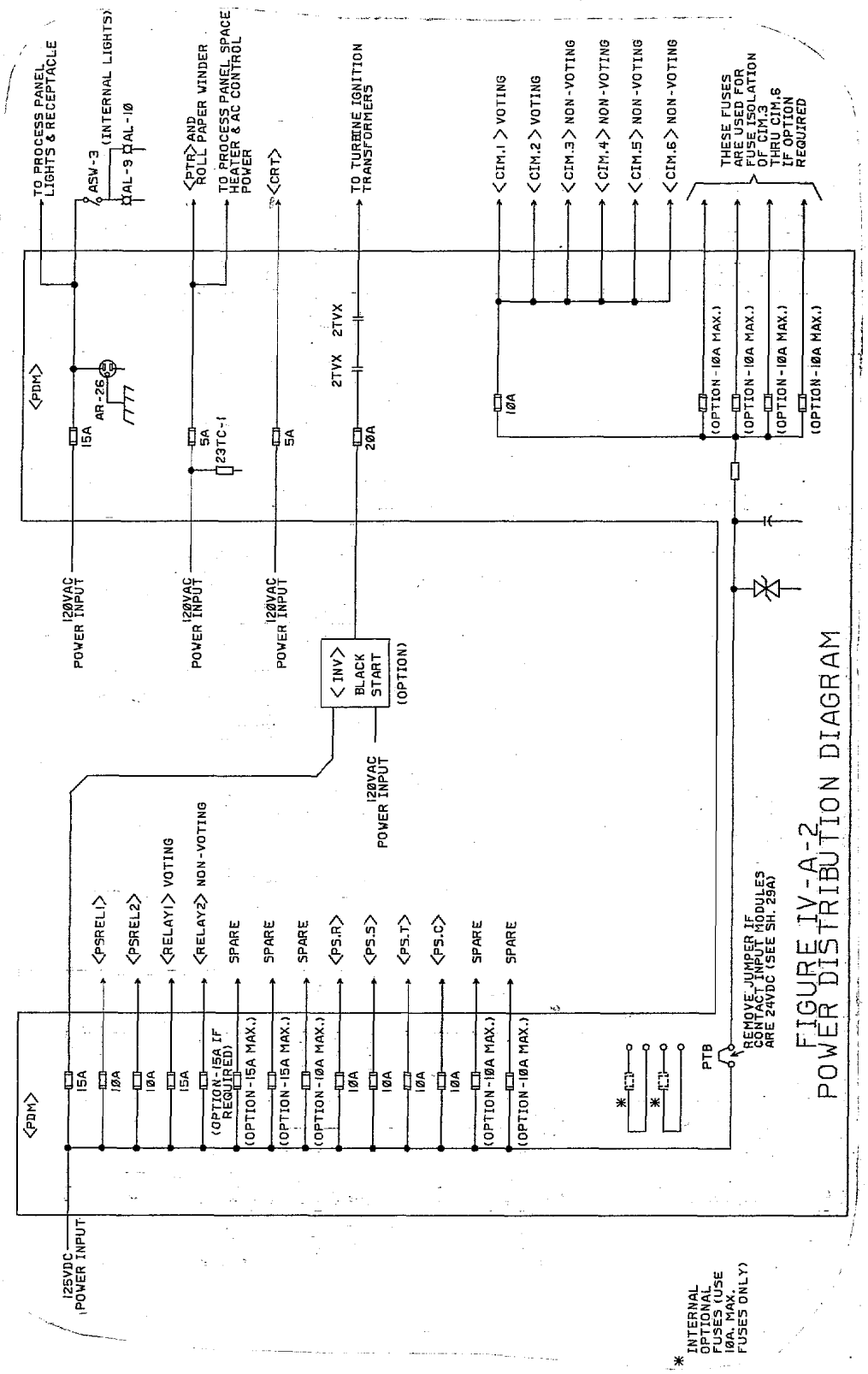


Figure IV-A-2: POWER DISTRIBUTION DIAGRAM

B. GROUND SYSTEM

All control commons are wired to a single termination bar at the top of the right side wall which is mounted on insulated stand-offs. Shields are wired to shield bars on the individual I/O modules. They are then wired to a single termination bar at the bottom of the right side wall which is insulated from the panel case with stand-offs. Ground straps connect the shield and control common termination bars to a single ground bar which is located directly below the shield termination bar at the bottom of the right side wall. This bar is bolted to the metal case, and its ground lug must be connected to the plant ground grid with a cable up to #4 AWG. Earth ground at the installation site is an electrical mat or grid buried in moist earth or a number of conducting rods driven an adequate distance into the earth and electrically tied together. It is used to shunt stray electrical currents from machinery, building structures and electrical equipment to ground for the protection of personnel and equipment. Capacitive noise filters are connected to their module's metal case which is then grounded with star washers fastened to each corner of the module. Safety grounds are wired directly to the panel case, and all hinged surfaces are connected to the case with ground straps.

Figure IV-B-1 illustrates the 125VDC ground detection circuit. The bus is fused in the power distribution module, [PDM], and fed to relay module #1, [RELAY1]. A SIXK card splits the bus into +62.5VDC and -62.5VDC segments with respect to ground, and a voltage divider reduces the voltage to 5.09VDC. Both channels are fed to input amplifiers on the HAFA card in the Communicator, [C], and then connected to the HAIC card for the analog to digital conversion. Algorithm GROUND_LEAK calculates the line to line voltage, i.e., S125P_N, and compares it to a setpoint to determine if the bus level is normal, i.e., L27DN = "1", or if an undervoltage condition exists, L27DN = "0". The ground calculation adds the positive bus to the negative bus which will equal "0" at point "A" if there is no ground and the software equivalent of 125VDC if a ground exists. The equation for point "E" is "E" = 60 [(125VDC/"A" absolute) - 1]/2, which will normally be infinity with no ground causing output L64D to be a "0"; a solid ground causes "E" to equal "0" which is less than setpoint "D" causing L64D to be a "1".

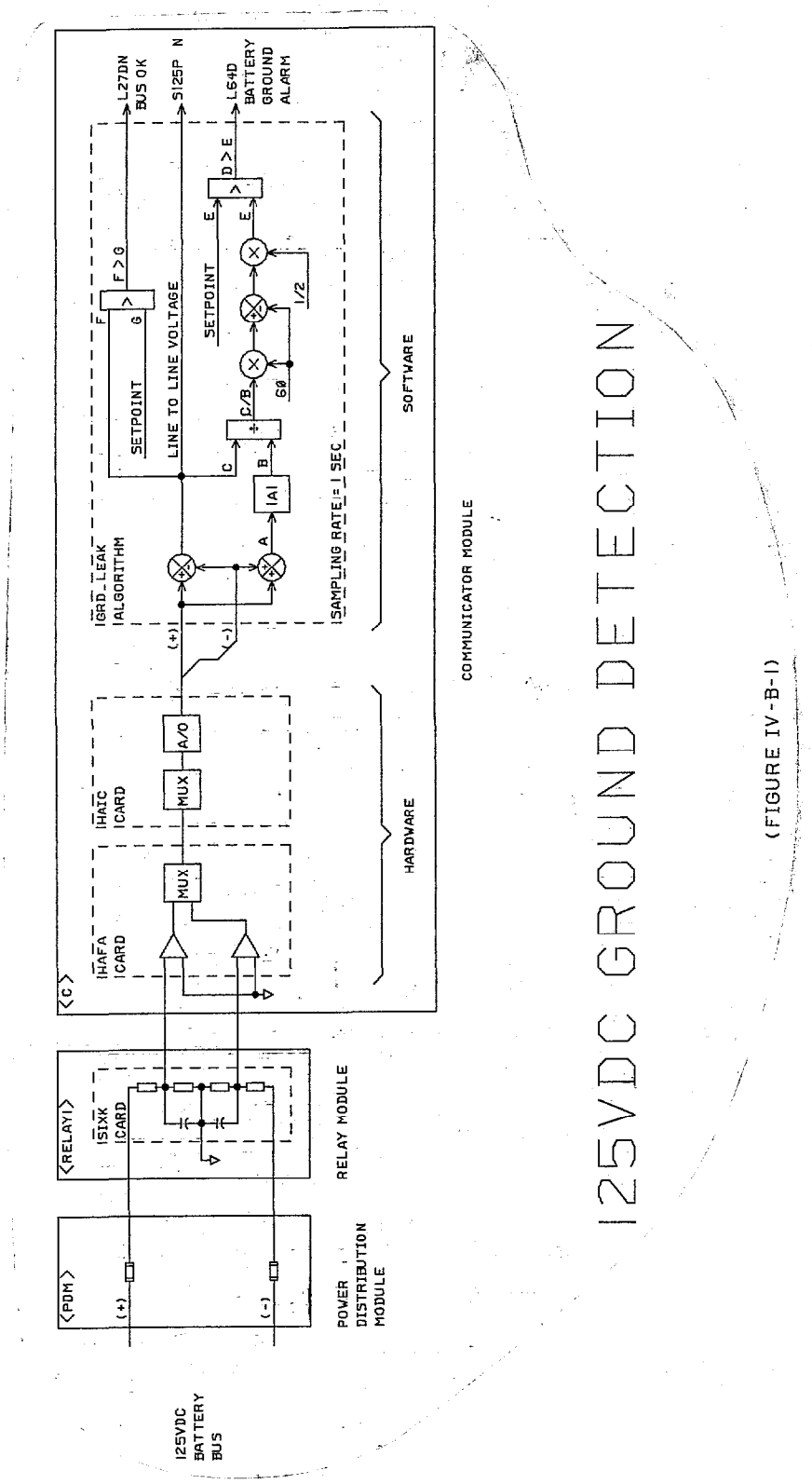


Figure IV-B-1: 125VDC GROUND DETECTION

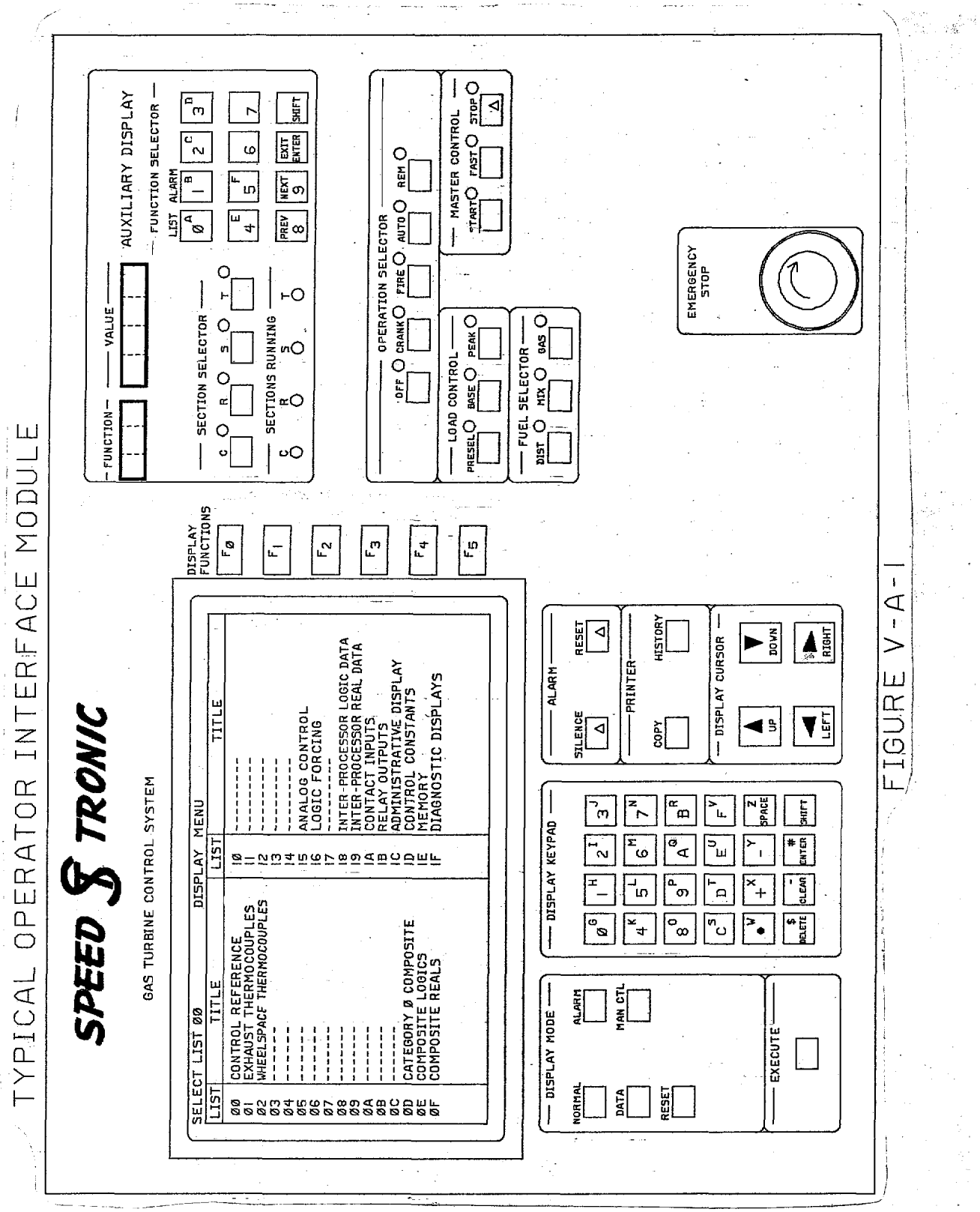


Figure V-A-1: OPERATOR INTERFACE MODULE (TYPICAL)

V. OPERATOR INTERFACE

A. THE OPERATOR INTERFACE MODULE [OPM]

The Operator Interface Module is the primary medium for interfacing with the control. Refer to Figure V-A-1 for a photograph of the module. The module is mounted on the left front door and is functionally divided between the left side which interfaces with the Video Display, [CRT], and the right side which interfaces with the control.

Operator pushbuttons are shown on the lower right hand side. A 7 X 4 matrix of membrane switches is covered by a graphic mylar overlay which assigns a function to each switch and "blacks out" unused switches. An HPIB card interfaces with the switches and Communicator [C], and the Controllers, [R], [S] and [T]. When a switch is pressed a beeper provides an audible acknowledgement of the command and a red LED begins to blink. The operator then has three (3) seconds to confirm the command by pressing the EXECUTE button in the lower left hand corner. If this button is not pressed in time, the red LED will stop blinking, and the control will wait for a new command. This procedure protects the control against accidental pressing of operator switches. Functional groups of switches, such as the operator selection switches and the EXECUTE switch, are color coded to highlight the commonality of control function. All switches in the 7 X 4 matrix interface with [C] through the HPIB card, except for three switches. The stop switch is placed on the second row on the far right-hand side and interfaces with [R], [S] and [T] for added reliability when initiating a normal shutdown sequence. And two (2) switches on the fourth row on the far right-hand side are connected to [R], [S] and [T], via the HPIB card. These are commonly used for manually raising and lowering the speed/load setpoint on compressor drives. Normally pressing the EXECUTE button is not required prior to or during a raise/lower command, but the EXECUTE feature is available if required. In the event of Communicator [C] failure, the stop and raise/lower buttons would not require pressing the EXECUTE button prior to initiating the command. Power for the switch matrix comes from the Communicator's 5V bus except the voted pushbuttons which are powered by a separate supply, the NPSM card. A shrouded EMERGENCY STOP pushbutton has a contact wired in series with the master protective "4" relays in Relay Module #1, [RELAY1], to directly de-energize the fuel solenoids independent of the computers. The status of the the EMERGENCY STOP pushbutton is monitored by the HAIC cards in [R], [S] and [T]. These cards, and the HAIC card in [C] provide all digital interface with the Operator Interface Module. The HVDB card in [C] is used to drive the Video Display [CRT].

The AUXILIARY DISPLAY provides back-up display and acknowledgement capability for key system parameters. It is used for accessing the computers' respective operating systems. Four (4) SECTIONS RUNNING LED's continuously monitor the running status of [C], [R], [S] and [T], and four (4) SECTION SELECTOR switches are used to select which computer's data will be displayed. Power for the AUXILIARY DISPLAY is supplied by the NPSM card.

A. THE OPERATOR INTERFACE MODULE [OPM] (continued)

AUXILIARY DISPLAY Operator Commands:

1. Touching SHIFT followed by EXIT causes the AUXILIARY DISPLAY to enter the mode select state which is indicated by a "1" rotating through the display field. Any display mode can be entered at this point.

2. Touching SHIFT followed by NEXT causes the display to advance.

3. Touching SHIFT followed by PREV causes the display to decrement.

4. Touching SHIFT followed by MODE causes the mode number, i.e., 1-6, to be displayed in the FUNCTION field for as long as MODE is pressed.

AUXILIARY DISPLAY Modes:

1. LIST (key pad point "0")

   This mode displays a pre-defined list of data points which are displayed for two (2) seconds each as they rotate through the display. Pressing NEXT or PREVIOUS will freeze the display and step the display one point at a time in either direction. Touching SHIFT followed by exit returns the display to the Mode Select state.

2. ALARM (key pad point "1")

   This display shows the number of all alarms in the order of their occurrence with unacknowledged alarms blinking and acknowledged alarms constant. Initially the first unacknowledged alarm appears. Pressing NEXT acknowledges the alarm and advances the display. Pressing RESET below the CRT deletes acknowledged alarms which are no longer present. Touching SHIFT followed by EXIT returns the display to the Mode Select mode.

Switches on the left side of the Operator Interface Module are dedicated to [CRT] interface and powered from [C]. The exceptions are the ALARM SILENCE and ALARM RESET switches which interface with [R], [S] and [T] through the HPIB card.

B. DISPLAYS AND LOGGING

Reference the "Display Organization" chart shown in Figure V-B-1. Five categories of displays are selected by pushbuttons located in the bottom left-hand corner of the Operator Interface Module, [OPM], in the DISPLAY MODE group.

1. NORMAL DISPLAYS

   The NORMAL pushbutton in the DISPLAY MODE group is used to select the Normal Display which contains all vital turbine parameters, reference Figure V-A-1. Three (3) different Normal Displays are dedicated to the Start-up, Running and Shutdown modes of operation and are

B. DISPLAYS AND LOGGING (continued)

1. NORMAL DISPLAYS (continued)

automatically changed when the operating mode changes. Parameters are displayed in engineering units and highlighted with inverse-video. A SELECT field on the bottom right-hand side of the display is used to monitor any three (3) control parameters simultaneously. These signals can be control or logic type signals, and they can be called by their functional name as it appears in the Control Elementary with out referencing the specific location in memory where the signal resides. An ALARM field on the bottom left-hand side of the display shows the number of acknowledged alarms and the most recent three (3) unacknowledged alarms in inverse video.

2. ALARM DISPLAYS

The ALARM pushbutton in the DISPLAY MODE group selects the alarm display containing the time and date of up to 19 alarm messages and a system queue with 57 messages. A new alarm appears flashing in inverse video at the bottom of the list, and is accompanied by an audible alarm. The alarm can be acknowledged by pressing the ALARM ACK pushbutton at right of the CRT. Alarm SILENCE and RESET pushbuttons are provided in the ALARM group below the CRT. An ALARM REVIEW pushbutton at the right of the CRT recycles the messages one at a time starting with the first alarm.

Diagnostic alarm messages can be observed by pressing the DIAGNOSTIC ALARM pushbutton at the right of the CRT. This display contains up to 19 messages on the initial display with a capability for 40 messages in the system queue and ALARM ACKNOWLEDGE, ALARM REVIEW and ALARM RESET pushbuttons at the right of the CRT. A detailed description of this display is provided.

An Historical Log of the alarm messages that have occurred during the last 72 hours can be obtained by pressing the HISTORY pushbutton in the PRINTER group. A print of the current CRT display can be obtained by pressing the COPY pushbutton in the PRINTER group. An option for automatic alarm logging is available which prints the alarm message when the alarm condition occurs and when it clears.

A lockout feature is provided to prevent a dithering field condition from repetitively setting a given alarm drop, and consequently, logging the same alarm repetitively. To lockout an alarm, press ALARM LOCOUT, the alarm number and ENTER. To unlock an alarm, press ALARM UNLOC, the alarm number and ENTER. Time tags have a resolution of 0.125 seconds which corresponds to the sampling rate of the sequencing code. An example of a typical alarm message display is shown in Figure V-B-1.

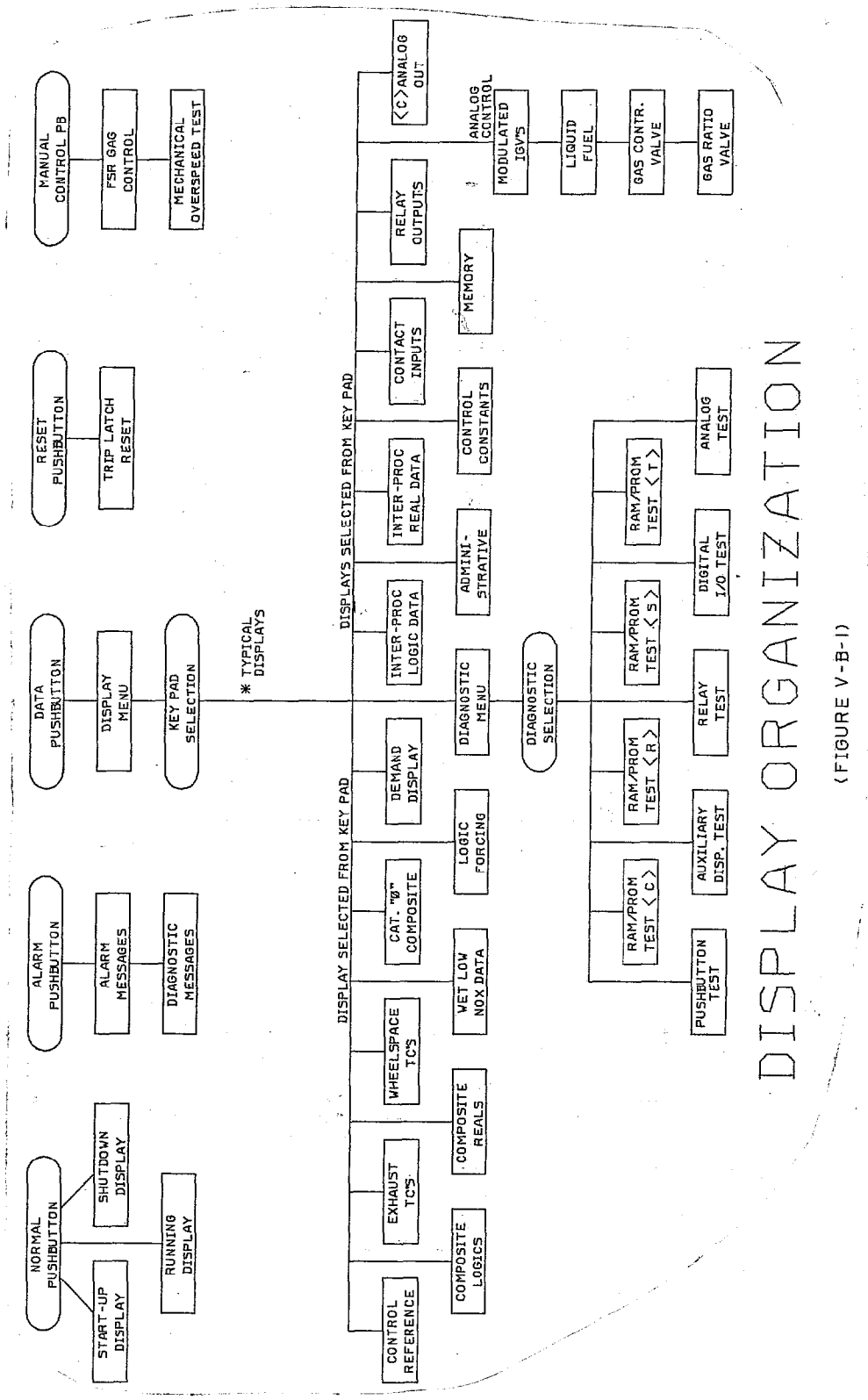


Figure V-B-1: DISPLAY ORGANIZATION (TYPICAL)

B. DISPLAYS AND LOGGING (continued)

ALARM STATUS 29 APRIL 83 18:16:09

DATE	TIME	DROP	ALARM
29 APRIL 83	17:40:56	31	GEN AIR FILTER - HIGH PRESS DROP
29 APRIL 83	18:15:15	32	DIESEL LUBE PRESSURE LOW
29 APRIL 83	18:16:02	68	BATTERY GROUND

3. TRIP-LATCH RESET DISPLAY

Some trip conditions are latched to prevent a restart until the operator manually resets the latch. This is accomplished by pressing the RESET pushbutton in the DISPLAY MODE group and then pressing the RESET pushbutton at the right of the CRT.

4. MANUAL CONTROL DISPLAYS

Manual Control displays are used primarily for maintenance functions and to supplement the 7 X 4 matrix of operator pushbuttons. Each display is dedicated to a function such as FSR GAG, MECHANICAL OVERSPEED TEST or MODULATED IGV and it contains all of the parameters relevant to that function. The right side of the display assigns a function to one (1) or more of the six (6) membrane switches located to the right of the CRT which can be pressed to initiate some control action. To access a display, press the MAN CTL pushbutton in the DISPLAY MODE group. Then press the NEXT PAGE pushbutton at the right of the CRT until the required display appears.

5. HISTORICAL LOG

An historical log will be printed automatically if a turbine trip occurs. It can be manually printed by pressing the HISTORY pushbutton. The log includes the primary turbine control parameters and the status of all of the alarm messages which have occurred during the past 72 hours as they appeared at ten (10) points in time.

6. DISPLAY MENU

A display menu can be obtained by pressing the DATA pushbutton in the DISPLAY MODE group. Then the DISPLAY KEYPAD is used to enter the list number of the desired display. A brief description of typical displays is provided for your reference.

CONTROL REFERENCE (List 00)
Contains key control parameters.

EXHAUST THERMOCOUPLES (List 01)
Displays all exhaust thermocouples and their average value.

B. DISPLAYS AND LOGGING (continued)

6. DISPLAY MENU (continued)

WHEELSPACE THERMOCOUPLES (List 02)
Shows the temperature of all wheelspace thermocouples.

LUBE OIL THERMOCOUPLES (List 03)
This display shows all lube oil thermocouple readings.

TIMERS AND EVENT COUNTERS (List 04)
The current values of all timers and counters such as FIRED TIME and TOTAL STARTS are shown on this display. Data is stored in EEPROM for permanent memory retention.

WET LOW NOX DATA (List 07)
All data required by the Environmental Protection Agency to verify compliance with low nitrous oxide emission regulations is contained in this display. A copy of the display is automatically printed every hour, and unacceptable emission levels are alarmed at ten (10) minute intervals.

CATEGORY 0 COMPOSITE (List 0D)
Contains all signals which are not logics or reals, i.e., integers.

COMPOSITE LOGICS (List 0E)
Shows all logic signals.

COMPOSITE REALS (List 0F)
Displays all real, i.e., integer, signals.

TRIP DISPLAY (List 10)
This optional display shows the CURRENT VALUE of various trip signals in [C], [R], [S] and [T]. It also shows the FROZEN VALUE of the trip signals at the time the trip occurred.

[C] ANALOG OUT (List 11)
This optional display contains up to eight (8) parameters of any type which are field selectable. They can be output as 4 to 20MA or 0 to +10VDC from HCVA cards mounted in Communicator [C]. The values of these parameters are usually updated at 4HZ due to the sampling rate of the algorithms which calculate the value of each parameter.

DEMAND DISPLAY (List 12)
Provides two (2) optional displays of 32 points each which can be used to display any control or logic signal. Automatic logging is available with a time setting of up to one (1) day.

HCVA SERVICE DISPLAY (List 14)
This optional display is used to force 4 to 20MA or 0 to +10VDC analog outputs from HCVA card channels which are dedicated to specific functions in [C], and/or [R], [S] and [T]. This display is normally enabled when the control is in the CRANK mode of operation.

B. DISPLAYS AND LOGGING (continued)

6. DISPLAY MENU (continued)

ANALOG CONTROL (List 15)
Servos can be automatically calibrated from the ANALOG CONTROL display corresponding to the specified servo. Each display contains a NEXT PAGE pushbutton to the right of the CRT for sequentially "stepping" through each display, i.e., page. Typical displays are MODULATED INLET GUIDE VANE CONTROL, LIQUID FUEL CONTROL, GAS FUEL VALVE CONTROL and GAS RATIO VALVE CONTROL. The calibration procedure is:

a. Select CRANK.

b. Enter the security code to access the calibrate mode.

c. Position the actuator to the minimum stroke, and:
   i. Adjust the LVDT to give 0.7VRMS.
   ii. Measure the displacement at the valve stem.
   iii. Push ADJUST MODE until LOCATE LOW appears in inverse video.
   iv. Push RAISE/LOWER until the correct valve position is displayed.

d. Position the actuator to the maximum stroke, and:
   i. Measure the displacement at the valve stem.
   ii. Push ADJUST MODE until LOCATE HIGH appears in inverse video.
   iii. Push RAISE/LOWER until the correct valve position is displayed.
   iv. Push CALIBRATE.
   Both CAL OFFSET and CAL GAIN will now change to new values.

e. The actuator may be forced via GSADJ anytime the display is in the calibrate mode.

LOGIC FORCING (List 16)
Enables forcing of up to 18 logic signals simultaneously including forcing of a given signal in [R], [S] and [T], independently. In most applications, forcing can be applied at any time except when the control is in the REMOTE mode of operation; however, a security code is required to initiate forcing. Real, i.e., integer, type values can be displayed with the logic signals, but only logic signals can be forced. Manually overriding any logic signal initiates an alarm message "MAINTENANCE FORCING MODE ENABLED", which provides a permanent record that a signal is being forced. If a communication failure is detected between [C] and any controller, the forcing for that controller is cancelled, and the display shows "....." for the data from that controller.

BRAM COPY/VERIFY (List 17)
This display can be used in conjunction with a PROM programming module for permanent memory retention of field edited application software, i.e., sequencing and control constants. The PROM programming module is not supplied by General Electric Company; however, many suitable types of modules are available such as the Pro-log Corporation model

B. DISPLAYS AND LOGGING (continued)

6. DISPLAY MENU (continued)

M980 which can be used for "gang" programming several PROM's. The following procedures can be used for programming:

a. On the CRT:
   i. Select the BRAM COPY/VERIFY DISPLAY (List 17).
   ii. Set the 2400 baud rate.
   iii. Select the appropriate mode on the CRT.

BRAM COPY/VERIFY (List 17)

b. On the Pro-log PROM Programming Module:
   i. Connect the serial adapter, and set the baud rate (8) "2400".
   ii. Turn on the Pro-log.
   iii. Connect the RS232 cable from the HMPF card in [C] slot 1M to the Pro-log modem connector.

c. Copy the low byte to the Pro-log buffer:
   (Reference the Pro-log manual section 15, page 1.)
   i. Set switches AUDIO, BUFFER, ON-LINE and MODEM on.
   ii. Set all other switches off.
   iii. Push the RESET pushbutton.
   iv. Push the 8 pushbutton . . . . . 9818 01 is displayed
   v. Push the 1, 0 and ENTER pushbuttons . 10 AAA is displayed
   vi. Push the START softswitch next to the CRT, and verify that the WRITE LED blinks and "1" flashes on the display until the transfer is complete.

d. Program PROM from the buffer to copy the PROM:
   (Reference the Pro-log manual section 6, page 6-7.)
   i. Insert the PROM in the COPY socket.
   ii. Set switches AUDIO, BUFFER, COPY, ON-LINE and MODEM on.
   iii. Push the RESET pushbutton
   iv. Push the DUP pushbutton 000 FFF is displayed
   v. Push the ENTER pushbutton 0000 is displayed
   vi. Push the ENTER pushbutton D AAA is displayed
   vii. When the programming is finished D *E1 is displayed

   * E1 is displayed for a single PROM, F is displayed if several PROM's are ganged.

e. Copy from the master PROM to the RAM buffer:
   (Reference the Pro-log manual section 6, page 6-3.)
   i. Move the PROM from the COPY socket to the MASTER socket.
   ii. Set switches AUDIO, BUFFER, COPY, ON-LINE, MASTER and MODEM on
   iii. Push the RESET pushbutton
   iv. Push the DUP pushbutton 000 FFF is displayed
   v. Push the ENTER pushbutton 0000 is displayed
   vi. Push the ENTER pushbutton D AAA is displayed

B. DISPLAYS AND LOGGING (continued)

6. DISPLAY MENU (continued)

BRAM COPY/VERIFY (List 17) (continued)

vii. When the transfer is complete D F is displayed, and the WRITE light goes off.

f. Verify the Pro-log buffer with the [C], BRAM: (#3 on CRT display) (Reference the Pro-log manual section 15, page 15-1.)

i. Set switches AUDIO, BUFFER, ON-LINE and MODEM on.
ii. Push the RESET pushbutton
iii. Push the 8 pushbutton 9818 01 is displayed
iv. Push the 0, 5 and ENTER pushbuttons 05 AAA is displayed
v. Push the ENTER pushbutton 0000 1FFF is displayed
vi. Push the 0, 1, 0, 0, 0, F, F and F 0100 0FFF is displayed
vii. Push the START softswitch next to the CRT.
viii. Push the ENTER pushbutton within 10 seconds.
ix. The CRT displays VERIFY COMPLETE.

g. Repeat steps c through f for the high byte PROM.

INTER-PROCESSOR LOGIC DATA (List 18)
Contains all logic data transmitted between [C], [R], [S] and [T].

INTER-PROCESSOR REAL DATA (List 19)
Contains all real, i.e., integer, data transmitted between [C], [R], [S] and [T].

CONTACT INPUTS (List 1A)
This display shows the status of all contact inputs as logic "1" or "0" as seen by the computers. A "1" corresponds to a closed field contact connected to a screw terminal on a Contact Input Module. A "0" corresponds to an open field contact. Shown below is an example which simulates the first eight (8) field contacts terminated at Contact Input Module #1, [CIM1], being closed and the second eight (8) being open. Since the same contact is seen by [R], [S] and [T], the display shows the logic three (3) times as seen by each Controller.

Point	Identifier	Status	Point	Identifier	Status
0020	CIM1.AR1	1111 1111	0021	CIM1.AR9	0000 0000
002C	CIM1.AS1	1111 1111	002D	CIM1.AS9	0000 0000
0038	CIM1.AT1	1111 1111	0039	CIM1.AT9	0000 0000

RELAY OUTPUTS (List 1B)
The status of all computer outputs to the relay drivers is displayed as a logic "1" or "0". The following example shows outputs from [R], [S], and [T] to the relay drivers where the first eight (8) drivers are receiving commands to turn-on, logic "1", and the second eight are receiving commands to turn-off, logic "0". Note that the first output from [R], [S] and [T] drives the Master Protective "4" relays and the second drives relay #1, "R1".

B. DISPLAYS AND LOGGING (continued)

6. DISPLAY MENU (continued)

RELAY OUTPUTS (List 1B) (continued)

Point	Identifier	Status	Point	Identifier	Status
0028	RLY1.AR1	1111 1111	0029	RLY1.AR9	0000 0000
0034	RLY1.AS1	1111 1111	0035	RLY1.AS9	0000 0000
0040	RLY1.AT1	1111 1111	0041	RLY1.AT9	0000 0000

ADMINISTRATIVE DISPLAY (List 1C)
Provides operator interface to set the time tag, English or Metric units and security codes. The first security code is embedded in PROM, and it must be entered before the second security code can be changed. This enables field changing of the second security code by authorized personnel only.

Two (2) PROM set catalog numbers appear at the bottom of this display which document the application software currently loaded in the panel.

CONTROL CONSTANTS (List 1D)
Displays all control constants for gains, time constants, setpoints, etc., and it enables changing any constant at any time after the security code has been entered.

MEMORY (List 1E)
Maintenance personnel use this display to access and edit computer memory. It is coded in hexidecimal, and it is not oriented toward operator use. A security code is required for access.

DIAGNOSTIC DISPLAYS (List 1F)
A menu of diagnostic displays can be obtained by selecting list "1F" on the DISPLAY KEYPAD. The menu is:

RAM/PROM CHECK FOR C	(On-line)
RAM/PROM CHECK FOR R	(On-line)
RAM/PROM CHECK FOR S	(On-line)
RAM/PROM CHECK FOR T	(On-line)
PUSHBUTTON TEST	(Off-line)
AUXILIARY DISPLAY TEST	(Off-line)
RELAY TEST	(Off-line)
DIGITAL I/O TEST	(Off-line)
ANALOG TEST	(Off-line)
DIAGNOSTIC DATA	(On-line)

Selection of any diagnostic display from the menu requires the operator to enter the security code. RAM/PROM checks for the Communicator, [C], and the Controllers, [R], [S] and [T], are continuously performed. The displays are continuously up-dated. A PASSED/FAILED message is displayed next to each card name and chip socket number. Reference Chapter VI, Section G - DIAGNOSTIC OVERVIEW for a description of on-line diagnostic alarm messages.

B. DISPLAYS AND LOGGING (continued)

6. DISPLAY MENU (continued)

DIAGNOSTIC DISPLAYS (List 1F) (continued)
Off-line diagnostics are enabled when the turbine is shut down via permissive L3DIAG. This typically becomes a logic "1" when the master protective "4" relays are de-energized, i.e., L4 = 0, and the turbine is stopped, i.e., L14HR = 1.

The PUSHBUTTON TEST displays the name of each pushbutton as it is pressed. Control pushbutton operation is verified by pressing a pushbutton and observing its LED blink. Note that the EXECUTE pushbutton should NOT be pressed while an LED is blinking to avoid a response by the control system.

The AUXILIARY DISPLAY TEST is performed by writing a sequence of outputs to the auxiliary display to confirm that all hexidecimal digits are operative.

The RELAY TEST verifies the operation of all relays, relay drivers and digital output ports. Relays which are controlled by Communicator [C] are automatically energized sequentially and verified by observing each relay LED light-up and then go out. Relays which are activated by Controllers [R], [S] and [T] are tested by turning on all outputs from one controller simultaneously and then sequentially driving each output from a second controller. This test is repeated three (3) times to test all voting combinations of outputs from [R], [S] and [T]. During normal control operation, failure of an output from one (1) controller will be out-voted on the relay driver card HRDB, and it will not be visible in the gas turbine control system. Consequently, periodic maintenance checks should be performed to confirm the viability of the 2/3 voting protection. Field use of this test must consider contact interface to other equipment prior to performing the test. Each relay module terminal board can be unplugged to disconnect the control from the field interface without removing each wire. The breaker closure relay in the Synchronizing Interface Module, [SYNCH], can be tested with this display and verified by observing its response from the auxiliary display.

Digital I/O is tested by connecting a loop-back cable from a card's output connector to its input connector and selecting that card on the DIGITAL I/O TEST DISPLAY. A successful test will be indicated by a TEST PASSED message, and an unsuccessful test will be indicated by a display of the I/O test results including the malfunctioning port address.

The ANALOG TEST diagnoses hardware failures in the analog control system. It uses an analog test module DS3820ATMA to condition signals and supply test signals, the INTERPROCESSOR DISPLAY to observe the test results and the ANALOG CONTROL DISPLAY to force the HSAA D/A setpoint and establish the initial conditions for the vibration and pulse rate test signals.

The DIAGNOSTIC DATA DISPLAY provides an on-line overview of the key signals used in the computer communication system.

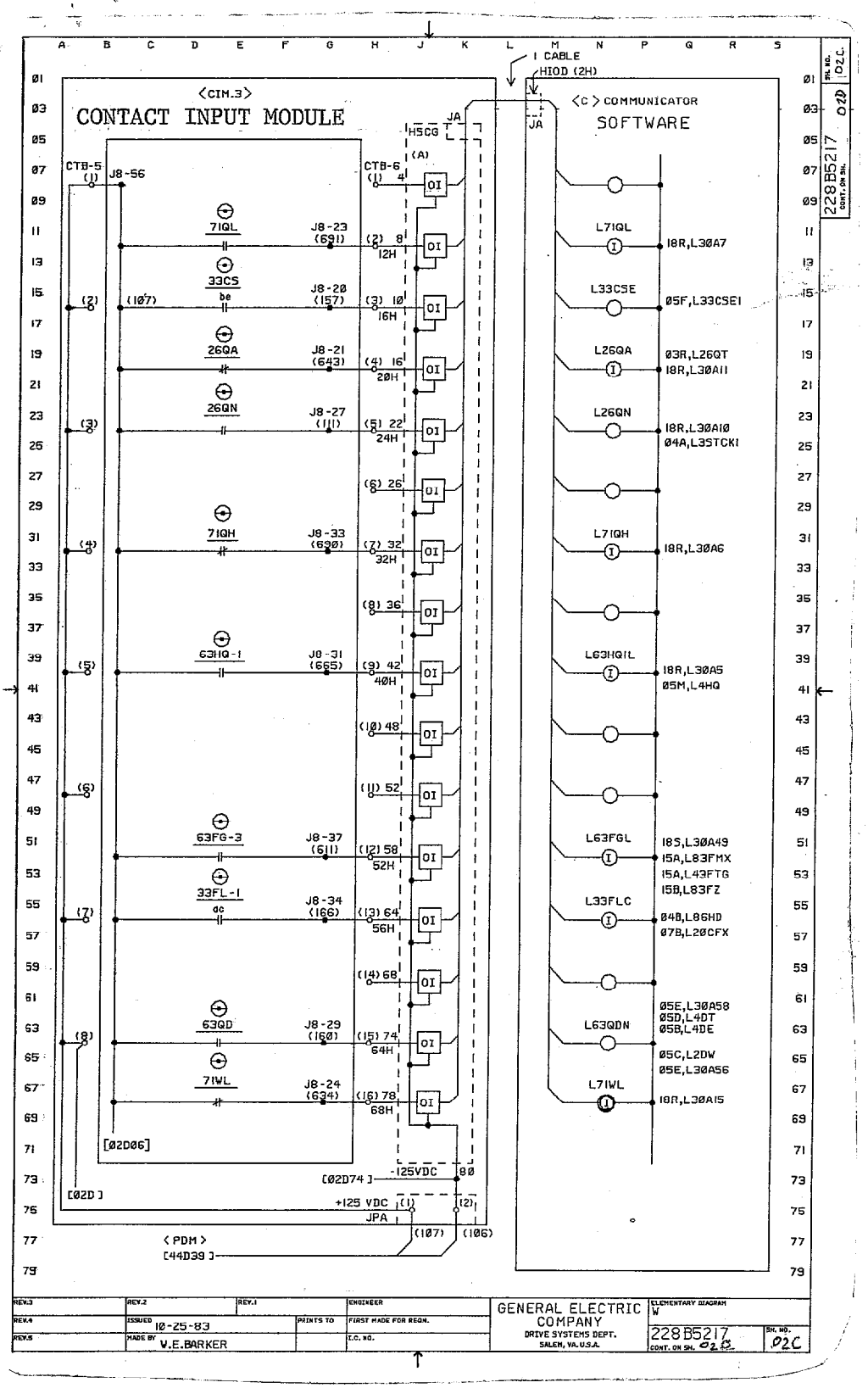


Figure VI-B-1: CONTACT INPUTS

VI. APPLICATION SOFTWARE

A. GENERAL

Software consists of application software customized to the needs of particular applications and fixed software dedicated to running the computer's operating system. Examples of application software are sequencing and control algorithms. And an example of operating system software is the computer memory map which dedicates areas of memory to particular functions such as logic inputs. The Control Elementary is used to elevate the readability of application software and hardware/software interface to a level where non-programmers can understand the control system without referring to the actual computer programs. Cross-references are used to coordinate between inputs, outputs, sequencing, control algorithms, alarm message displays, etc.

B. CONTACT INPUTS

Consider the case of a gas fuel pressure switch, 63FG-3, which is mounted on the turbine's accessory base and used to automatically transfer to liquid fuel if the gas fuel pressure drops below the setpoint. Figure VI-B-1 shows a typical contact input sheet with switch 63FG-3 located at coordinate D/52, i.e., column D, row 52. The switch is wired through connector J8 pin 37 to Contact Input Module #3, [CIM3], terminal board "A" point 12 by wire (611). An HSCG card optically isolates the signal which is then connected to the Communicator, [C]. The input port is scanned eight (8) times per second, inverted as noted by the "I" in the "software" coil and named L63FGL. An inverted input such as L63FGL becomes TRUE, i.e., logic "1", when the field contact opens, and a non-inverted input such as L33CSE becomes TRUE when the field contact closes. The status of all contacts connected to the module can be displayed simultaneously on the contact input display, List 1A, which reads the inputs ahead of the inversion mask. Any input like L63FGL can be displayed by calling it by its functional name without referring to the computer input point number. The first "L" in L63FGL stands for "logic", the "63FG" ties the signal name to the name of the field pressure switch and the second "L" tells the user that the signal is TRUE when the gas fuel pressure is "low". Note that the diagram deliberately omits the maze of computer address, data and control lines used to read this signal and concentrates on the application. If hardware troubleshooting is necessary, the Panel Connection Diagram is used for wiring details.

Now that we have seen the signal entered into the computer, we need to know where it is used. A cross-reference appears next to each "software" coil, and we can see that the "software" contacts from L63FGL are used on sheet 15A in sequencing rungs L83FMX and L43FTG.

B. CONTACT INPUTS (continued)

Gas turbine field device nomenclature and their corresponding software names are defined by the following standards:

DEVICE #	FUNCTION	SOFTWARE EXAMPLE
"1"	Master Element	L1X = Master Control Start Logic
"2"	Sequence Timer	L2TV = Turbine Vent Timer Logic
"3"	Checking Relay	L3 = Complete Sequence Logic
"4"	Master Relay	L4 = Master Protective Logic
"5"	Stopping Devices	L5E = Emergency Stop Logic
"6"	Starting Circuit Breaker	
"8"	Control Power Switch	
"10"	Unit Sequence Switch	
"12"	Overspeed Device	L12H = HP Elect. Overspeed Trip Logic
"13"	Synchr. Speed Indicator	
"14"	Speed Relay	L14HS = HP Shaft - Operating Speed Logic
"15"	Speed or Freq. Matching	
"18"	Accel. or Decel. Device	
"20"	Solenoid Valve	L20FGX = Gas Fuel Stop Valve Logic Output
"21"	Distance Relay	
"23"	Temp. Control Device	
"25"	Synchr. Check Device	L25 = Breaker Close Command Logic
"26"	Temp. Sensing Device	L26QN = Lube Tank Temp. Normal Logic
"27"	Undervoltage	L27DN = Battery Voltage Normal Logic
"28"	Flame Detector	L28FD = Flame Detected Logic
"30"	Annunciator Relay	L30A22 = Alarm Message #22 Logic
"32"	Directional Power Relay	
"33"	Position Switch	L33CSE = Starting Clutch Engaged Logic
"34"	Master Sequence Device	
"37"	Undercurrent/power Relay	
"38"	Bearing Protective Device	
"39"	Mech. Condition Monitor	L39VA = High Vibration - Alarm Level
"40"	Field Relay	
"41"	Field Circuit Breaker	
"43"	Manual Selector Device	L43C = Turbine Cranking Mode Selected
"45"	Atmospheric Monitor	L45FTX = Fire Detected Trip Logic
"46"	Reverse Phase Cur. Relay	
"47"	Phase Seq. Voltage Relay	
"48"	Incomplete Sequence Relay	L48 = Turbine Incomplete Sequence Logic
"49"	Thermal Relay	L49X = Auxiliary Motor Overload Logic

B. CONTACT INPUTS (continued)

Nomenclature (continued)

DEVICE #	FUNCTION			SOFTWARE EXAMPLE
"50"	Inst. Overcurrent Relay	L50QVN	=	Elec. Prec. Current Normal Logic
"51"	AC TD Overcurrent Relay			
"52"	AC Contactor or Breaker	L52CR	=	Cranking Motor Contactor Logic
"55"	Power Factor Relay			
"57"	SC or Grounding Device			
"59"	Overvoltage Relay			
"60"	Volt./Cur. Balance Relay	L60SYN1	=	System Voltage is Present Logic
"62"	TD Stopping/Opening Relay	L62CD	=	Turbine Cooldown Timer Logic
"63"	Liq/Gas Pressure Relay	L63QA	=	Low Lube Oil Pressure Logic
"64"	Ground Protective Relay	L64D	=	125VDC Ground Logic
"65"	Governor, i.e., ex. servos			
"66"	Notching/Jogging Device			
"67"	AC Dir. Overcurrent Relay			
"68"	Blocking Relay			
"69"	Permissive Switch	L69GV	=	IGV Temp. Control Perm. Logic
"70"	Elect. Operated Rheostat	L70R	=	Raise Speed/Load Setpoint Command
"71"	Liquid or Gas Level Relay	L71QH	=	Lube Oil Level High Logic
"72"	DC Contactors/ Breakers	L72QEX	=	Emergency Lube Oil Pump Logic
"75"	Position Changing Mech.			
"77"	Pulse Transmitter			
"80"	Liquid or Gas Flow Relay			
"81"	Frequency Relay	L81GY	=	Generator Underfrequency Logic
"82"	DC Reclosing Relay			
"83"	Automatic Control Relay	L83B	=	Base Load Logic
"84"	Operating Mechanism	L84TG	=	Total Gas Fuel Logic
"85"	Receiver Relay			
"86"	Lockout Relay	L86CB	=	Compressor Bleed Failure Lockout
"87"	Diff. Protective Relay			
"88"	Aux. Motor or Motor Gen.			
"89"	Line Switch			
"90"	Regulating Device	L90LR	=	Load Regulator Raise Setpoint
"91"	Voltage Directional Relay			
"93"	Field-changing Contactor			
"94"	Tripping Relay	L94X	=	Turbine Normal Shutdown Command
"95"	Miscellaneous			
"96"	Transducer			

C. SEQUENCING

Figure VI-C-1 shows sequencing rungs as they actually appear in the Control Elementary Diagram. Sequencing is shown in a "ladder Diagram" format, and is executed at 8 HZ. The term, ladder, is used because the graphical representation resembles a ladder with the rungs of the ladder being the individual sequencing calculations which determine each logic state. The top of sheet 15A identifies the contents of the sheet as [R], [S] and [T] software, and the left-hand column defines the function of each rung. Logic input L63FGL is shown as a normally-closed contact originating in Communicator [C] on sheet 02C. It appears in rungs L83FMX and L43FTG. The actual sequencing code is a very high level language which is entered in [R], [S] and [T], exactly as it appears on sheet 15A and scanned eight (8) times per second. Rung L43FTG has the following code:

```
LDF L3GFLT      ;Load FALSE the first contact L3GFLT
OR  L84MIXT     ;Or TRUE contact L84MIXT
AN  L43G        ;And TRUE contact L43G
LDF L83REC      ;Start a new STRING, and load FALSE contact L83REC
AN L43FTG       ;And TRUE contact L43FTG
ORS             ;Or the new STRING with the previous STRING
LD L43FTGC      ;Start a new STRING, and load TRUE contact L43FTGC
AN L83REC       ;And TRUE contact L83REC
ORS             ;Or the new STRING with the previous STRING
ANF L43D        ;And FALSE contact L43D
ANF L63FGL      ;And FALSE contact L63FGL
STO L43FTG      ;STORE the result in L43FTG
```

Sequencing commands are:

LD - LOAD TRUE; push stack and LOAD contact TRUE to the stack top
LDF - LOAD FALSE; push stack and LOAD contact FALSE to the stack top
AN - AND TRUE; AND contact TRUE with the stack top
ANF - AND FALSE; AND contact FALSE with the stack top
OR - OR TRUE; OR contact TRUE with the stack top
ORF - OR FALSE; OR contact FALSE with the stack top
ANS - AND STRING; AND the current STRING with the previous STRING
AND - the stack top with the stack top + 1
ORS - OR STRING: OR the current STRING with the previous STRING
OR - the stack top with the stack top + 1
STO - STORE; STORE the result of the rung (stack top) in memory
ESC TMV K____ T____ - ESCAPE to TIMER subroutine, # = 8 X seconds, T = IDATA symbol name where the current value is stored
ESC CTV E____ K____ C____ - ESCAPE to COUNTER subroutine, E = LDATA symbol name where the old enable bit is stored, # = counts, C = IDATA symbol name where the current value is stored
ESC CMP ____ ____ - ESCAPE to COMPARATOR subroutine, i.e., A $\geq$ B.

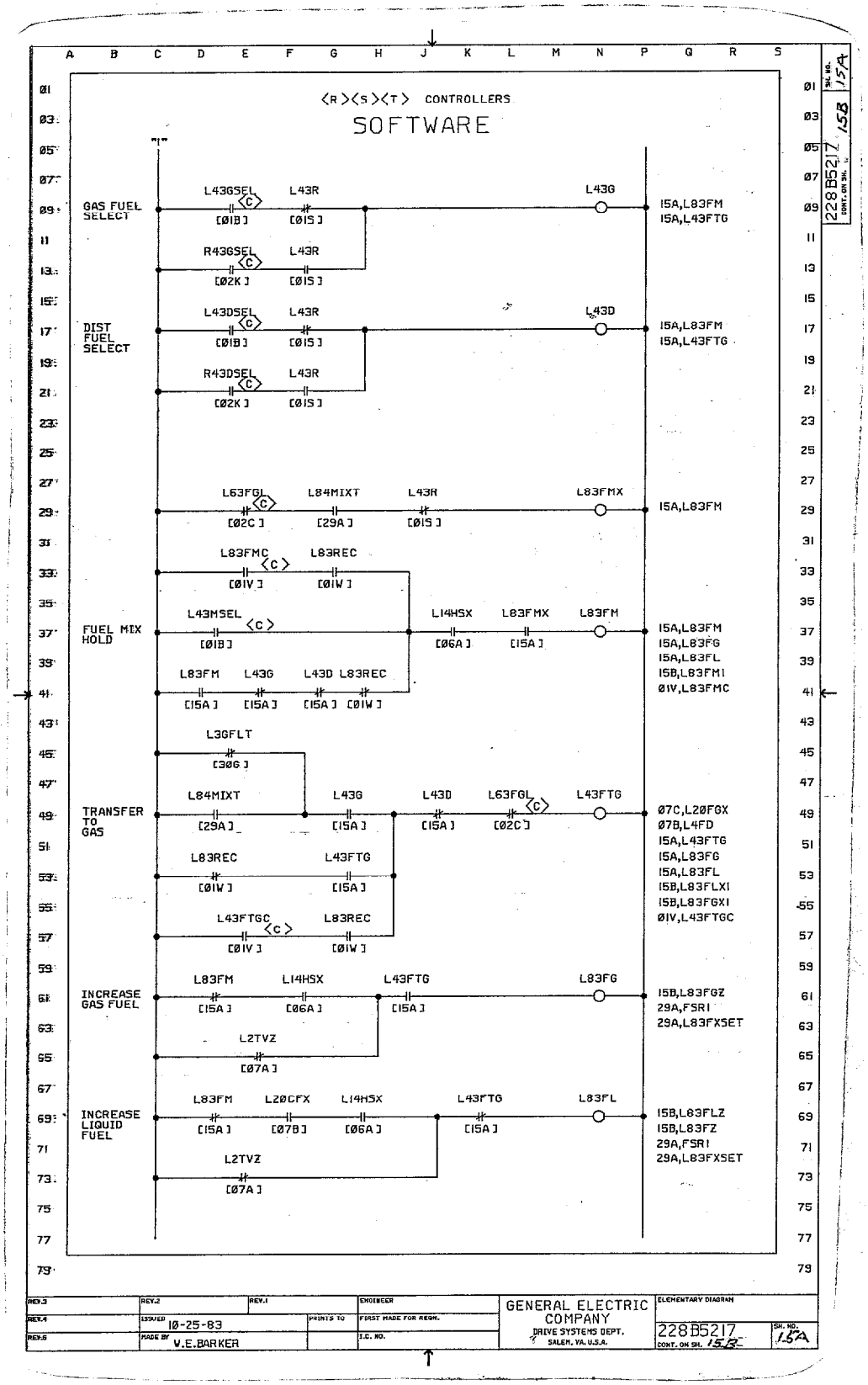


Figure VI-C-1: SEQUENCING LADDER DIAGRAM

C. SEQUENCING (continued)

Sequencing can be edited by connecting a "dumb" terminal to the HMPF card in slot 1M of Communicator [C] via an RS232 interface and a null modem. The null modem is supplied in the panel service kit. Files containing the Communicator sequencing and the Controller sequencing are contained in [C], which enables the terminal interface to [C] to change either file; however, controller sequencing changes are not transferred from [C] to [R], [S] and [T] until the controllers are re-initialized. This "downloading" process can be performed at any time. The change is retained in BRAM, i.e., battery-backed RAM, for up to ten (10) days at 25 degrees centigrade if power is lost. The memory retention feature is enabled when the "freeze" control constant FREEZE_L is set to a logic "1" on the CONTROL CONSTANTS DISPLAY page 00. Setting FREEZE_L to a logic "0" will cause all sequencing and control constant changes to default to their original values which are stored in PROM if power is lost.

A typical sequencing change on a terminal can be made by:

1. Typing "CONTROL" and simultaneously typing "Z" which will cause a prompt ">" and a request for the "PASSWORD".
2. Type the security code, and a CARRIAGE RETURN to acquire a new prompt "#".
3. Select the sequencing editor file by typing EDISEQ C_SEQ or RST_SEQ and a CARRIAGE RETURN which will cause a new prompt ">".
4. Select the sequencing rung to be changed by typing "R" and the rung name such as R L1X and a CARRIAGE RETURN to display the sequencing code and the memory line numbers for each instruction in the rung.

```
Example:  0000 LD   L1S
          0001 LDF  L83REC
          0002 AN   L1X
          0003 ORS
          0004 LD   L83REC
          0005 AN   L1XC
          0006 ORS
          0007 AN   L1XX
          0008 ANF  L3SFLT
          0009 STO  L1X
```

5. Now select the required editor command(s) from the following list:

   >D "LINE NUMBER" - This command allows you to DELETE existing instructions. If the LINE NUMBER is omitted, a deletion will occur at the current line pointer position. Each instruction is displayed before deletion to enable a confirmation that the intended instruction is being deleted. At this point, a CARRIAGE RETURN will complete the deletion, and a "CONTROL" "Z" will inhibit the deletion. Type "CONTROL" "Z" to exit the insert mode.

C. SEQUENCING (continued)

>E "LINE NUMBER" - This command allows you to EDIT existing instructions. If the LINE NUMBER is omitted, the instruction at the current line pointer position will be edited. The instruction to be edited is displayed first, and then the new instruction is typed. E commands can be used to step through a sequence of instructions one at a time and display the instruction with no changes. If a carriage return is entered without entering a new instruction, the instruction remains unchanged, the current line pointer is advanced and the next instruction is displayed. Type "CONTROL" "Z" to exit the insert mode.

>F This command lists all sequencing commands. To abort the printing of the sequencing file before completion, type "CONTROL" "Z".

>H The HELP command lists all EDISEQ commands.

>I "LINE NUMBER" - This command allows you to INSERT a new instruction at any point in the program. The insertion always takes place immediately before the specified LINE NUMBER in order for the current instruction and all of the following instructions to be "pushed down" in the sequencing to make room for the new instruction(s). If the LINE NUMBER is omitted, the insertion will occur at the current line pointer position. Type "CONTROL" "Z" to exit the insert mode.

>N The NEXT command is used in conjunction with the "R" RUNG command or the "W" WHERE USED command to display the NEXT sequencing rung in the sequencing file.

>P The PREVIOUS command is used in conjunction with the "R" RUNG command or the "W" WHERE USED command to display the PREVIOUS sequencing rung in the sequencing file.

>R "RUNG NAME" - This command is used to display a string of instructions which comprise a rung. The RUNG NAME, if entered, is the storage point in memory where the results of the rung calculation are stored. If the RUNG NAME is not specified, the rung which is designated by the current line pointer is displayed. A new rung display will set the current line pointer to the first instruction in the rung to enable editing and deleting with the default modes of the "E" and "D" commands. A semicolon (;) is used at the end of rung to separate it from the next rung and enhance the readability.

C. SEQUENCING (continued)

>S The STATUS command is used in conjunction with the "R" rung command to review the status of the currently displayed rung on the printer. For example: prompt " > " and command "R L1S" and "CARRIAGE RETURN" results in

```
0000 LD  L1START
0001 OR  L1FAST
0002 ANF L430
0003 STO L1S
>
```

then type "S" and "CARRIAGE RETURN" for the status 1 0 0 1

This means that:

```
0000 LD  L1START  =  1  the contact is closed
0001 OR  L1FAST   =  0  the contact is open
0002 ANF L430     =  0  the contact is closed
0003 STO L1S      =  1  rung L1S is a "1"
```

>TI This command generates a list of all integer signal names in IDATA.

>TL This command generates a list of all logic signal names in LDATA.

>W "LOGIC NAME" - This is a WHERE USED command which which searches for the sequencing rungs where the specified LOGIC NAME is used. It initially locates the first place in the sequencing file where the logic name is used, and then the next time a "W" and "CARRIAGE RETURN" are typed, the WHERE USED program will search for and print the next place where the previously declared LOGIC NAME is used.

6. A "CONTROL" "Z" command is used to exit the sequencing editor, and another "CONTROL" "Z" is used to exit the remote terminal interface mode with [C].

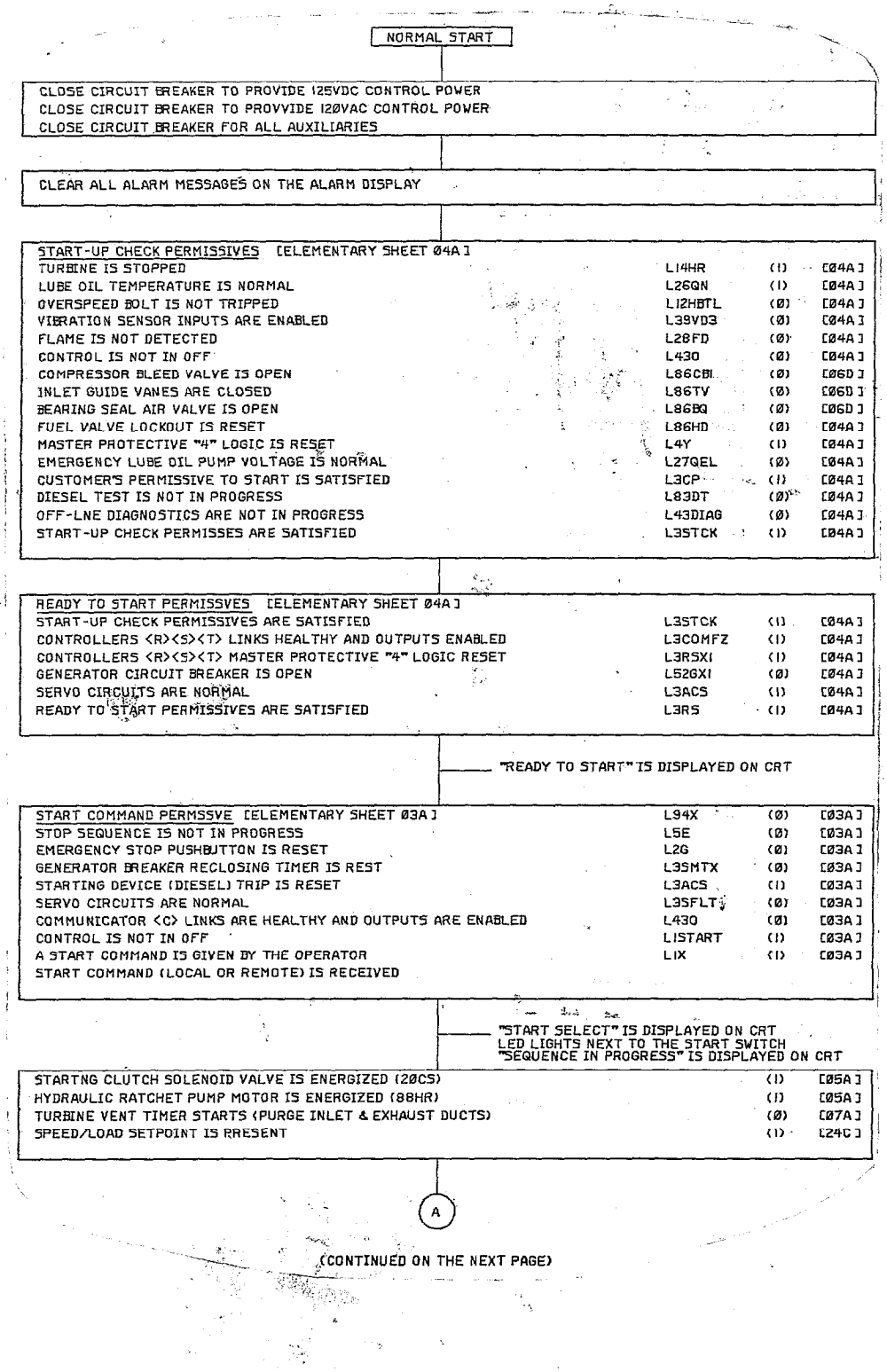


Figure VI-C-2: TYPICAL MS6001 START-UP SEQUENCE BLOCK DIAGRAM

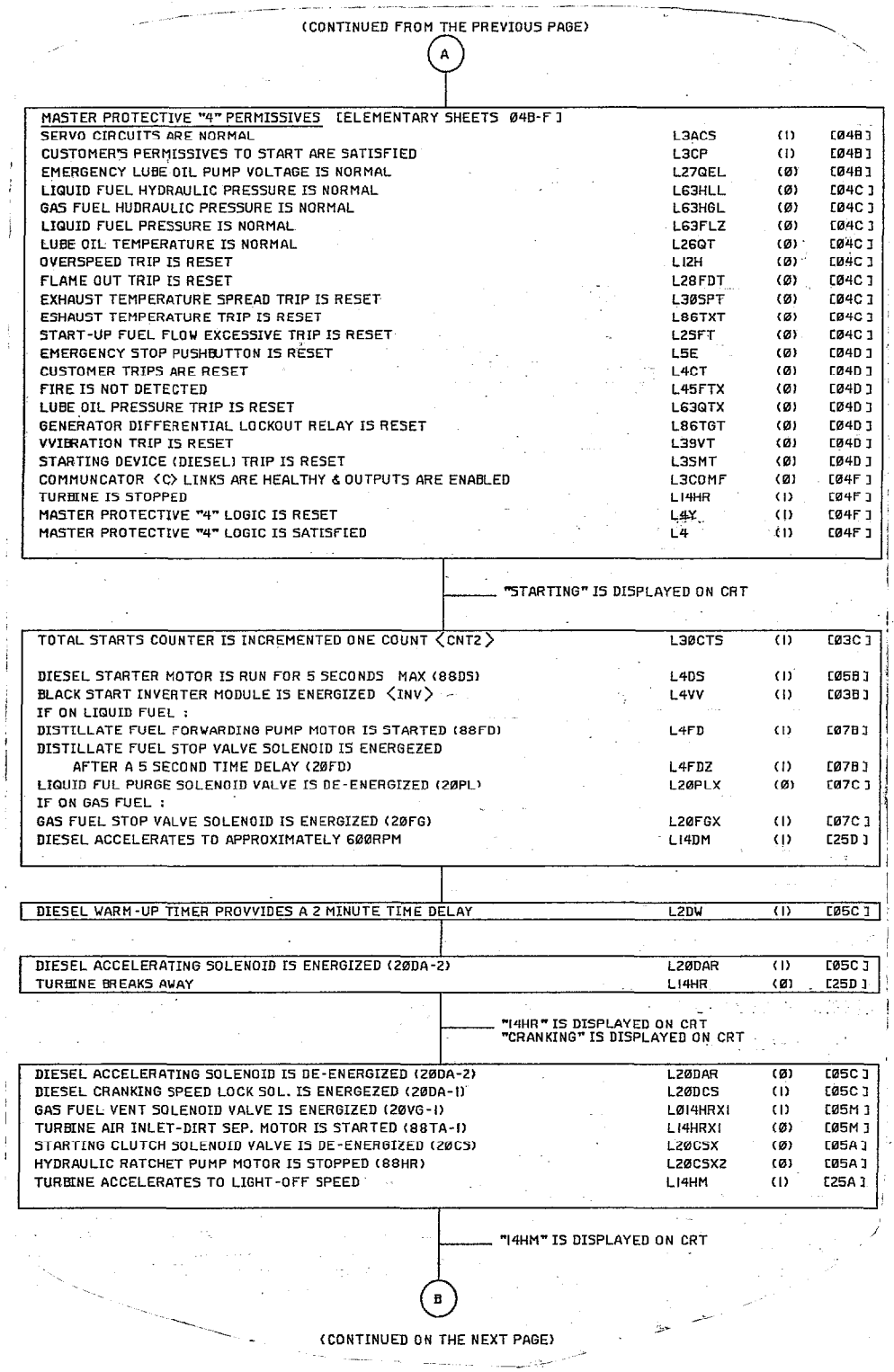


Figure VI-C-3: TYPICAL MS6001 START-UP SEQUENCE BLOCK DIAGRAM

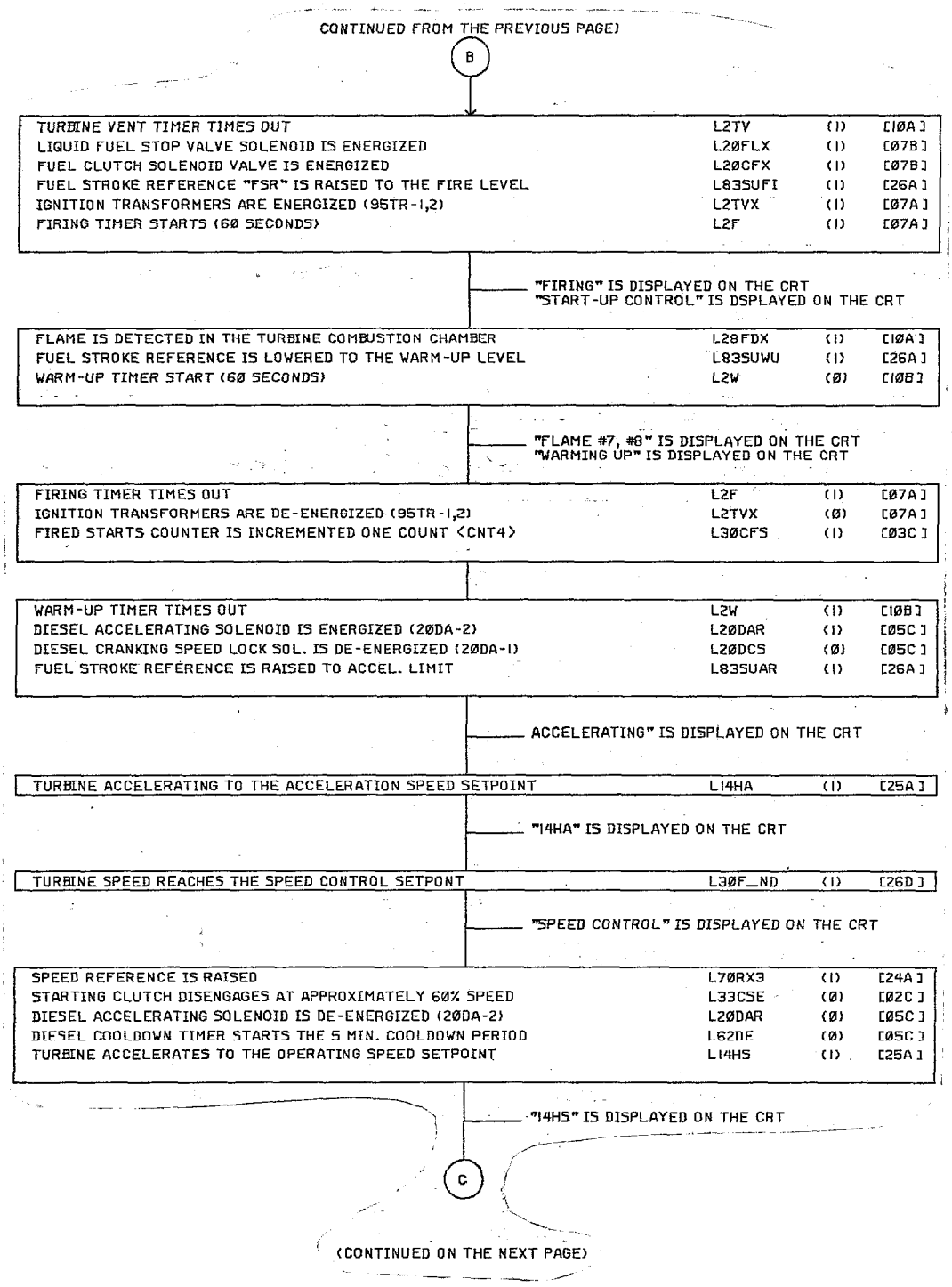


Figure VI-C-4: TYPICAL MS6001 START-UP SEQUENCE BLOCK DIAGRAM

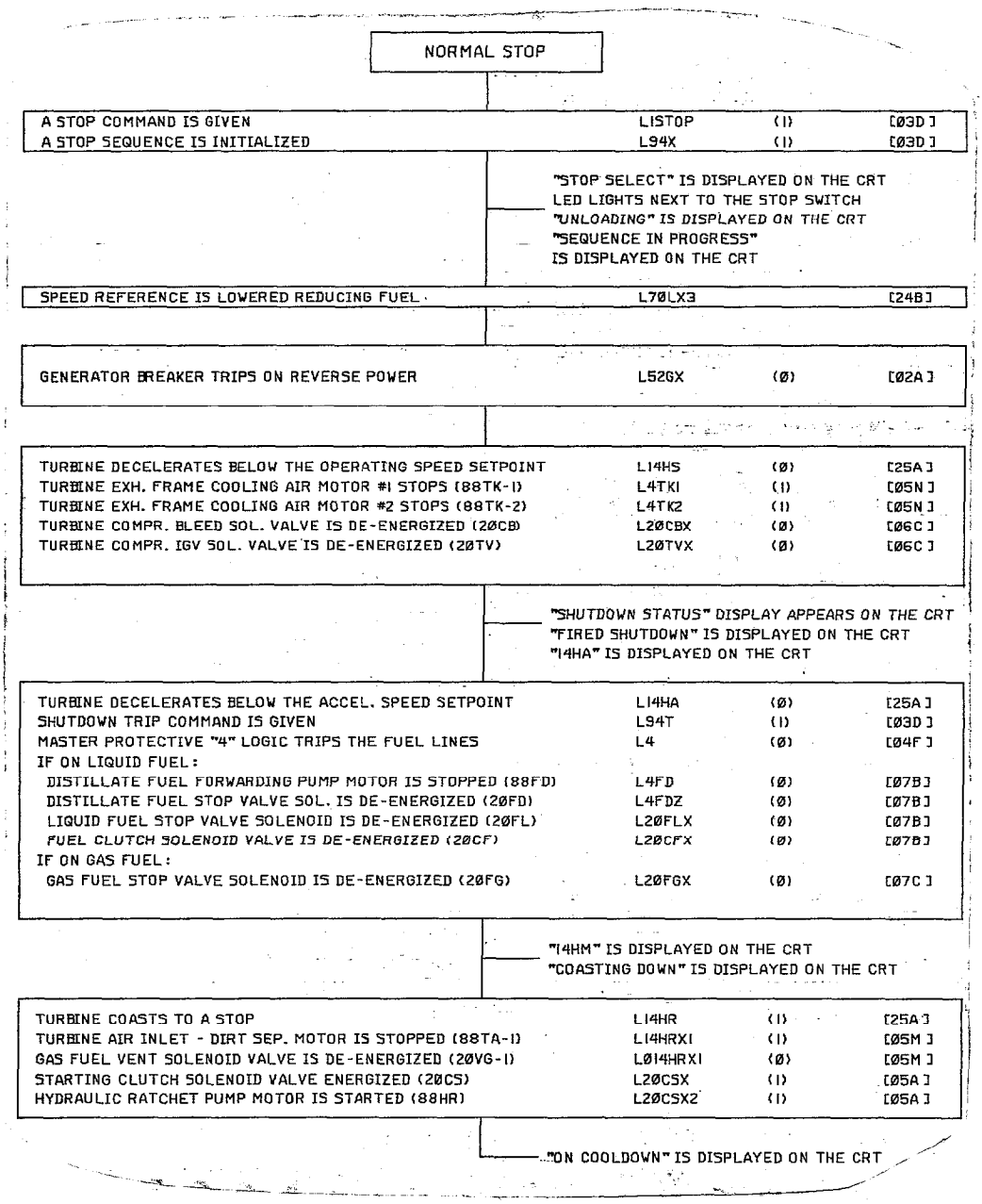


Figure VI-C-6: TYPICAL MS6001 SHUTDOWN SEQUENCE BLOCK DIAGRAM

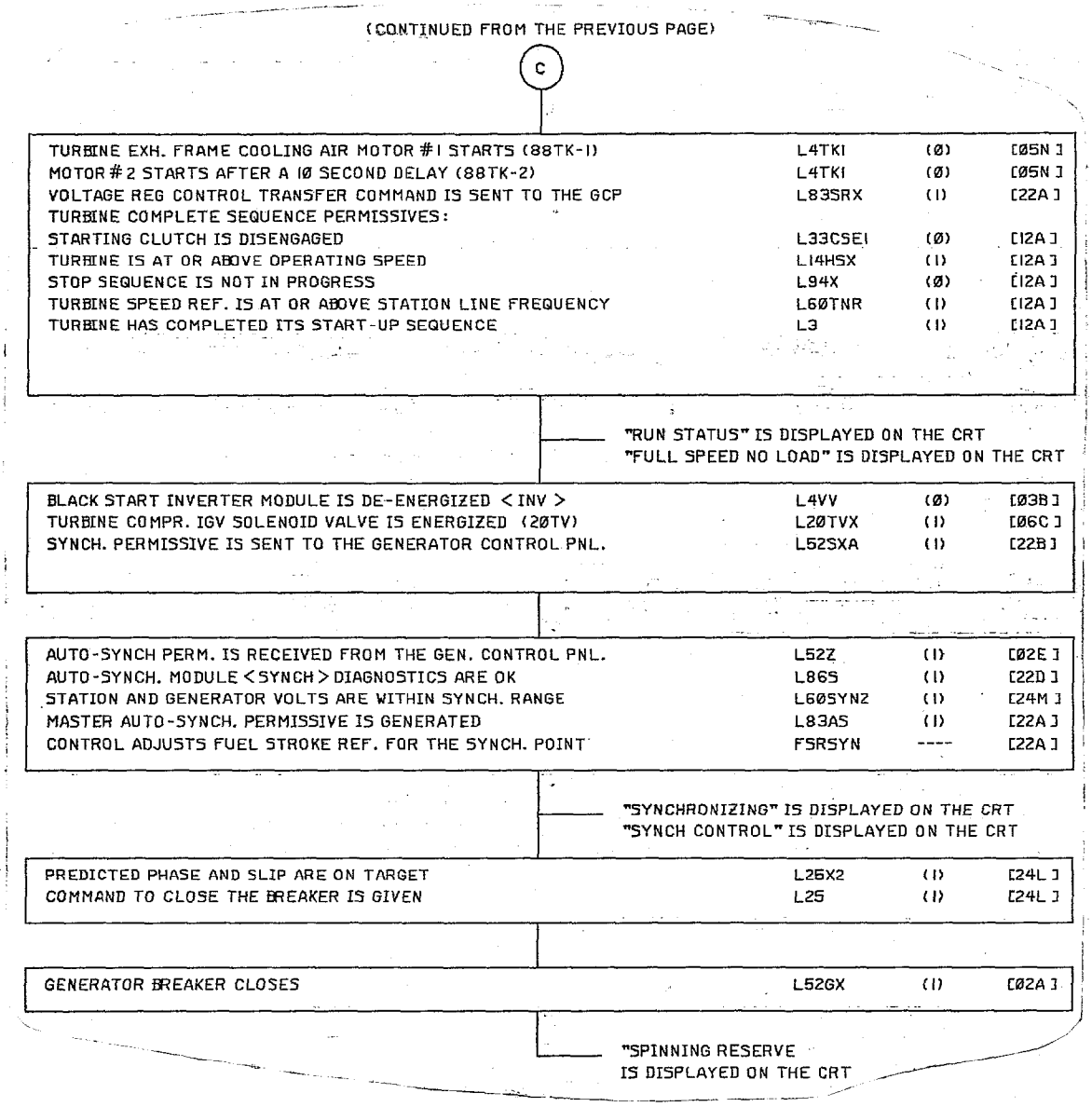


Figure VI-C-5: TYPICAL MS6001 START-UP SEQUENCE BLOCK DIAGRAM

D. ANALOG INPUTS

Figure VI-D-1 illustrates analog input processing. Analog inputs are terminated on one (1) of the analog I/O modules, [AIO], or one (1) of the thermocouple modules, [TCM]. These I/O modules contain no active components and are used for noise filtering only. Ribbon cables are used for connections between the I/O modules and the cards mounted in the electronics modules. Each card is dedicated to a specific function and is used to multiplex the various input signals prior to feeding them to a single output amplifier. Each amplifier has a range of -10VDC to +10VDC corresponding with the maximum value of the parameter being monitored. For example, an increasing exhaust temperature would drive the amplifier in the positive direction; however, an open thermocouple would drive the amplifier into negative saturation. Each card is wired to the HAIC which multiplexes the individual card inputs and performs the analog to digital conversion.

A software module named CONVERT fetches each input and places it in a comprehensive file, GET_ANALOG_IN, containing all analog inputs. Thermocouple inputs require an additional program named ADOS_CALC which provides the cold junction compensation. A temperature sensor is mounted on each thermocouple input module which provides a reference on the first input channel. The last input, i.e., #16, has a jumper wired between the screw terminals to provide a short circuit reference. Program ADOS_CALC calculates the difference between the cold junction reference and the short circuit reference which is added to the individual thermocouple readings in GET_ANALOG_IN. Thermocouple inputs require an additional conditioning process to linearize the values according to the type of thermocouple used. This is performed in software module, TC_LINEARIZE, which also normalizes the values. Normalize means that each value is scaled to 1 CNT B11 = 1 degree Fahrenheit. The data base stores the values of the analog inputs which are called by control algorithms and display routines. Control algorithms use the data in their calculations, and then return the results to the data base for storage.

E. CONTROL ALGORITHMS

Refer to Figure VIII-E-1 for a diagram of the Overtemperature Alarm and Trip algorithm as it appears in the Control Elementary Diagram. The top of the sheet defines the function as [R], [S] and [T] software, and the dotted lines contain the algorithm. Inputs to the algorithm are shown on the left and cross-referenced to each signal's origin. Outputs from the algorithm are cross-referenced to the sheet where they are used and the name of the function in which they are used. Control constants, such as TTKOT3, are shown as inputs contained in ovals. They are most frequently used as adjustable offsets, setpoints, gains and time constants, and they can be changed with the Control Constants Display, List 1D, after entering a security code. Any constants which are self-contained in the algorithms are shown within the dotted lines and are not available for field adjustment.

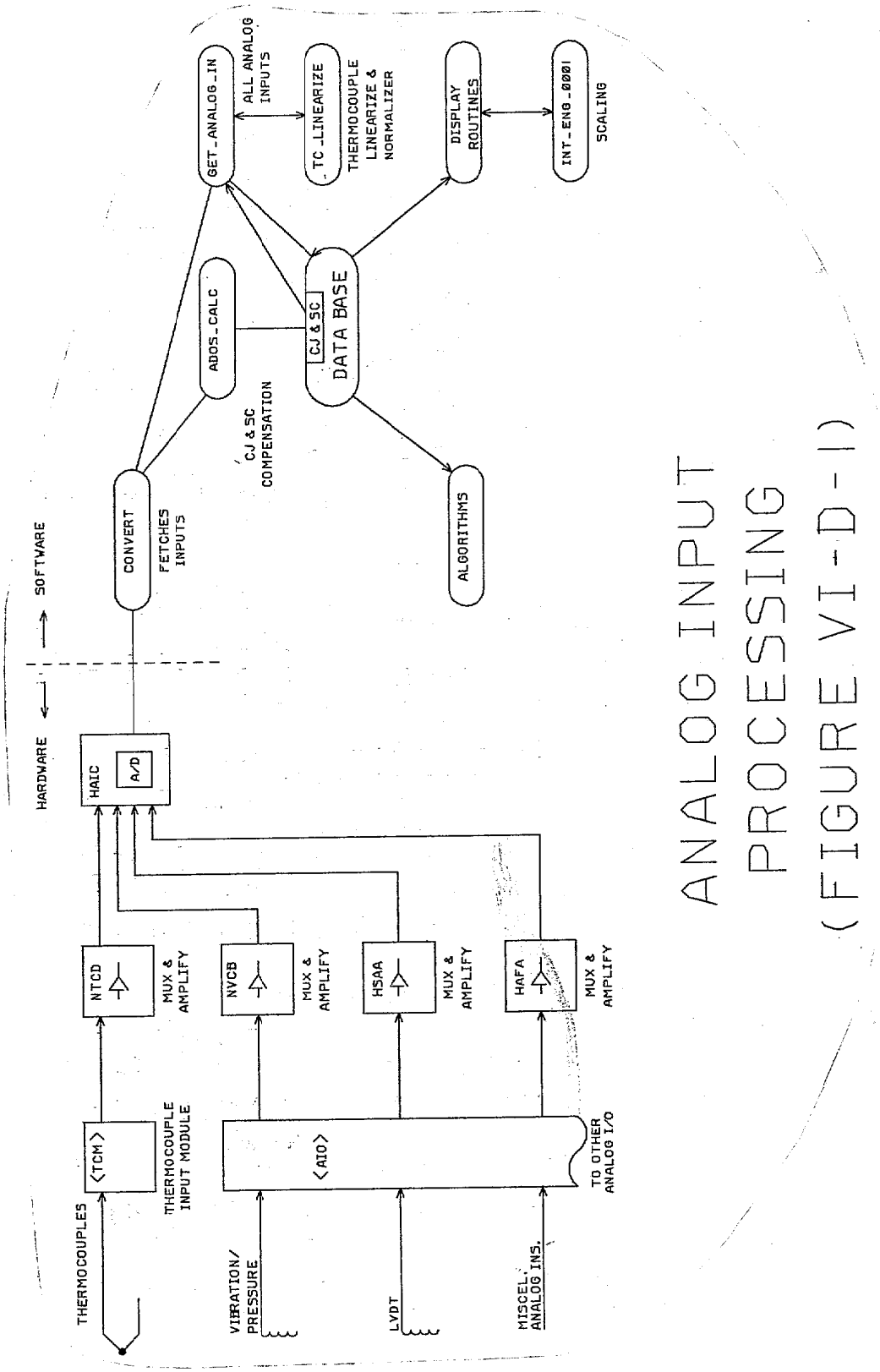


Figure VI-D-1: ANALOG INPUT PROCESSING

E. CONTROL ALGORITHMS (continued)

Algorithms are written in PL/M-86, a high-level language designed for systems and application programming of the microprocessor. The program name appears in the upper left-hand corner of the algorithm symbol, and it contains suffixes to identify the version, modification and revision. Each name contains up to 32 characters and may contain a "_" or "$" as delimiters to allow the use of meaningful names. Algorithm name L86TXTV2_ 00XX defines the function as L86TXT, the version as V2, the modification as 00 and the revision as XX because it is not relevant. Version numbers designate application differences between the algorithms. Modification numbers document changes to programs of a given version which preclude the use of older vintages of the program. Revision numbers identify minor changes to programs which do not affect functional operation. Occasionally, a special edition of a version is needed for a customized application which is documented by an S_ following the version number. This stringent control of programs is used to ensure that the software is up-to-date and correct; consequently, the code is not available to field personnel for editing. Sampling rates for each program are shown in the bottom left-hand corner of the algorithm symbol. Figure VI-E-1 shows typical symbols used to represent algorithm calculations.

F. ANALOG OUTPUTS

There are two (2) types of analog outputs from the turbine control panel. The first type is a DC milliampere output from Controllers [R], [S] and [T] to servo valves for functions such as liquid fuel flow control, gas fuel flow control, modulated inlet guide vane control, etc. Application software for these functions is contained in control algorithms which are customized for each function. A description of the software for the previously mentioned functions is contained in chapter VII. A detailed description of the servo valve driver system is provided in Chapter VII, Section H, and a description of the servo auto-calibrate procedure is given in Chapter V, Section B.6.

4 to 20MA and 0 to +10VDC analog outputs are available from Controllers [R], [S] and [T] and Communicator [C]. Controller outputs are generally process signals, such as inlet heating. They are calculated with the dedicated control algorithms that are shown in the control elementary. Their software output references are fed to digital to analog converters on HCVA cards in each controller. There are two (2) isolated output channels per card which supply a signal to Analog I/O Module #4, [AIO4], where a median select is performed with non-active components to determine the final process output signal. These outputs can be calibrated with the HCVA SERVICE DISPLAY, i.e., list 14, similar to the servo outputs.

Communicator [C] provides analog outputs primarily for monitoring. Jumpers on the cards are set for the required current or voltage output, and [C], ANALOG OUT DISPLAY, i.e., list 11, is used to select any signal in the data base for field calibration. The selection of the signal and the scaling require no customized application software.

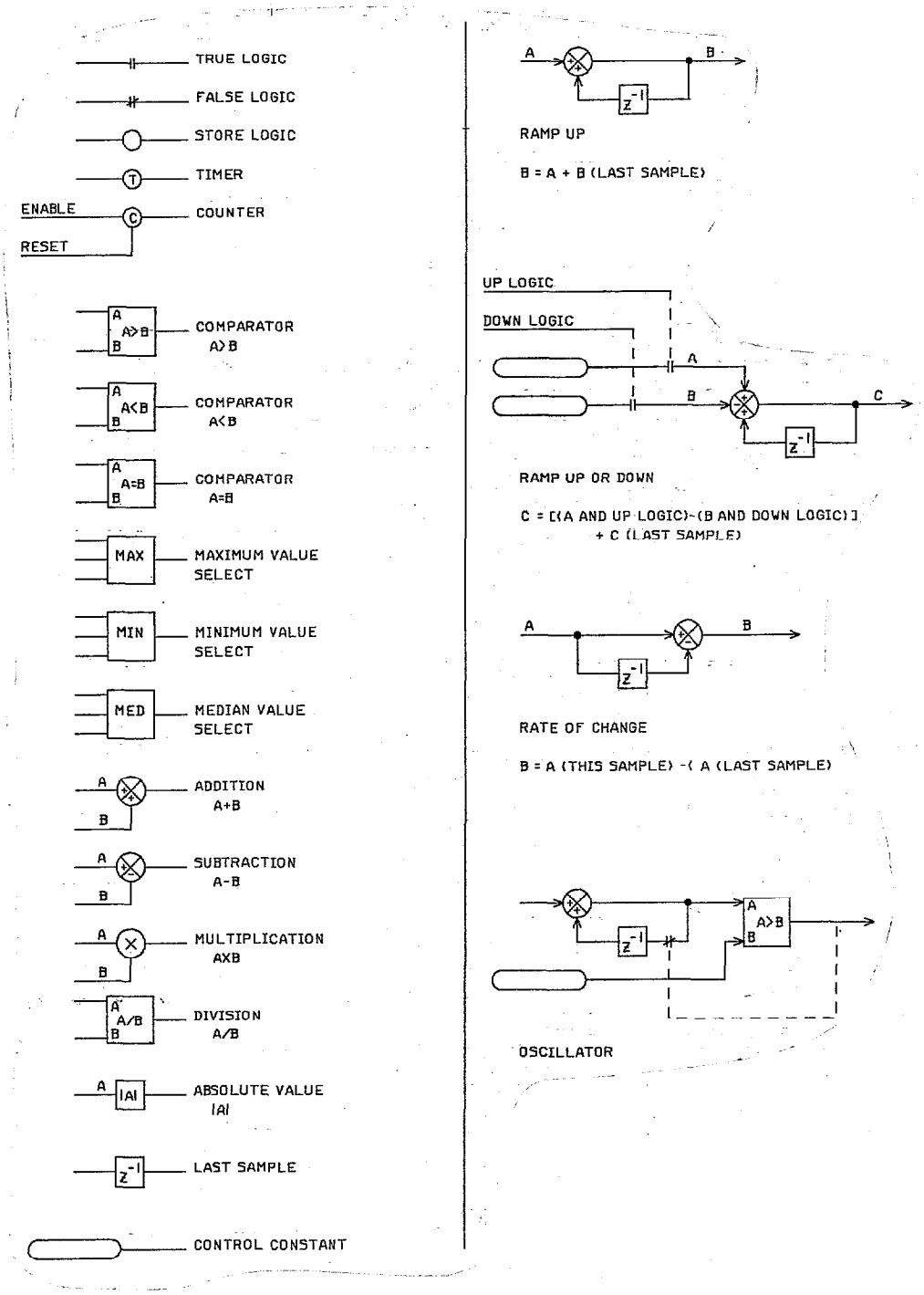


Figure VI-E-1: TYPICAL ALGORITHM SYMBOLS

G. DIAGNOSTIC OVERVIEW

Diagnostics consist of three (3) categories:

1. Built in on-line diagnostics which continuously monitor the health of the control system with the computer and turbine running.
2. Built in off-line diagnostics which analyze the control system with the computer on, the turbine stopped and the pluggable I/O terminal boards disconnected.
3. Remote diagnostics designed to test the control panel with the turbine stopped and the I/O terminal boards disconnected.

Built in on-line diagnostics include:

PROM Test	RS232 Link Test
BRAM Test	Communication Error Rate Test
[C], RAM Test	Idle/Task Status
Voting Reals Test	D/A and A/D Test
Power Supply Test	Task Status Test
Thermocouple Test	Link Failure Reset Information

A failure of one of these components initiates an alarm, and a message appears on the CRT describing the problem. In some cases the message may be sufficiently specific to immediately locate the source of the problem, such as a low voltage condition on the [R] power supply. This type of problem can be fixed by adjusting the supply or replacing the supply while [S] and [T] continue to run the turbine. Other fault conditions which are less specific can be exemplified by an erroneous reading of the [T] speed input. A diagnostic alarm indicates a discrepancy between the [T] input and the inputs from [R] and [S]. This could be due to a failure of the pulse rate to digital card HPRB in [T], or possibly a defective magnetic pick-up. Observation of the pulse rate input signal at the terminal board on the analog I/O module, [AIO3], determines whether the magnetic pick-up is the problem, and if it is not, the next most likely suspect is the HPRB card. These examples illustrate the advantage of having redundant computers which can be continuously compared to alarm any differences, and then fix the defective component while continuing to run the turbine from the other computers. The latter example of the defective pulse rate reading shows the limitations of the built in on-line diagnostics which can locate the fault area but can not determine the difference between a defective sensor and a defective I/O card with out external assistance.

Built in off-line diagnostics consist of an analog I/O test, digital I/O test, relay test, pushbutton test and auxiliary display test. These tests are performed with the computers on and the turbine stopped and disconnected from the appropriate control panel I/O. For example, all relays can be automatically operated in sequential order from the Operator Interface Module, [OPM], after entering the security code; however, pluggable terminal boards on the relay modules must be removed to prevent inadvertent operation of field devices as the relays are sequentially energized. These pluggable terminal boards are used on all I/O modules to enable disconnecting an entire terminal board with out removing each field wire on each screw terminal. Loop-back ribbon cables connect card outputs to inputs to provide automatic systems tests.

G. DIAGNOSTIC OVERVIEW (continued)

For example, HIOD cards in Communicator [C] interface with the contact input modules and the relay driver module. A loop-back cable connects the HIOD digital output connector to its input connector and the resultant test automatically tests the entire card. Analog I/O tests are performed in a similar manner with the assistance of an analog test card NATD which conditions the outputs to electrically match the input requirements. Diagnostic tests of the analog I/O modules are the lowest priority of troubleshooting because the modules contain no active components and consist of high frequency R-C filters only, which are mounted on flow-soldered back-planes, i.e., printed circuit cards. This shifts the probability of failure to the cards in the electronics modules which can be inserted or removed in pre-assigned slots. Note that cards should not be inserted or removed with power on.

Typical diagnostic alarm messages are:

MODULE	ALARM
BRAM_CHECK_SUM_0100	CHECKSUM ERROR ON HMPF BOARD
COM_ERROR_DIAG_0100	DEVICE [1,2,3], INCREASED UART ERROR COUNT
DA_AD_DIAG_0100	DA/AD LIMIT CHECK: [REFERENCE], [FEEDBACK]
DIAG_ALM_0000	PRINTER INOPERATIVE
HCMA_DIAG_0100	[HCMA2J, HCMA2L], [CPU ERR81, CPU ERR82, DPU ERR83, PRR ERR84, CFG ERR85]
IDLE_DIAG_0000	IDLE TIME LOW
PROM_DIAG_0100	PROM FAILURE [LOCATION]
RAM_DIAG_0100	RAM FAILURE [LOCATION]
STATUS_DIAG_0000	CONTROL TASK ACTIVE AND BID
SUPPLY_C_DIAG_0100	PWR SUPPLY [SIGNAL NAME], LIMIT CHECK
SUPPLY_Q_DIAG_0100	PWR SUPPLY [SIGNAL NAME], LIMIT CHECK
TC_DIAG_0000	THERMOCOUPLE FAILURE [SIGNAL NAME] INCREASED SC OFFSET [SIGNAL NAME] CJ SPREAD EXCESSIVE
TC_Q_DIAG_0000	THERMOCOUPLE FAILURE [SIGNAL NAME] INCREASED SC OFFSET [SIGNAL NAME]
VOTE_LOGIC_DIAG_0100	VOTING MISMATCH [SIGNAL NAME]
VOTE_REAL_DIAG_0100	VOTING MISMATCH [SIGNAL NAME]
INITIALIZE_PARTY_LINE_0000	UNIT [NUMBER], CHANNEL [NUMBER], PARTY LINE FAILED
MASTER_WATCHDOG_0000	PARTY LINE DRIVER OR HMHA FAILURE
XMIT_LIST_DEF_EXC_0000	UNABLE TO DOWNLOAD LIST [NUMBER], TO UNIT [NUMBER]

A typical diagnostic alarm display appears as follows:

DIAGNOSTIC ALARM STATUS 29 APRIL 83 18:16:09

DATE	TIME	DIAGNOSTIC ALARM
29 APRIL 83	17:40:56	31 1 [R] VOTING MISMATCH FSRS1

G. DIAGNOSTIC OVERVIEW (continued)

1. GROUP 00, 01, 02, 03 - PROM DIAGNOSTICS test each PROM by calculating a checksum and comparing it with the original checksum. The diagnostic message identifies the processor, card, slot location and specific chip which is faulty as shown in the following example: [R] PROM FAILURE HMPF1M U22
2. GROUP 10 - BRAM DIAGNOSTICS calculate a checksum to indicate a failure and provide the following message: [C] CHECKSUM ERROR ON HRMB BOARD
3. GROUP 20, 21, 22, 23 - RAM DIAGNOSTICS test each RAM, except for dual-ported RAM's, a word at a time, by exercising bit patterns while the interrupts are disabled. The diagnostic message identifies the processor, card and slot location of the defective chip as follows: [R] RAM FAILURE HXPD1K
4. GROUP 31, 32, 33 - ANALOG VOTING DIAGNOSTICS compare the values of analog signals in [R], [S], and [T], with the median value in [C], and alarm if any of the individual signals vary from the median value by more than the allowed tolerance. The resultant alarm message specifies the controller and signal name which failed the test as shown in the following example: [R] VOTING MISMATCH "BB1"
5. GROUP 41, 42, 43 - LOGIC VOTING DIAGNOSTICS compare the values of logic signals in [R], [S], and [T], with the median value in [C] and alarm if any of the individual signals vary from the median value. The resultant alarm message specifies the controller and signal name which failed the test as shown in the following example: [R] VOTING MISMATCH "L63QT"
6. GROUP 50, 51, 52, 53 - POWER SUPPLY DIAGNOSTICS compare each power supply bus with the ideal voltage level and generate an alarm message if a bus deviates from the ideal value by more than the allowable tolerance. A sample alarm message is: [C] POWER SUPPLY "P15V" LIMIT CHECK
7. GROUP 60 - HCMA DIAGNOSTICS test the status byte set by the on-board processor and alarm the card, slot and failure according to the following example: "HCMA2J" CPU ERR81"
8. GROUP 70, 71, 72, 73 - UART ERROR DIAGNOSTICS test the error counters maintained by the UART drivers and alarm any increased error rate as shown in the following example:

   DEVICE 1 INCREASED UART ERROR COUNT
   [R] HMPF INCREASED UART ERROR COUNT
9. GROUP 80, 81, 82, 83 - THERMOCOUPLE DIAGNOSTICS tests for failed thermocouples, i.e., negative saturation, short circuit tolerance and cold junction proximity to other cold junctions. Sample messages are:

   CJ SPREAD EXCESSIVE
   INCREASED SC OFFSET "SC-NAME"
   THERMOCOUPLE FAILURE "TC-NAME"
10. GROUP 90, 91, 92, 93 - IDLE TIME DIAGNOSTICS check that the idle time is above the minimum threshold level, i.e., that the processor is not overloaded. A sample alarm message is: [T] IDLE TIME LOW
11. GROUP A1, A2, A3 - DA/AD DIAGNOSTICS test the two D/A converters on each HSAA card by comparing the output reference to the HSAA with the feedback from the HAIC A/D converter and alarming any disagreement as shown in the following example: [T] DA/AD LIMIT CHECK: "OUT" "BACK"

G. DIAGNOSTIC OVERVIEW (continued)

DIAGNOSTIC ALARM STATUS 29 APRIL 83 18:16:09 (continued)

DATE	TIME	DIAGNOSTIC ALARM
29 APRIL 83	18:15:15	A1 0 [S] DA/AD LIMIT CHECK:FQROUT FQRBAK
29 APRIL 83	18:16:02	31 1 [R] VOTING MISMATCH FOR

The DIAGNOSTIC ALARM DISPLAY is accessed from the control system ALARM DISPLAY, and it lists all existing failures in chronological order with a maximum capacity of 40 messages. Softswitches on the display allow the operator to acknowledge and reset alarms. The REVIEW softswitch enables scrolling of the screen to look at the alarms one at a time.

Sample message:

```
29 APRIL 83 16:05:32  20      1      PROM FAILURE HMPF1M U23
     |          |      |      |              |
     |          |      |      |              |
   DATE       TIME   GROUP  STATUS         MESSAGE
```

When an alarm condition occurs, a message appears on the CRT with the date in the far left hand corner followed by the time when the fault was logged into the link list. A group number specifying which diagnostic test failed will blink in inverse video until the alarm is acknowledged. After acknowledging the alarm, it can be reset via the RESET ALARM softswitch if the actual alarm condition has cleared. This is indicated by the status flag.

The status flag can assume four (4) different values:

1. STATUS = 0 indicates that the failure cleared the last time it was tested.
2. STATUS = 1 indicates that the fault is still present. All alarms will come up in this mode the first time they appear.
3. STATUS = 2 This status mode will appear on the screen for any Controller [R], [S] and [T] alarm if the communication link failed, i.e., L3COMF = 1, after the alarm was annunciated. If STATUS = 2 remains on the screen after the controller recovers, it means that the alarm did not come back and an acknowledge and reset will erase that alarm message.
4. STATUS = 3 occurs if an alarm is followed by a [C] link failure, and when the link recovers, the alarm reappears.

The diagnostic programs are organized according to the type of diagnostics being performed. A brief description of the various GROUPS of programs are listed below:

G. DIAGNOSTIC OVERVIEW (continued)

12. TASK STATUS DIAGNOSTICS tests for the proper task completion and alarms any timing problems which would indicate that the system is bogging down. A sample message is: [S] CONTROL TASK ACTIVE AND BID
13. A/D DIAGNOSTICS test the A/D converter on the HAIC card and provide the following typical message: [R] HAIC A/D CONVERTER FAILED

H. APPLICATION SOFTWARE TEST

An in-house software test bed consists of hardware which is identical to the equipment shipped on every order. A copy of the customer's application software is loaded into the facility and evaluated in an environment simulating the installation requirements. Logic inputs and outputs are used to simulate field contact inputs and relay contact outputs. Signal generators simulate magnetic pick-ups, millivolt sources simulate thermocouple inputs, voltage sources simulate pressure transducers, etc. The system engineer assigned to the order performs the following tests:

1. Turbine start-up sequence
2. Turbine shutdown sequence
3. Governor control loops (static simulation)
4. Alarms are tested from their point of origin
5. Trips are tested from their point of origin
6. Analog input scaling is reviewed
7. Displays are reviewed
8. Running reliability test - loss of a computer

I. CUSTOMER RS232 INTERFACE

THIS SECTION WILL BE ADDED AT A LATER DATE.

NOTES

VII. THE GOVERNOR

A. CONTROL OVERVIEW

The gas turbine is controlled by regulating the fuel flow to the turbine's combustion chamber. A single parameter called the fuel stroke reference FSR is calculated in the [R], [S] and [T] Controllers as a function of speed, temperature and pressure feedbacks to determine the fuel limit. Figures VII-A-1 and VII-A-2 provide integrated software and hardware one-line diagrams of the control system with references inside the modules identifying on which sheet the modules are located in a typical control elementary. The key algorithm FSR selects the individual control loop, such as speed or temperature, calling for the least fuel. Since Controllers [R], [S] and [T] have individual FSR values, the Communicator selects the median value and corrects the other controllers. The resultant outputs drive three (3) coil fuel servos which are used to extend the range of the two-thirds voting concept to the turbine. Current from the three (3) coils induces a magnetic field in the torque motor which acts in conjunction with a permanent magnet to move the armature and open the valve.

Individual parameters used to determine the fuel stroke reference are:

FSRACC - Acceleration Control
FSRMAN - Manual Suppression of the fuel reference
FSRN - Speed Control
FSRSU - Start-up Control
FSRSYN - Synchronizing Control
FSRT - Temperature Control

At approximately 20% speed, the typical firing speed of the machine, a fuel to air mixture is selected which will produce approximately 1000°F temperature rise in the combustion chamber. This fuel to air mixture is sufficient to provide satisfactory ignition and cross-firing but not enough to cause any mechanical damage. The minimum fuel limit, commonly called the minimum "blowout fuel", is the least fuel which will maintain the flame but is not sufficient to keep the turbine at full-speed no-load.

As the turbine starts to accelerate, additional fuel is added to compensate for the increased air flow as programmed by the open loop start-up control. The turbine continues to accelerate until the governor goes on speed control which establishes the maximum fuel limit.

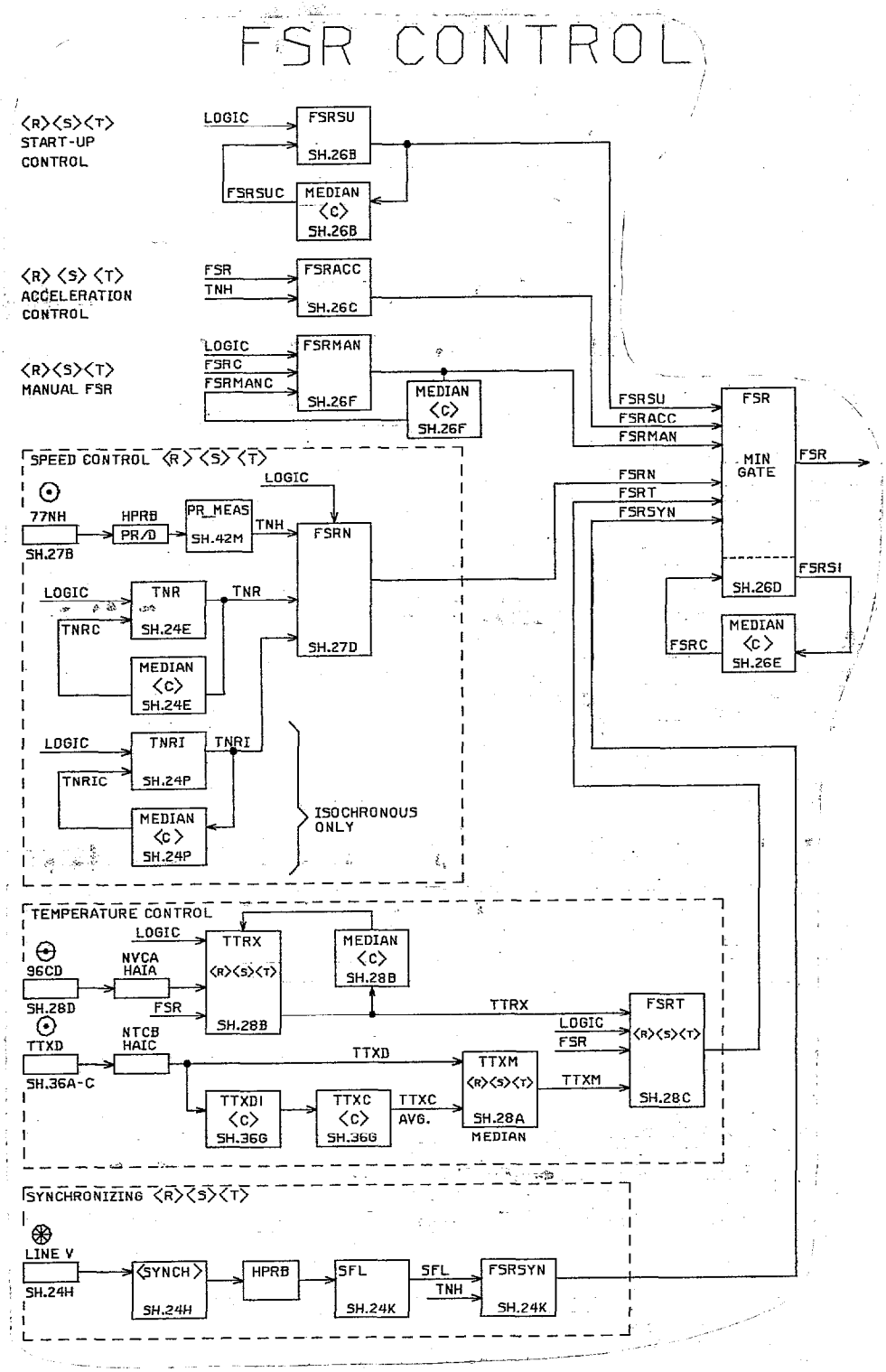


Figure VII-A-1: FUEL STROKE REFERENCE CONTROL

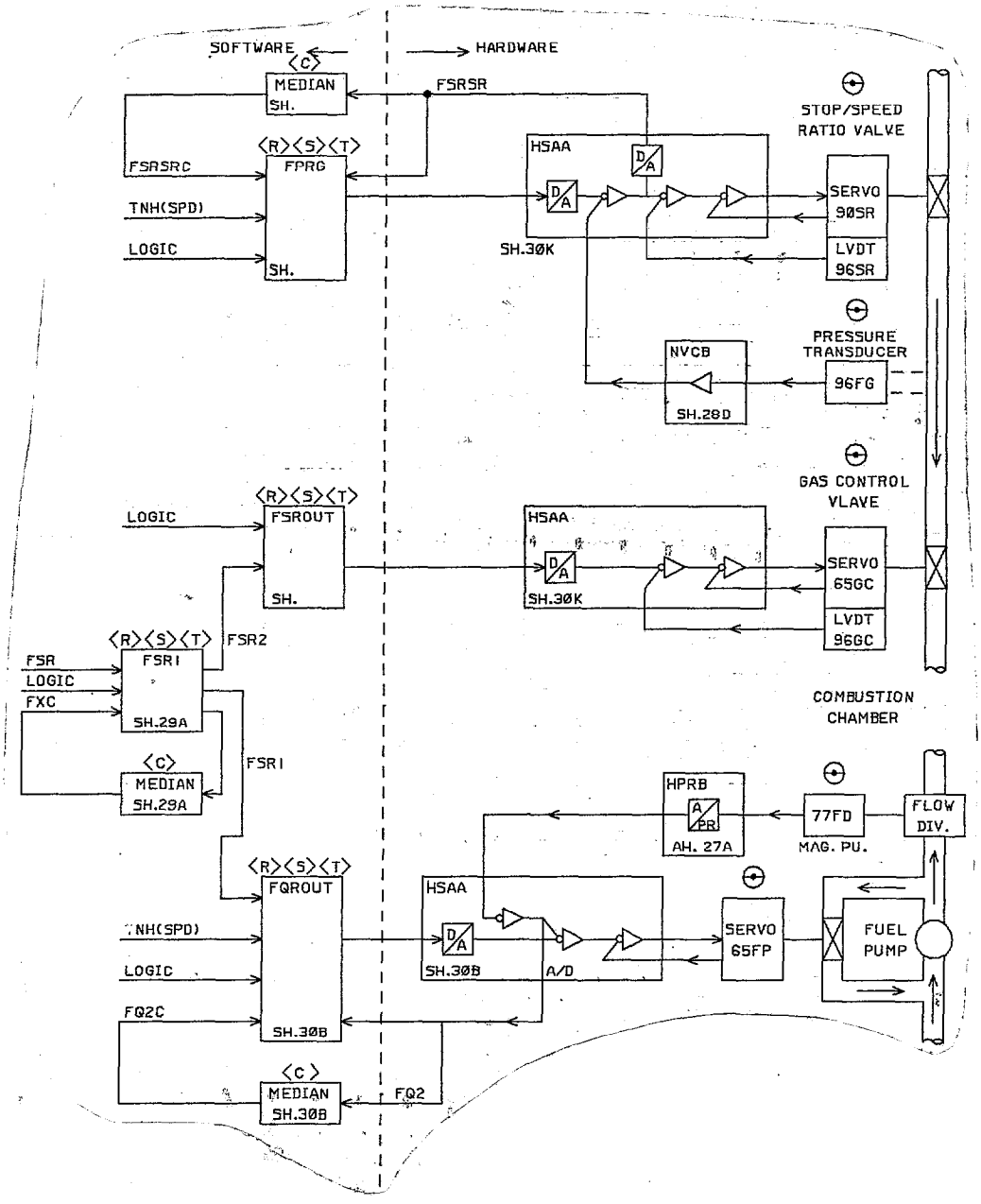


Figure VII-A-2: FUEL CONTROL, TYPICAL MS6001

B. START-UP CONTROL

The start-up control is an open loop system which increases the fuel stroke reference as the turbine start-up sequence progresses to pre-assigned plateaus. Figure VII-B-1 shows the start-up algorithm FSRSU located in Controllers [R], [S] and [T] and the median select algorithm MEDIAN_REF_SYNCH_C located in Communicator [C].

The algorithm output FSRSU is enabled when the master protective L4 logic becomes a "1". This occurs simultaneously with the energizing of the fuel stop valve solenoids. The fuel reference is held at zero (0) while the air flow is increased through the combustion chamber. Logic L83SUFI steps FSRSU to the fire level specified by constant FSKSU_FI when the ignition transformers are energized. Fuel is maintained at this level until approximately two (2) seconds after flame is detected. At this time, logic L83SUWU steps the fuel down to the warm-up level specified by constant FSKSU_WU in order to reduce the thermal shock to the turbine's hot gas path parts.

After the warm-up period expires, logic L83SUAR raises the fuel limit to the acceleration level defined by constant FSKSU_AR. A comparator detects whether the difference between the fuel level setpoint FSKSU_AR and the fuel command FSRSU is greater than the allowable deadband FSKSU__DB, and if it is, the comparator selects a ramp function to slowly increase the fuel reference to the acceleration level. This is accomplished by adding the value of FSRSU during the last sampling of the algorithm to the rate constant, FSKSU_IA, and using the sum as the new fuel reference. Since the algorithm is sampled four (4) times per second, the slope of the acceleration fuel schedule is 4 X FSKSU_IA per second. Eventually, the difference between the acceleration limit FSKSU_AR and the fuel reference FSRSU decreases to less than the deadband established by FSKSU_DB which causes the comparator to disable the ramp and use the acceleration limit as the fuel reference. The governor is now on speed control.

Logic L83SUMX will initiate another ramp to the maximum fuel limit determined by constant FSRMAX. This occurs when the breaker closes on generator drives and usually when the turbine start-up sequence is complete on mechanical drives.

Logic L4REC is used to preset FSRSU to the median value FSRSUC as calculated by Communicator [C] when a Controller, i.e., [R], [S] or [T], is restarted.

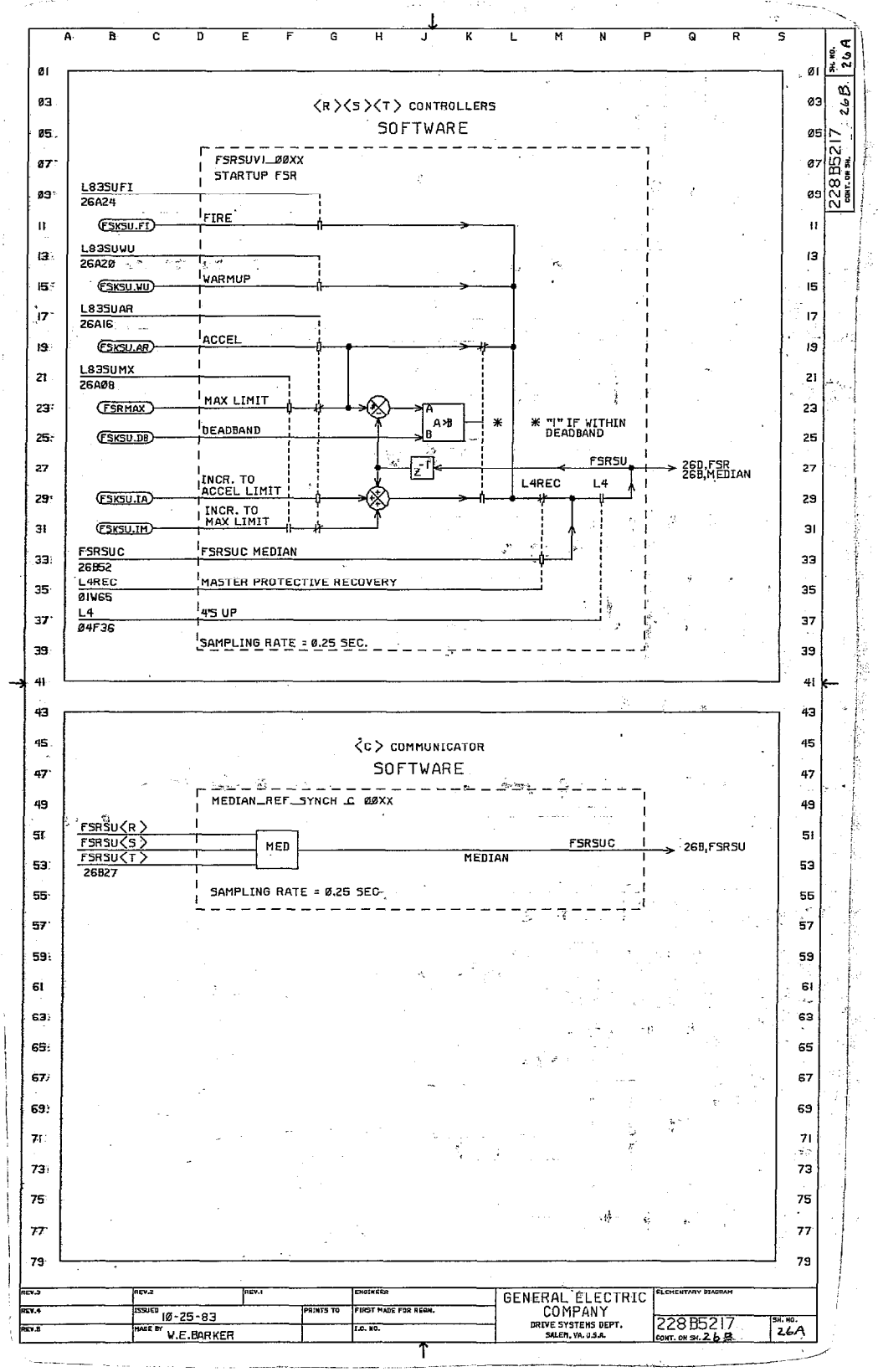


Figure VII-B-1: START-UP CONTROL

C. ACCELERATION CONTROL

The acceleration control algorithm FSRACC adjusts the fuel stroke reference according to the rate of change of turbine speed to reduce the thermal shock to the hot gas path parts of the machine. The algorithm subtracts the high pressure set speed, TNH, being read during this sample from the speed which was read during the last sample, and an offset FSKACCR1 is added defining the allowable acceleration. Logic input L86ACR2 provides the ability to select a second allowable acceleration rate as contained in control constant FSKACCR2. Another logic input signal L30F_ACN determines whether FSRACC is inversely proportional to the acceleration, when "not on" acceleration control or inversely proportional to the rate of change of acceleration when "on" acceleration control. For example, when the turbine is accelerating and the governor is "not on" acceleration control, TNH from the current sample will be at a higher speed than TNR from the last sample causing FSRACC to decrease as long as it is between the maximum and minimum limits established by constants FSRMAX and FSRMIN, respectively. FSR is added to the acceleration calculation; therefore, when the governor is "not on" acceleration control FSRACC = FSR + FSKACC2 [FSKACC1 - TNH this sample + TNH last sample], and when the governor is "on" acceleration control the equation is: FSRACC = FSR + [(FSKACC1 - TNH this sample + TNH last sample) X (FSKACC3 this sample - FSKACC4 last sample)]

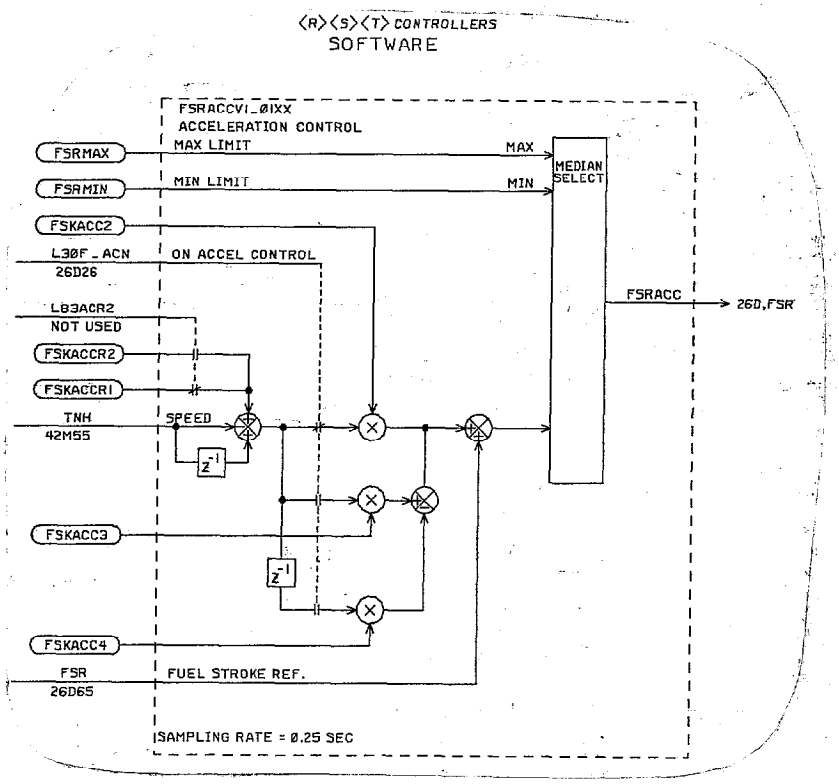


Figure VII-C-1: ACCELERATION CONTROL

D. SPEED CONTROL

1. APPLICATION

There are two (2) types of turbine speed control systems. Constant speed systems are applied when the turbine load is a system of large capacity such as power generation applications where the load is determined by the system and not by the turbine. Variable speed systems, such as pumpers, change their speed to adjust the load. In addition, the power generation applications are divided into droop applications and isochronous applications. Large power systems run one turbine isochronous to establish the frequency and the other machines droop with a load proportional to the difference between the called for speed and the system frequency. Consequently, an isochronous machine sets its speed reference to establish the system frequency, and a droop machine varies its speed reference to compensate for the speed droop and hold the frequency. An increase in the system load will be absorbed by the isochronous machine which increases its output to hold the system frequency while the droop machines pick up no load, i.e., steady-state, because their output is determined by the difference between their called for speed and the system frequency only. Figure VII-D-1, below, shows the relationship between % speed and % load. The isochronous machine's speed reference is defined as 100% speed, and the typical 4% droop machines with a 103% speed setpoint will output 75% load.

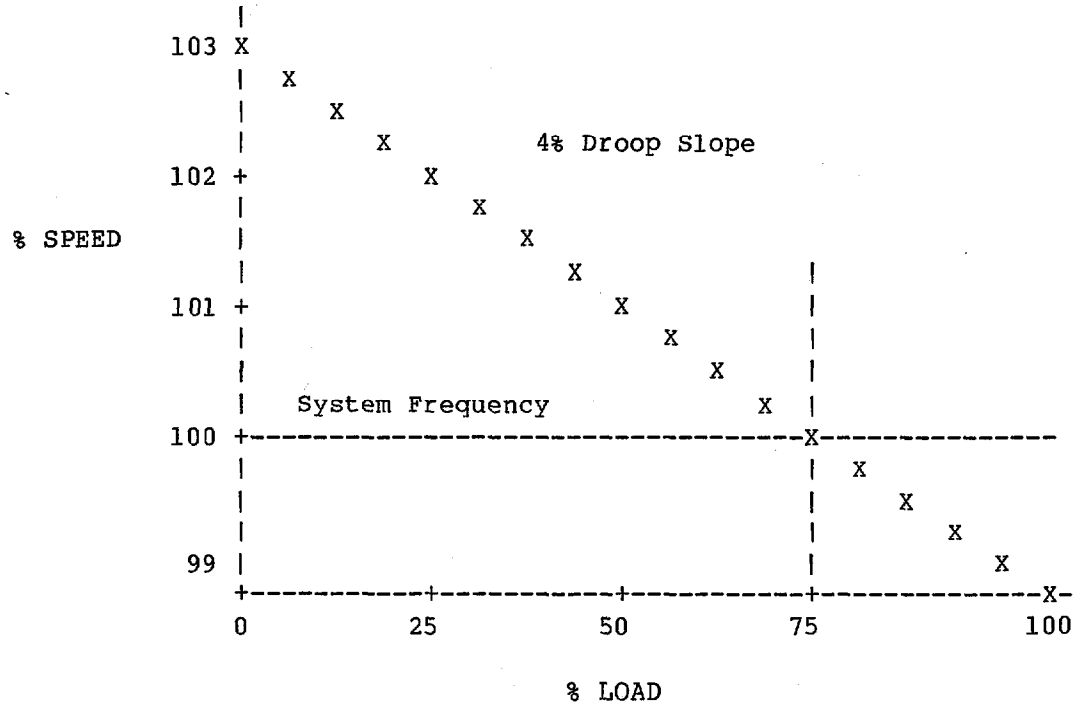


Figure VII-D-1: %SPEED VERSUS %LOAD CHARACTERISTICS

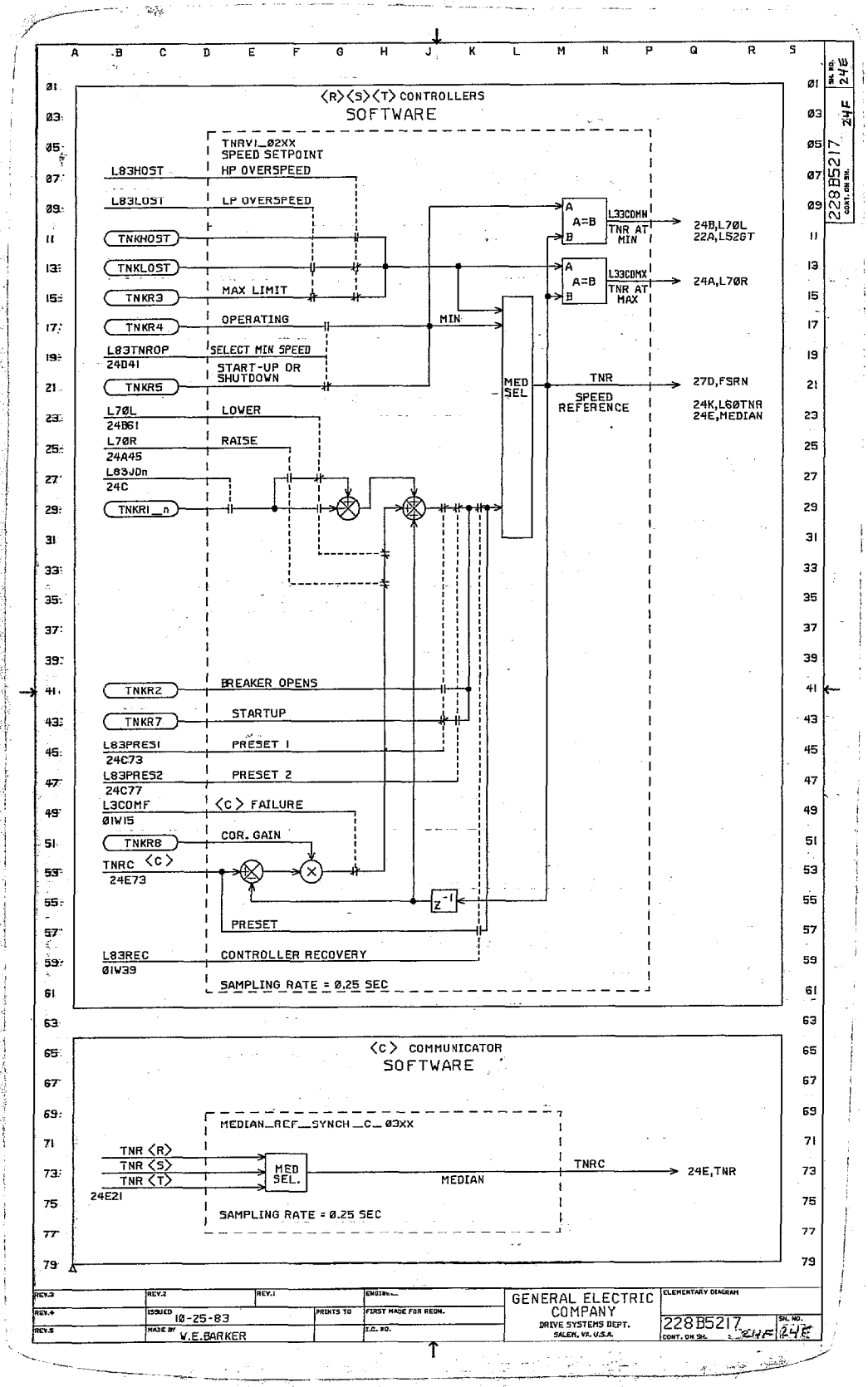


Figure VII-D-2: SPEED REFERENCE ALGORITHM - TNR

D. SPEED CONTROL (continued)

2. SPEED REFERENCE

The speed reference algorithm, TNR, is located in Controllers [R], [S] and [T], and is shown in Figure VII-D-2. Its output, also named TNR, is calculated by ramping or stepping to assigned levels defined by constants. A turbine start command causes logic L83PRES2 to become a "1" which presets TNR to the start-up level defined by constant TNKR7. L83PRES2 resets to a "0" after the machine breaks away, and TNR remains at the preset level until the governor goes on speed control. At this point, a raise command is given by L70R which begins to raise the setpoint at a rate selected by L83JD(0-7) and defined by constants, TNKR1(0-7). The (0-7) suffix defines an array of eight (8) possible rate selections. The ramp function is implemented by adding an amount of speed specified by TNKR1(0-7) to TNR from the last sample at a rate determined by TNKR1(0-7) and the inherent sampling rate, 0.25 seconds, of the algorithm. Lower command calculations subtract the ramp setpoint from the previous TNR value by the same method. Note that the algorithm contains no comparator to disable the ramp when it reaches the desired level. This is accomplished by speed/load feedbacks to the external control which disables the raise, L70R, and lower, L70L, commands at the appropriate time.

A median feedback, TNRC, from Communicator [C] is compared with the local speed reference, TNR, and the error is used to bias the TNR calculation proportional to the gain defined by constant TNKR8. L3COMF inhibits this correction when a [C] failure occurs. A controller recovery occurs when L83REC = "1" which presets the TNR calculation to the median value TNRC.

L83HOST and L83LOST raise the maximum speed setpoint for testing the mechanical overspeed bolt for the high pressure set or the low pressure set respectively. These functions are manually selected from the MANUAL CONTROL DISPLAY for the MECHANICAL OVERSPEED TEST, and they raise the the electrical overspeed limits while increasing the maximum speed reference setpoint to the limits specified by constants, TNKHOST and TNKLOST.

3. ISOCHRONOUS SPEED REFERENCE

The isochronous speed reference algorithm is named TNRI with the "I" designating "isochronous". It is functionally similar to the TNR algorithm described previously. Both algorithms are used in control systems with droop/isochronous control selection, and both feed the speed control fuel stroke reference algorithm FSRN. Selection of the droop or isochronous mode of operation is made by sequencing and fed to the algorithms by logic inputs. In order to execute a bumpless transfer between the two (2) modes, tracking algorithm L70TRACK is used to monitor both setpoints. It is shown in Figure VII-D-5. The isochronous speed setpoint, TNRI, is subtracted from the primary speed setpoint, TNR, prior to subtracting the calculated droop, DWDROOP. This value is compared against an established deadband, TNKLDB, to determine if the TNR setpoint needs to be raised, L70TKR, or lowered, L70TKL.

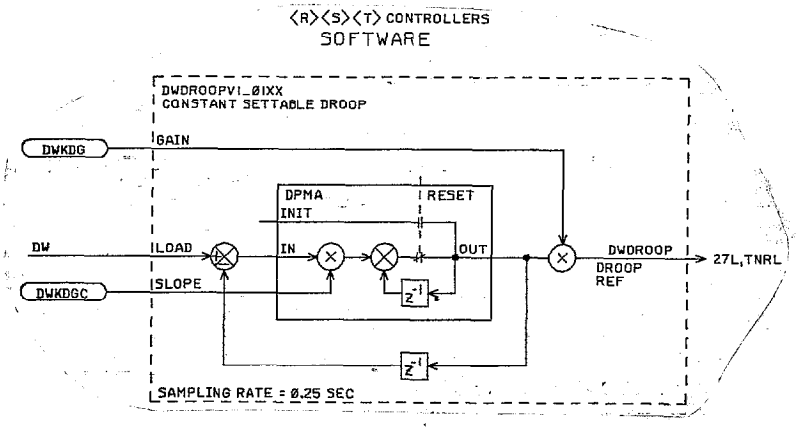


Figure VII-D-3: CONSTANT SETTABLE DROOP

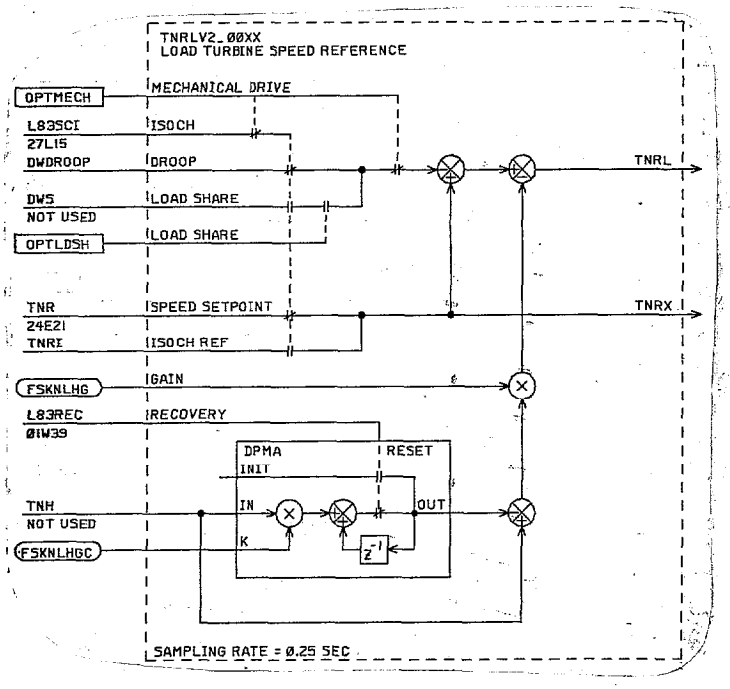


Figure VII-D-4: LOAD TURBINE SPEED REFERENCE

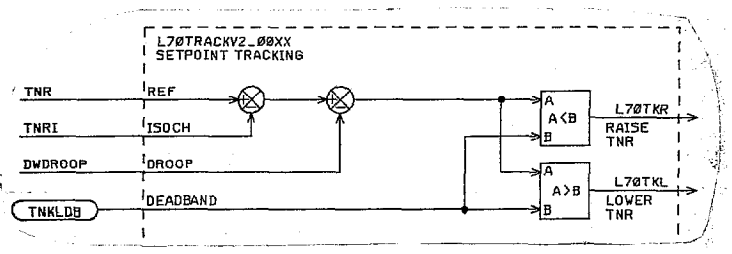


Figure VII-D-5: SETPOINT TRACKING

D. SPEED CONTROL (continued)

4. LOAD TURBINE SPEED REFERENCE

The load turbine speed reference algorithm, TNRL, is shown in Figure VII-D-4. It can be used for various applications:

a. Constant setable droop on load inputs, DWDROOP.
b. Isochronous control for two (2) shaft gas turbines as selected by logic L83SCI which enables the isochronous speed reference input, TNRI.
c. Isochronous load sharing as defined by the non-adjustable control constant, OPTLDSH, which enables the load sharing input bias, DWS.
d. Some single shaft mechanical drives which require biasing the speed reference, TNRL, as a function of the high pressure set speed, TNH.

5. SPEED INPUTS

The turbine speed is monitored by magnetic pickups mounted near a 60 tooth wheel attached directly to the turbine shaft. The pickups generate voltage pulses as the teeth of the wheel pass under them, and the resultant signals are wired to analog I/O module #3 [AIO3]. Three (3) separate pickups are used to monitor speed on a given shaft for redundancy. Each signal is monitored by one of the HPRB cards in Controllers [R], [S] and [T], which contain the pulse rate to digital converters. Since some pulse rate inputs are a function of wheels which do not have 60 teeth, an algorithm, PR_MEAS, scales each input according to the number of teeth on its wheel. Typical inputs are high pressure set speed, low pressure set speed, starting device speed and liquid fuel flow from the flow divider magnetic pickup. Each input is scaled by adjustable control constants which are displayed on the CRT and protected by the security code. The scaled inputs are then called by various control algorithms as required.

Turbine sequencing requires several speed check points to determine when the machine has reached pre-defined levels. Algorithm L14H compares the high pressure set speed, TNH, with control constants which establish the speed levels. Algorithm L14L is used to monitor speed levels for the low pressure set. The logic outputs are:

L14HR = "1" when the high pressure set is stopped
L14HM = "1" when the high pressure set is at or above minimum speed
L14HA = "1" when the high pressure set is at or above accel. speed
L14HS = "1" when the high pressure set is at or above operating speed
L14LR = "1" when the low pressure set is stopped
L14LS = "1" when the low pressure set is at or above operating speed

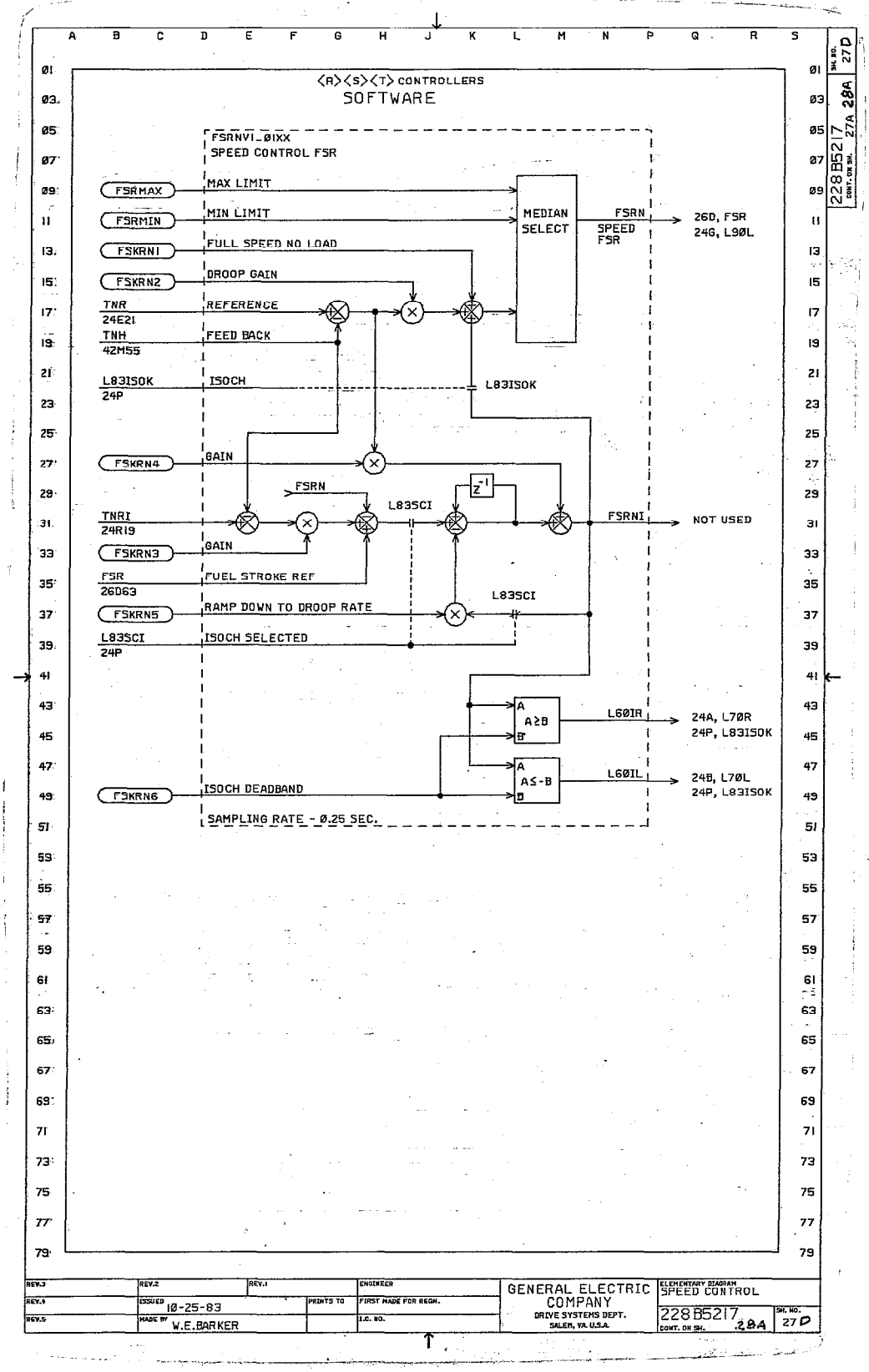


Figure VII-D-6: SPEED CONTROL FSR - GENERATOR DRIVES

D. SPEED CONTROL (continued)

6. SPEED CONTROL FUEL STROKE REFERENCE (SINGLE SHAFT GENERATOR DRIVES)

The speed control fuel stroke reference algorithm FSRN version "1", is used for generator drives, and it is shown in Figure VII-D-6. The turbine high pressure set speed, TNH, is subtracted from the called for speed reference, TNR, multiplied by the system droop, FSKRN2, and offset by the full speed no load setpoint defined by constant, FSKRN1. If the control is not adding the isochronous bias, i.e., L83ISOK is "0", then the result of this calculation becomes the speed control fuel stroke reference, FSRN, provided that it is within the maximum and minimum limits established by constants, FSRMAX and FSRMIN, respectively.

If the control is in the isochronous mode, then the FSRN calculation is biased by the isochronous fuel stroke reference, FSRNI. Turbine speed TNH is subtracted from the isochronous called for speed reference TNRI, scaled by FSKRN3 and biased by FSR and FSRN. Note that if the governor is on speed control then FSR = FSRN, and the resultant controlling parameter will be the scaled error calculation. This value ramps up or down, is offset by the scaled droop error, and then used to bias the FSRN calculation. FSRNI is constantly compared with the defined isochronous deadband, FSKRN6. It will raise the TNR setpoint if FSRNI is found to be too high. L83ISOK will continue to add the FSRNI bias to the FSRN calculation as long as L60IL or L60IR is a "1" or as long as the control is in the isochronous mode, i.e., L83SCI is a "1". If the isochronous mode is not selected, then FSRNI will ramp down at a rate determined by constant FSKRN5.

7. SPEED CONTROL FUEL STROKE REFERENCE (MULTIPLE APPLICATIONS)

Speed control fuel stroke reference algorithm FSRN version "2", is used for various applications. It is shown in Figure VII-D-7. A turbine speed feedback is selected by the non-adjustable control constant, OPT1SFT, which enables either the high pressure set speed feedback, TNH, or the low pressure set speed feedback, TNL. Logic, L83HOST, is used to override OPT1SFT if a high pressure set overspeed bolt test is in progress. This feedback is compared with the called for speed reference, TNR or TNRL, depending on whether the constant setable droop option is being provided as defined by constant, OPTDWDR. The resultant error is multiplied by constant, FSKNG, and compared with the maximum limit prior to adding an FSR bias and finally establishing the speed control fuel stroke reference, FSRN. FSR is scaled by the gain constant, FSKNGC, and ramped by the DPMA subroutine until the subroutine output nullifies the original FSR input. Maximum and minimum limits restrict the range of FSRN via constants, FSRMAX and FSRMIN. The minimum limit can be a fixed parameter defined by constant, FSRMIN, or it can be a variable parameter whose calculation results in a variable input named FSRMIN. Non-adjustable constant, OPTVMN, selects the variable minimum if it is set to a logic "1". In addition, the maximum limit for the speed reference/feedback error calculation can be made variable if non-adjustable constant, OPTHPSOR, is a "1".

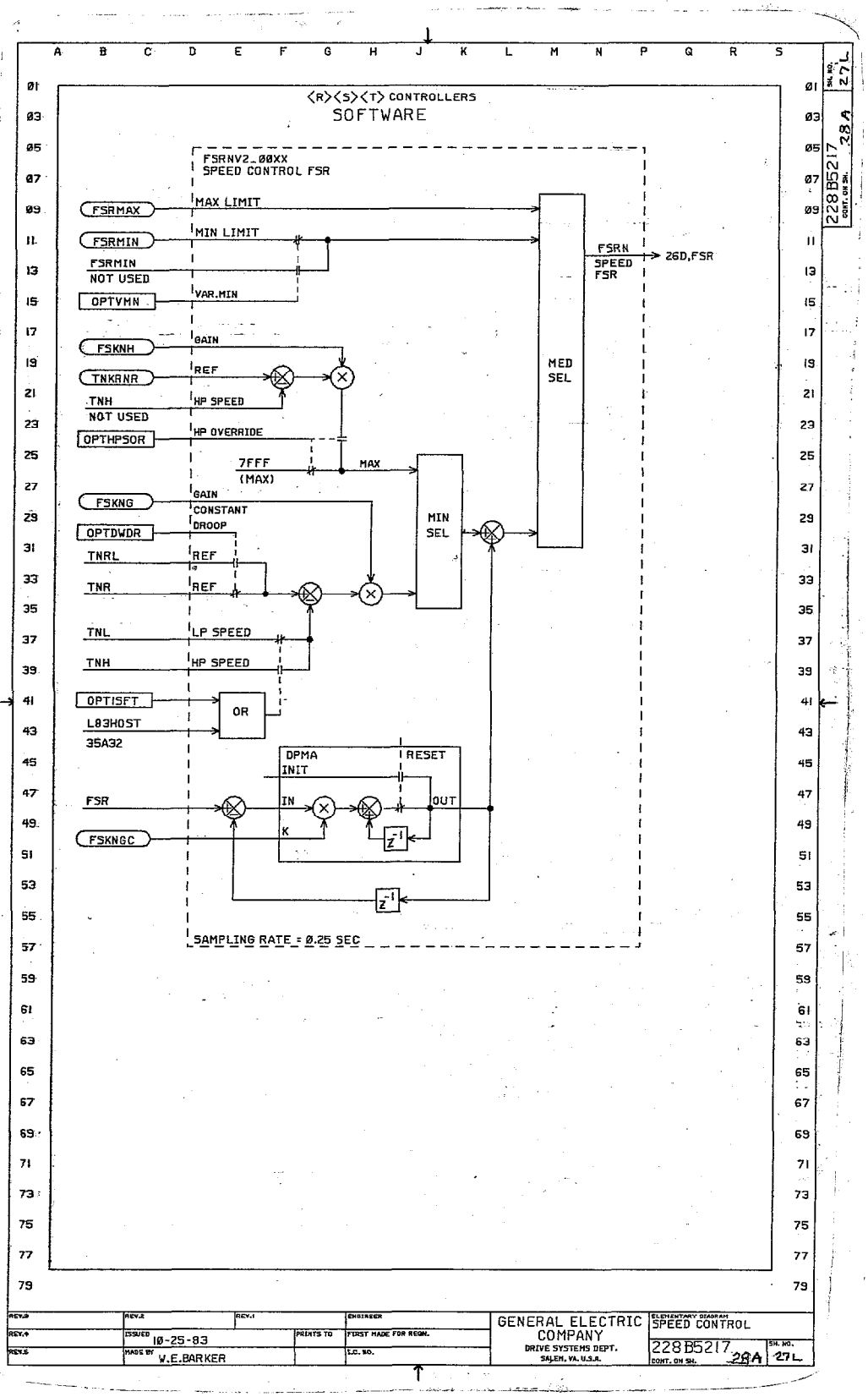


Figure VII-D-7: SPEED CONTROL FSR - MULTIPLE APPLICATIONS

E. NOZZLE CONTROL

1. APPLICATION

Two (2) shaft gas turbines have the ability to control two (2) parameters, such as load speed and exhaust temperature versus single shaft gas turbines which can control one (1) parameter only. Load speed can be controlled by regulating the fuel stroke reference, FSR, and exhaust temperature or high pressure set speed can be controlled by modulating the nozzle angle. The nozzles are curved blades which are mounted between the high pressure set and the low pressure set. An electrically modulated servo valve varies the nozzle angle, and consequently, varies the pressure drop across each set. Opening the nozzle, i.e., increasing the angle, increases the pressure drop across the high pressure set causing it to accelerate, and conversely, closing the nozzles causes the low pressure set to accelerate. During a start-up sequence, the nozzles are fully open to accelerate the high pressure set. When the governor begins to regulate speed and pick-up load, the nozzles begin to close, and the firing temperature increases.

2. HP TURBINE SPEED REFERENCE (TWO SHAFT TURBINES)

The high pressure set speed reference for two (2) shaft gas turbines is calculated in algorithm TNRH located in controllers [R], [S] and [T]. It is shown in Figure VII-E-1. TTRX supplies the base load exhaust temperature reference which includes the correction for the compressor discharge pressure. The difference between reference TTRX, and median temperature TTXM, is offset by TNKRHO and multiplied by TNKRHG. During a start-up sequence, the difference between TTRX and TTXM is a large positive value which causes TNRH to call for fully open nozzles. An error calculation subtracts TNRH from its median value, TNRHC, which is calculated in [C]. It scales the error with TNKRHCG before adding it to the local TNRH calculation. L3COMF interrupts this correction if the communication link with [C] fails. The DPMA subroutine is used to soften transients in TNRH proportional to the gain contained in constant TNKRHGC. Maximum and minimum limits are contained in constants, TNKRHHSS and TNKRHLSS, although the minimum can be modulated by the corrected speed and air flow setpoint, TNRHAQ, if the air flow option is supplied. Logic output L60NZTC is used to indicate whether the exhaust temperature is controlling TNRH or if TNRH is being limited by the modulated minimum limit.

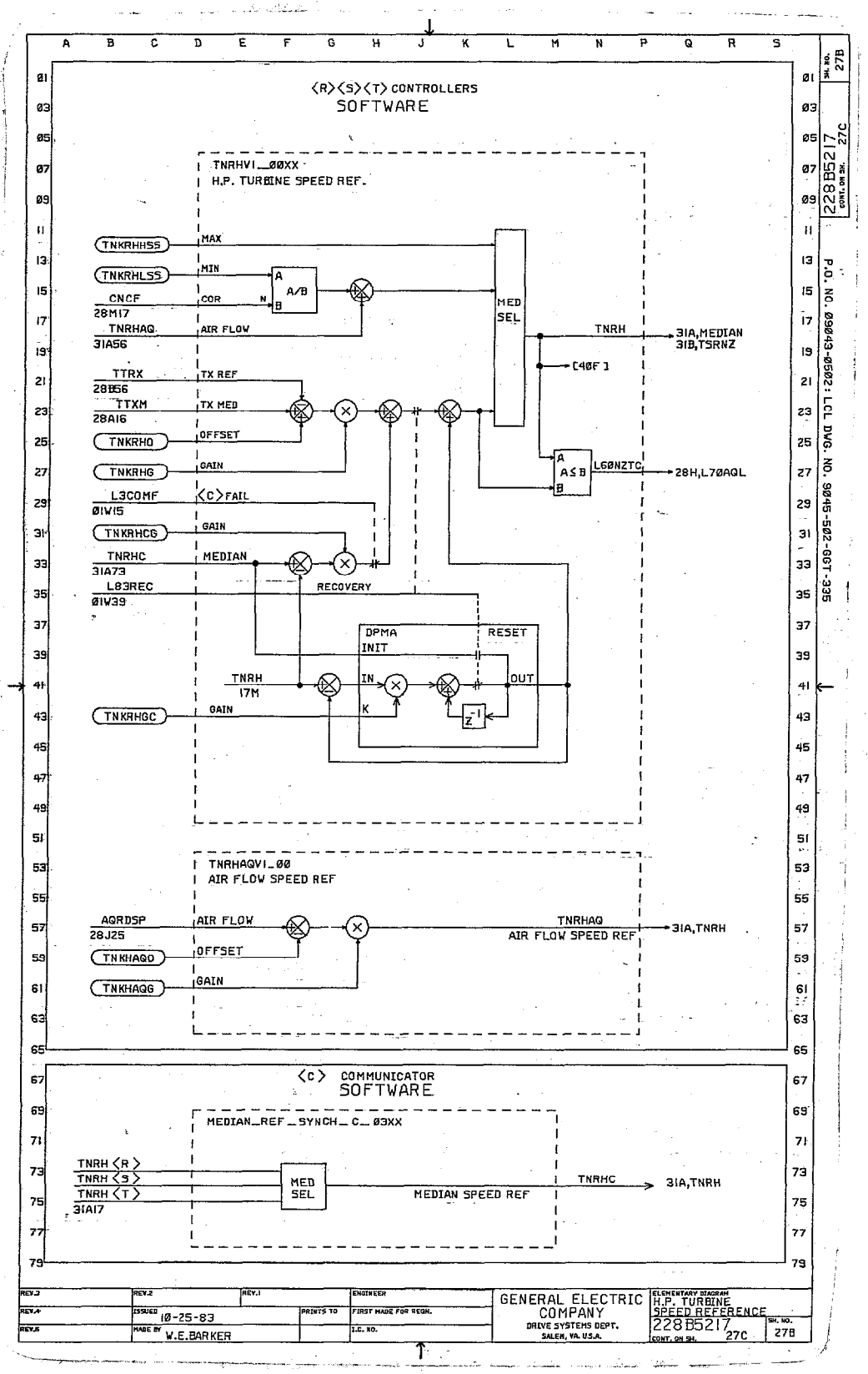


Figure VII-E-1: HP TURBINE SPEED REFERENCE (TWO SHAFT TURBINES)

E. NOZZLE CONTROL (continued)

3. The nozzle angle reference algorithm, TSRNZ, is shown in Figure VII-E-3. High pressure set speed TNH is subtracted from speed reference TNRH and offsets are added for the fuel stroke reference and a proportional gain option. FSRS1 is the local fuel stroke reference without the median correction. It is subtracted from minimum limit stored in constant FSRMIN, and multiplied by TSKRNZNG before being compared with its maximum and minimum limits, TSKRNZGM, and the fixed value, 00H. The called for nozzle angle is scaled by constant TSKRNZG prior to a correction bias determined by the difference between local nozzle angle reference TSRNZ, and median value TSRNZC. This calculation is performed similarly to the median correction just described for the speed reference algorithm, TNRH. Maximum and minimum limits for the nozzle angle are contained in constants TSKRNZMX and TSKRNZMN. If a test of the high pressure set overspeed bolt is performed, then logic L83HOST sets called for nozzle angle TSRNZ, to the maximum limit. The overspeed test can be selected via the MANUAL CONTROL DISPLAY called MECHANICAL OVERSPEED TEST.

4. NOZZLE CONTROL VALVE OUTPUT

The algorithm which drives the digital to analog converter on the nozzle control servo amplifier card, HSAA, is shown below in figure VII-E-2. It reads called for nozzle angle reference TSRNZ, multiplies by constant NXKOUTG, adds electrical offset NXKOUTO and adds a mechanical bias, NXKOUTB. Output TSRNZOUT can be calibrated by selecting the ANALOG CONTROL DISPLAY, i.e., List 15, for the NOZZLE CONTROL. This enables JADJ to select the nozzle servo, i.e., #14, and set TSRNZOUT to the value defined by GSADJ.

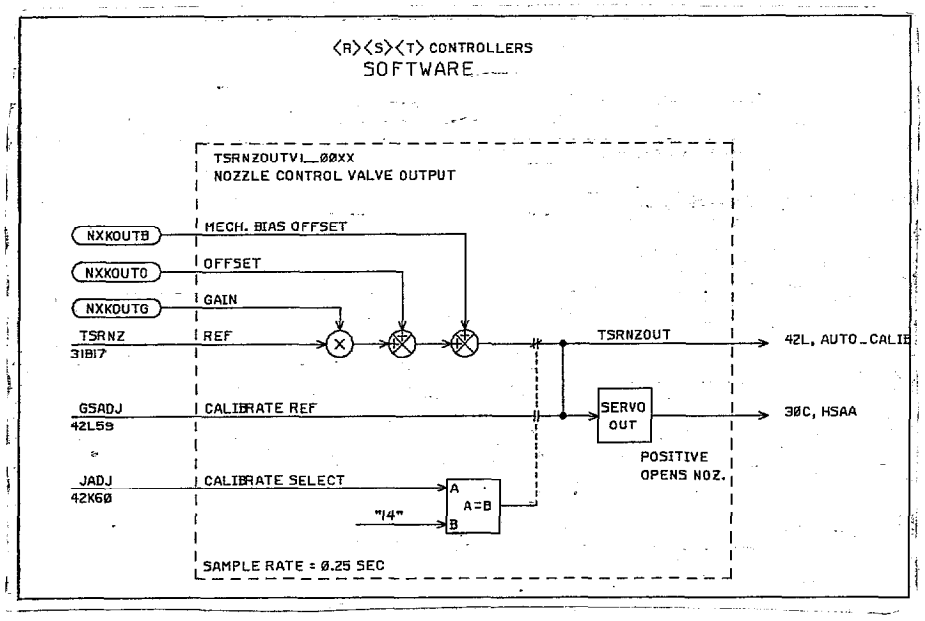


Figure VII - E - 2: NOZZLE CONTROL VALVE OUTPUT

F. TEMPERATURE CONTROL

1. APPLICATION

Temperature control is used to limit the turbine firing temperature by regulating fuel flow. The firing temperature is the hottest temperature in the turbine and the most difficult to monitor due to the heat and the thermal gradients. However, exhaust temperature is measurable and is proportional to the firing temperature. Chromel-alumel thermocouples are mounted in the exhaust duct and provide a temperature feedback proportional to the firing temperature with a time lag corresponding to the velocity of air flow through the turbine. Another consideration is the relationship between the ambient temperature and the firing temperature. Air is more dense on cool days causing the firing temperature to increase for a given high pressure set speed. The increased temperature will improve the turbine's efficiency but it must be factored into the temperature control calculation to prevent over-firing the machine. Fortunately, compressor discharge pressure downstream of the last compressor stage is a function of ambient temperature, compressor efficiency and the high pressure set speed, and is a measurable parameter. As the ambient temperature decreases, the more dense air causes the compressor discharge pressure to increase which electrically lowers the exhaust temperature reference.

2. TEMPERATURE CONTROL REFERENCE

Algorithm TTRX calculates the temperature control reference as a function of compressor discharge pressure CPD, and the fuel stroke reference, FSR. Figure VII-F-1 shows the algorithm located in Controllers [R], [S] and [T]. Logic, L83JT(0-7), selects which set of control constants the algorithm will use in its calculation as noted by one of eight possible suffixes, 0-7. For example, one set of constants would be used for a base load characteristic and another set would be used for a peak load characteristic. Constant TTKn_I, establishes the isothermal limit via the minimum select function which selects the lowest temperature reference called for by CPD, i.e., TTRXP, the FSR reference, i.e., TTRXS, or the isothermal limit. Normally CPD determines the called for reference and FSR is used as a back-up in case the CPD transducer fails low. In this case, TTRXP exceeds TTRXS and logic L60TRF would become a "1" indicating that the temperature reference had failed. Constants TTKn_C and TTKn_S define the corner point and slope of the characteristic, and the resultant value is subtracted from the isothermal limit. As the ambient temperature decreases and CPD increases, TTRXP will decrease causing the final called for temperature reference, TTRX, to lower. The FSR characteristic is calculated in the same manner.

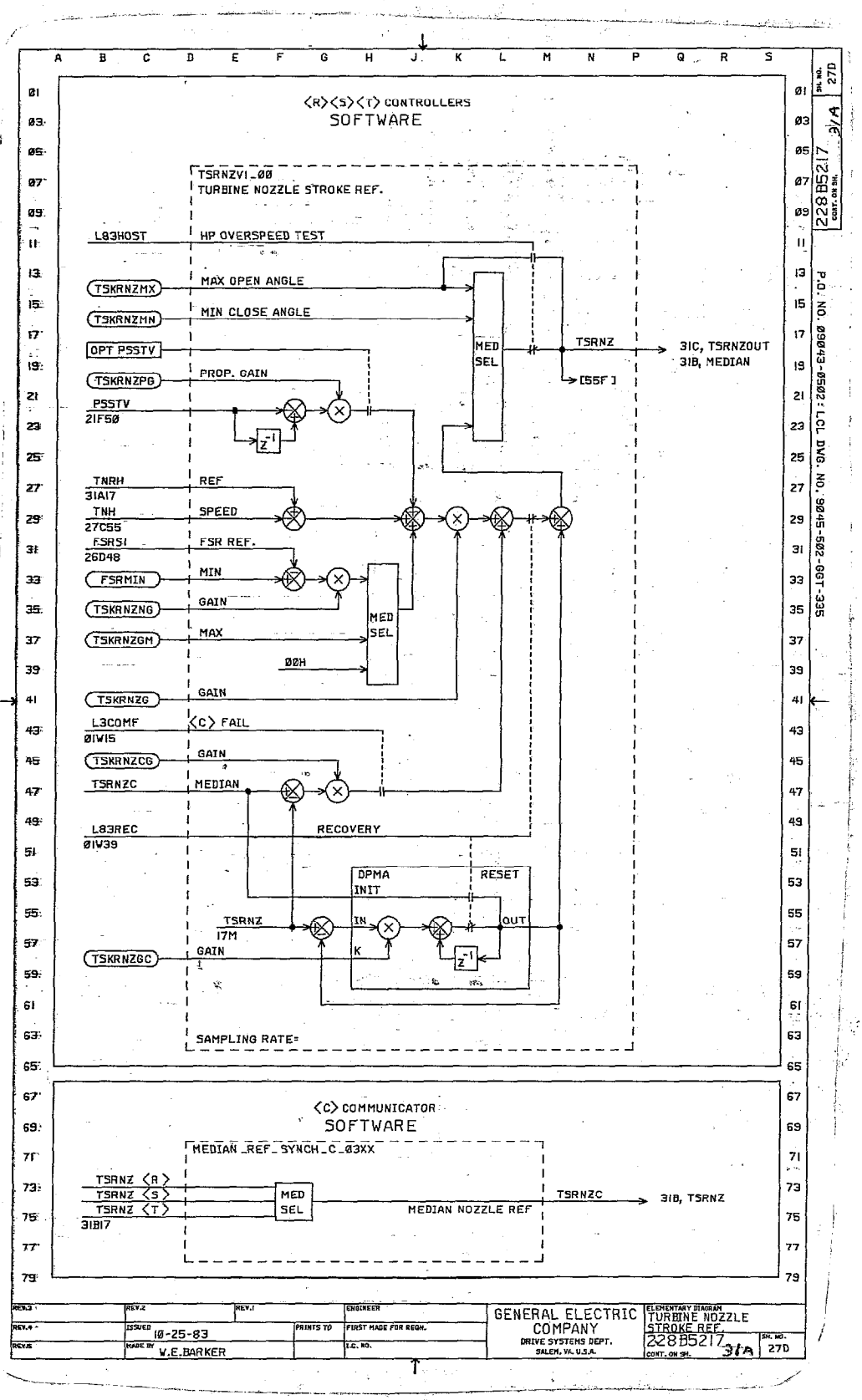


Figure VII-E-3: NOZZLE STROKE REFERENCE

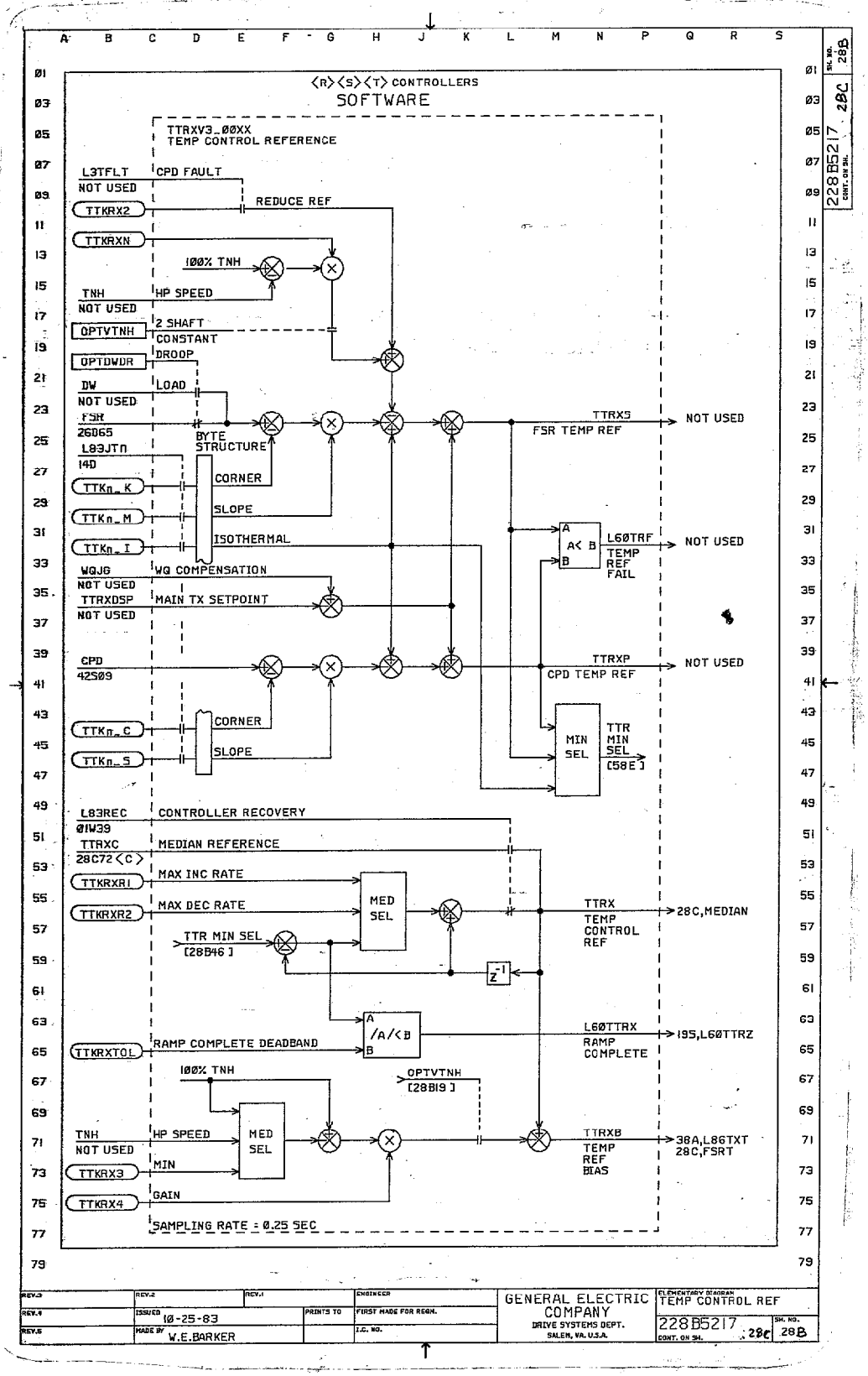


Figure VII-F-1: TEMPERATURE CONTROL REFERENCE

F. TEMPERATURE CONTROL (continued)

2. TEMPERATURE CONTROL REFERENCE (continued)

The result of the minimum select calculation is TTR which is compared with the final temperature control reference. The differential is added to the value of TTRX recorded during the last sample of the algorithm. TTRX continues to increase or decrease until there is no error between TTR and TTRX. At this time, the differential is less than the allowable deadband specified by TTKRXTOL and L60TTRX becomes a logic "1", indicating that the ramp is complete. Constants TTKRXR1 and TTKRXR2 restrict the maximum slope of the ramp in the positive and negative directions respectively. During a controller recovery, L83REC presets TTRX to the median value, TTRXC, calculated in [C] and shown in Figure VII-F-4.

Two (2) shaft gas turbines require an additional calculation to bias TTRX as a function of the high pressure set speed, TNH, to determine a new temperature reference, TTRXB. If the non-adjustable control constant, OPTVTNH, is a "1", then the bias is subtracted from TTRX, and if OPTVTNH is a "0", then TTRX = TTRXB. Constant setable droop applications require a second non-adjustable control constant, OPTDWDR, to switch from the fuel stroke reference, FSR, feedback to the watts transducer feedback, DW, as a back-up parameter to the compressor discharge pressure. Logic L3TFLT is used to reduce the temperature reference if a temperature system fault is detected.

3. TEMPERATURE FEEDBACK

The exhaust thermocouple inputs are divided between controllers [R], [S] and [T]. A description of the processing of thermocouple inputs is given in Section VI.D.

Three algorithms are used to calculate the temperature feedback. They are: (a) TTXD1, Exhaust Temperature Vector; (b) TTXC, System Exhaust Temperature Average, shown in Figure VII-F-2; and (c) TTXM, Exhaust Temperature Median Feedback shown in Figure VII-F-3. Since the thermocouple inputs to [R], [S] and [T] are divided between the controllers, algorithm TTXD1 combines the inputs in [C], and renames each signal to arrange the thermocouple signal names in numerical order according to their physical location on the gas turbine and the actual thermocouple device name. This provides the operator with meaningful nomenclature for the thermocouple displays. Constants TTKXM2R, S and T, define the number of thermocouples monitored by each controller, and constants TTKXDRn, Sn and Tn define the mechanical orientation of each thermocouple. Logics L3COMF0, 1 and 2, preset a group of inputs to "0" if a controller's link or outputs are disabled. The renamed outputs are called TTXD1_n. These outputs are then sorted from the highest value, i.e., suffix "0", to the lowest value and assigned the name, TTXD2_n. JXD_n is an array of the thermocouple device numbers sorted from the hottest (0) to the coldest (n-1).

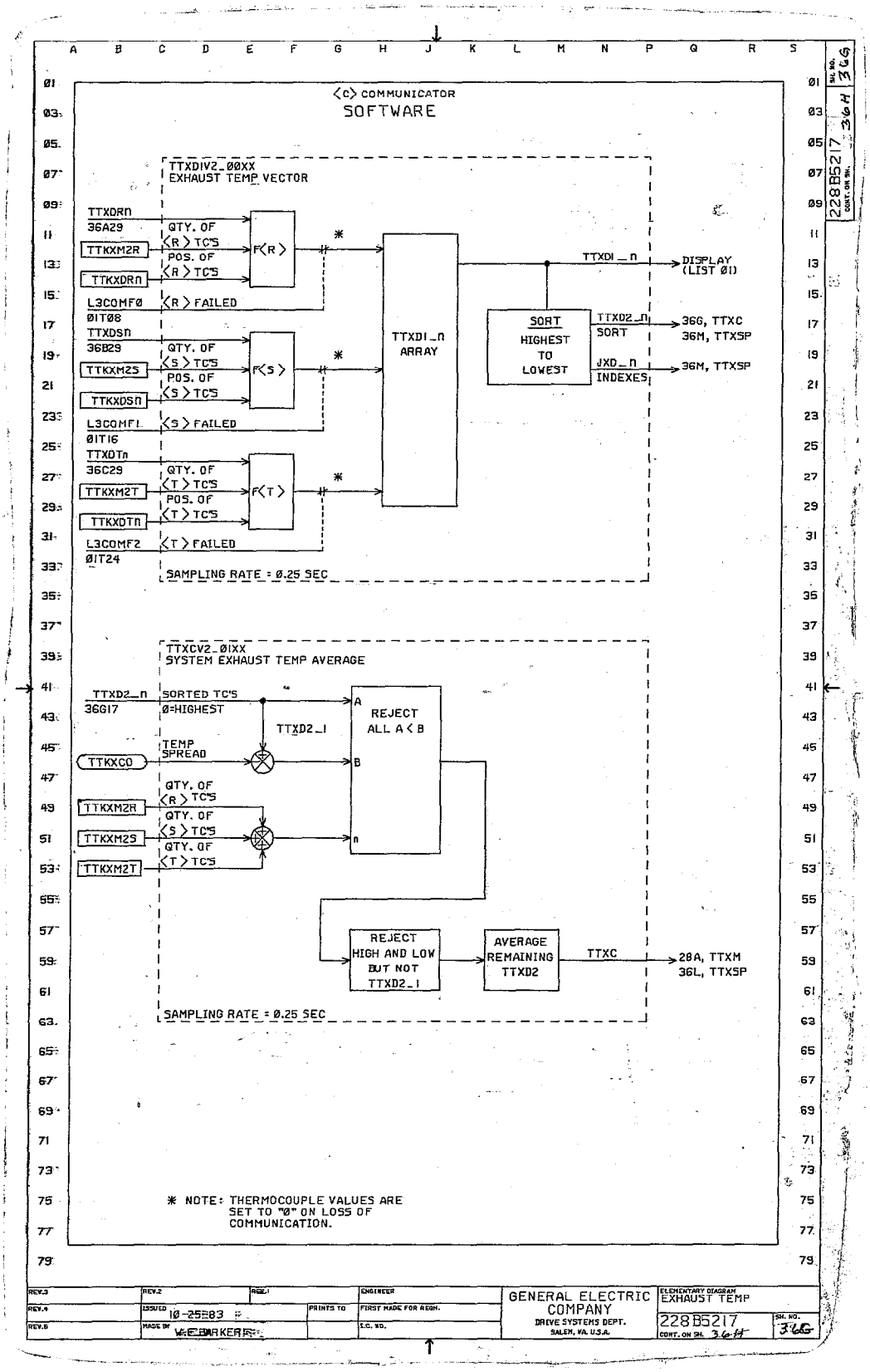


Figure VII-F-2: EXHAUST TEMPERATURE SORT AND AVERAGE

F. TEMPERATURE CONTROL (continued)

3. TEMPERATURE FEEDBACK (continued)

Algorithm TTXC reads the sorted thermocouple values and calculates an average temperature. Constant TTKXCO specifies the allowable temperature spread starting with the second highest reading, i.e., TTXD2_1, and constants, TTKXM2R, S and T, are added to determine the total number of thermocouple inputs. The highest input, i.e., TTXD2_0, and lowest input are automatically rejected from the array. Then the average of the remaining inputs is calculated as TTXC. If any controller fails, its thermocouple readings in the sorted temperature array are set to "0" by algorithm TTXD1, which automatically places them outside the allowable temperature spread defined by TTKXCO, and they are not included in the average calculation.

Algorithm TTXM is shown in Figure VII-F-3. It calculates the median temperature for "each" controller, TTXM1, compares it with the average temperature, TTXC, calculated in Communicator [C], and corrects its median calculation at a rate determined by constant TTKXM1. The comparison is stopped if the [C] link or outputs are disabled making the algorithm's output, TTXM, a function of the current local median, TTXM1, and the error calculated during the last sample before [C] failed.

Thermocouple inputs from each controller, TTXD( ), are sorted from the highest reading to the lowest. Median value TTXM1 is calculated from the sorted readings as a function of the number of thermocouple inputs specified for that controller by constant TTKXM2. This "local" median temperature is then compared with the average temperature of all the thermocouples, TTXC. The differential is used to correct TTXM provided that the differential is within the limits defined by TTKXM3. A DPMA subroutine is used to ramp the error correction at a rate determined by TTKXM1 assuming that Communicator [C] is healthy, i.e., L3COMF = "0". The resultant corrected median temperature, TTXM, is the sum of the local uncorrected median, TTXM1, and the error output from the DPMA subroutine. If TTXM drops below the minimum allowable temperature, TTKXM4, after the turbine has reached acceleration speed, i.e., L86TFINH = "0", then a latched trip signal, L86TFB, will be generated and an alarm message will appear. I.E., EXHAUST THERMOCOUPLES OPEN TRIP. Note that two (2) controllers must set L86TFB to a "1" before a turbine trip will occur.

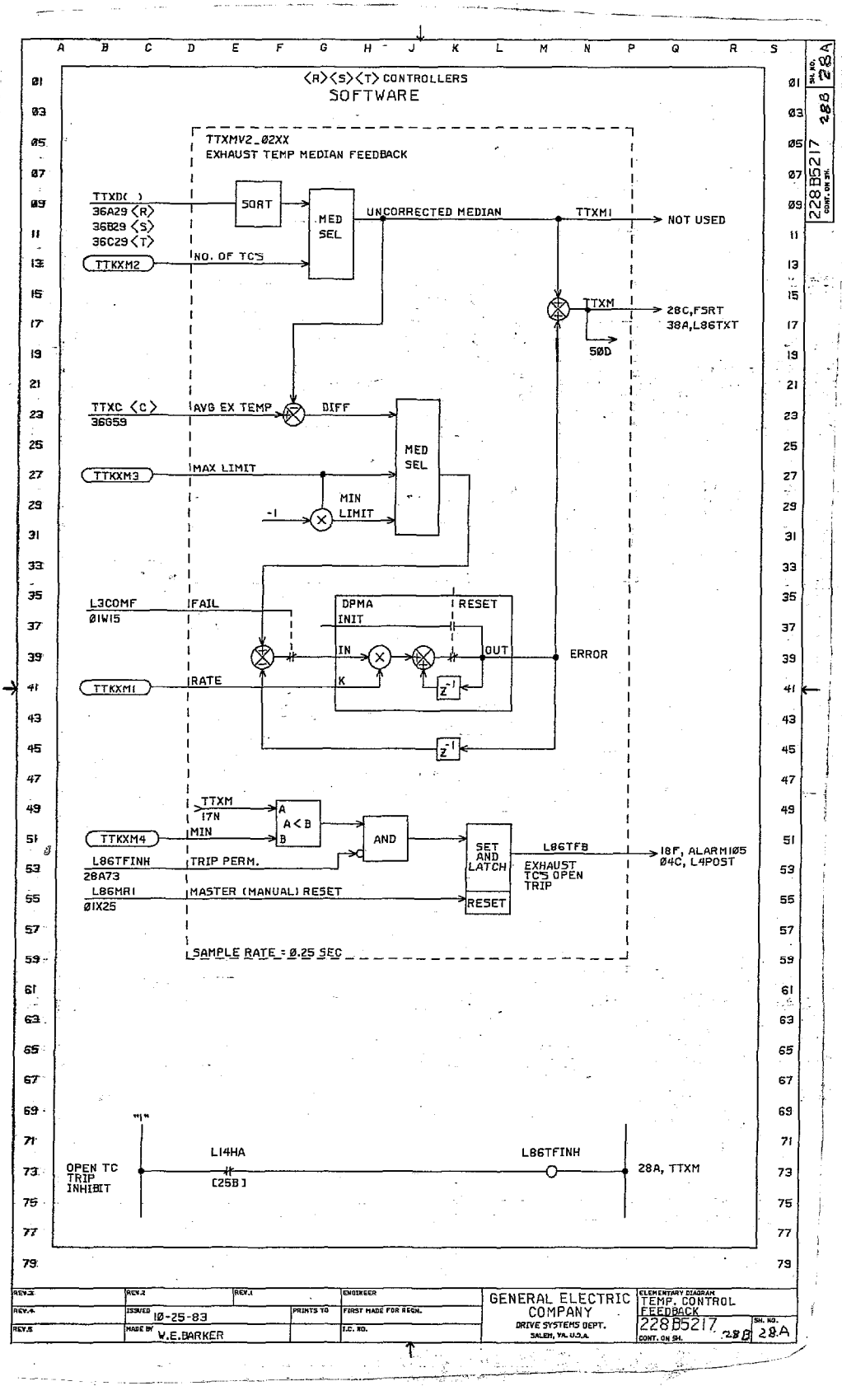


Figure VII-F-3: EXHAUST TEMPERATURE MEDIAN FEEDBACK

F. TEMPERATURE CONTROL (continued)

4. TEMPERATURE CONTROL FUEL STROKE REFERENCE

Algorithm FSRT calculates the temperature control loop fuel stroke reference in Controllers [R], [S] and [T], and is shown in Figure VII-F-4. The temperature differential is calculated between the called for temperature reference, TTRXB, and the median feedback, TTXM. Then logic L30F_TMP selects a gain defined by constants FSKTG1 or FSKTG2, depending on whether the governor is on temperature control. A conditioned FSR input is added to this differential to create the FSR temperature reference, FSRT, with the maximum and minimum limits defined by FSRMAX and FSKTMIN, respectively. If the called for temperature reference, TTRXB, is higher than the median feedback, TTXM, then FSRT will be higher than FSR, and temperature control will "not" be the control loop which is governing fuel. When the exhaust temperature reaches the called for reference, then the error is zero (0) and FSRT = FSR. The rate of change of the FSR bias is restricted by the DPMA subroutine and the FSKTGC constant.

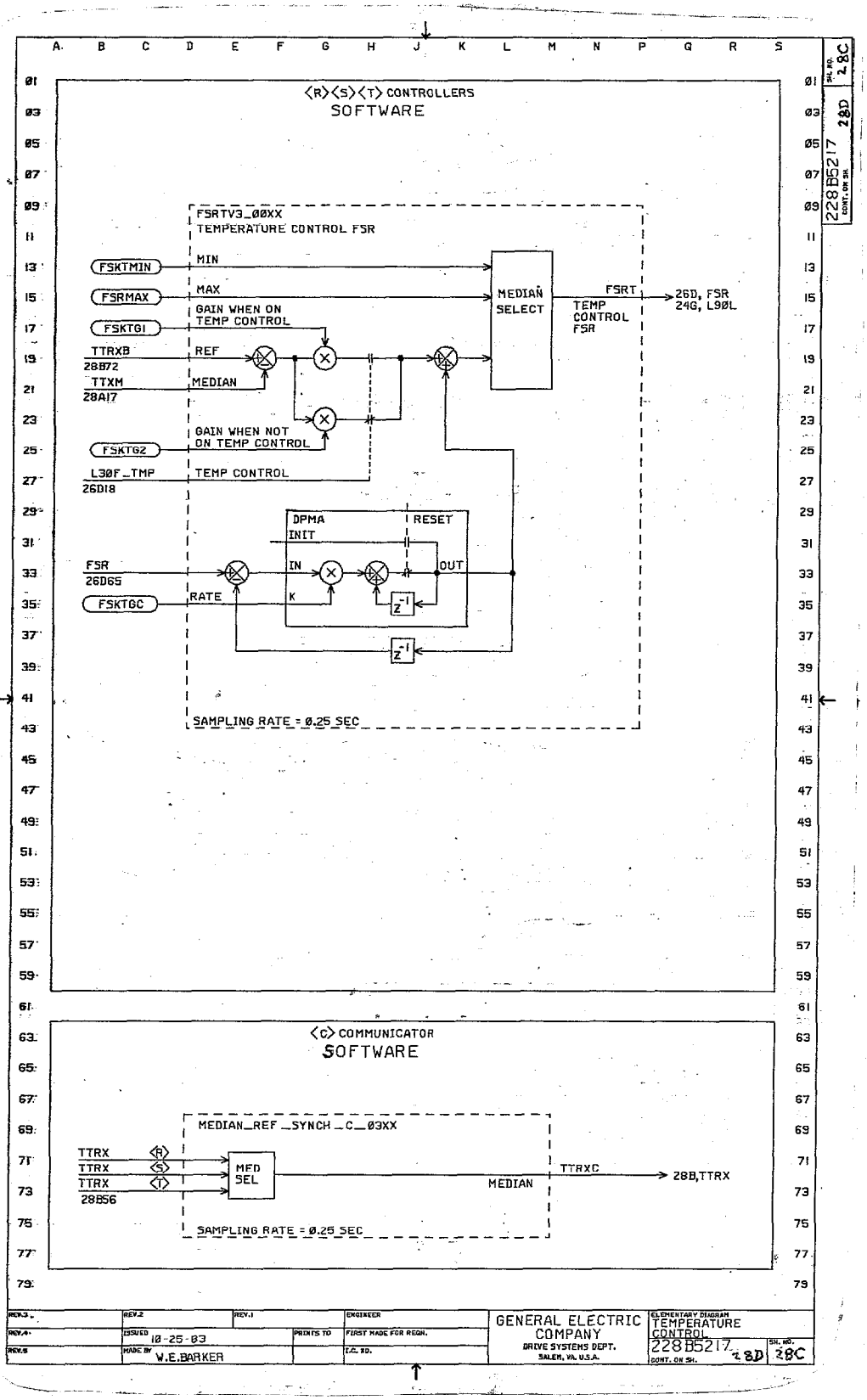


Figure VII-F-4: TEMPERATURE CONTROL FUEL STROKE REFERENCE

G. SYNCHRONIZING

1. APPLICATION

During a start-up sequence, turbine speed reference TNR is raised until turbine speed TNH reaches 100.3%, i.e., typical, of the station line frequency, SFL. The synchronizing fuel stroke reference, FSRSYN, now overrides the speed control fuel stroke reference, FSRN, and governs the turbine speed based on the called for full speed no-load setpoint and the error between TNH and SFL + 0.3%. A command to close the breaker is given when the generator to line phase differential and the slip are at the pre-determined synchronization point. When the breaker closes, FSRSYN ramps up until the governor's fuel stroke reference is overridden by speed control.

2. SYNCHRONIZING INTERFACE MODULE [SYNCH]

Figure VII-G-1 shows the Synchronizing Interface Module and its interface to the generator control panel and the computers. The module receives two (2) 120VRMS 60/50HZ inputs from the generator and line potential transformers. These inputs are isolation transformers inside the module which step-down the voltage to 8VRMS prior to conditioning the inputs with zero volt crossover detector circuits and rectifiers. Every time the voltage crosses the zero volt threshold in a positive direction, a TTL square wave output is generated in phase with the AC inputs. These outputs, i.e., GR, GS, GT, LR, LS and LT, and the outputs of the voltage level detectors, i.e., DVR, DVS, DVT, SVLR, SVLS and SVLT, are fed to the HPRB and HAIC cards in Controllers, [R], [S] and [T].

When the controllers calculate the appropriate time to close the breaker, the HPRB cards send logic outputs to the synchronizing interface module to energize the breaker closure relay, "25". A 2/3 vote is performed by the relay driver circuit which generates outputs, POSK1 and NEGK1, to drive the relay coil. Three (3) diagnostic circuits monitor the status of the relay driver circuit, and the HAFA card in Communicator [C], monitors each signal for a permissive to synchronize. 1DIAG1 is true when the relay contacts are closed. 0DIAG2 is true when NEGK1 is greater than +6.2V. 0GIAG3 is true when POSK1 is less than +20V.

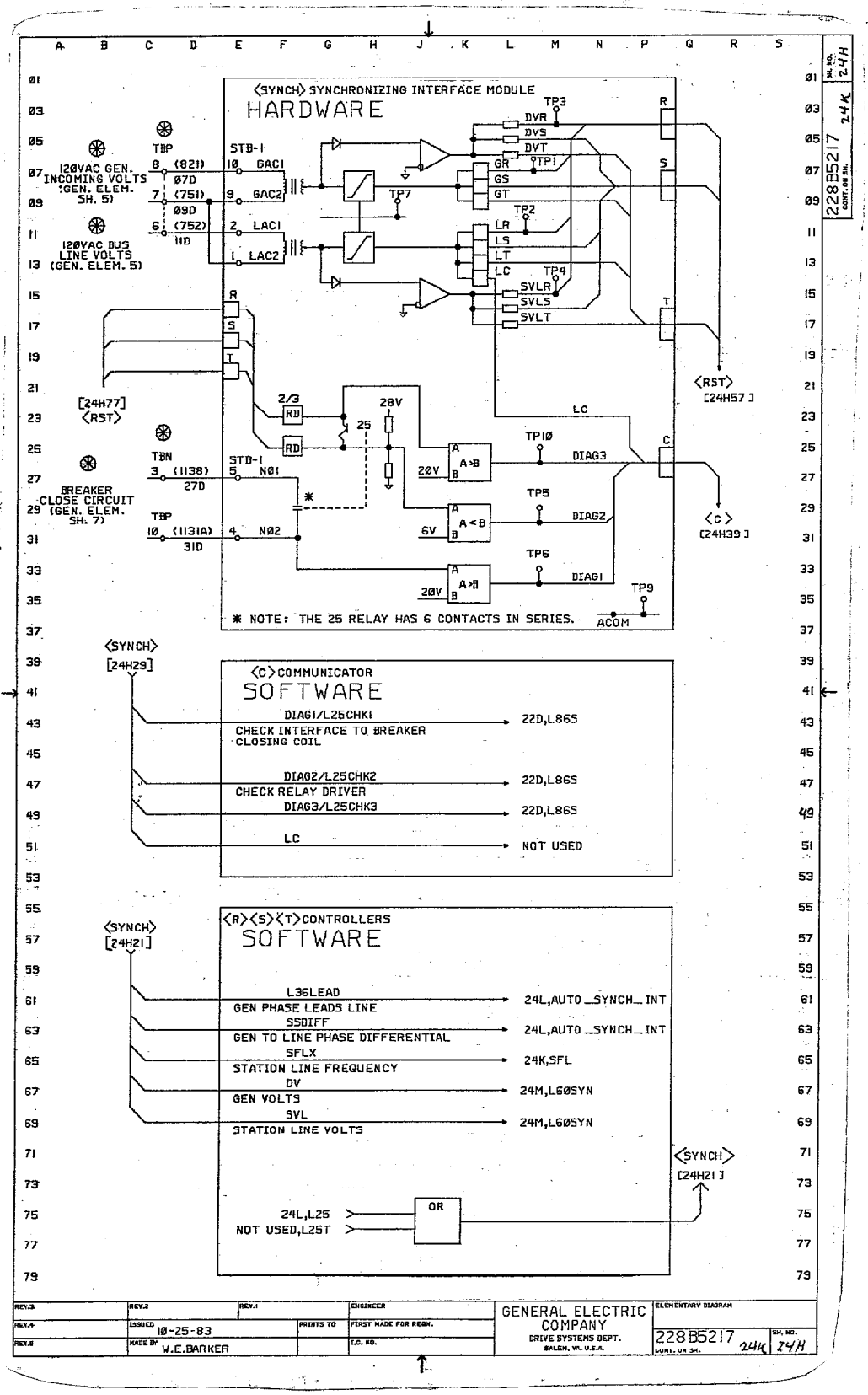


Figure VII-G-1: SYNCHRONIZING INTERFACE MODULE

G. SYNCHRONIZING (continued)

3. SYNCHRONIZING FUEL STROKE REFERENCE

The called for turbine speed reference, TNR, is automatically raised until FSRSYN overrides FSRN at 100.3% of the station line frequency, SFL. Algorithm L60TNR calculates this transition point and is shown in Figure VII-G-2. When turbine speed reference TNR exceeds station line frequency SFL plus the 0.3% offset stored in constant LK60TNR1, logic L60TNR, stops the raise command to TNR. L27BZ is used to disable this calculation if a line deadbus condition is detected and the raise command is stopped at 100% of TNR. L14HS enables the algorithm output when the turbine reaches its operating speed.

Algorithm FSRSYN is shown in Figure VII-G-2. It is enabled by the auto-synchronize permissive, L83AS, which is a function of the generator control panel synchronize permissive, L52Z, the voltage differential between the generator and the station line buses, L60SYN2, and the synchronizing interface module diagnostics, L86S. The full speed no-load FSR constant, FSKRND1, establishes the reference which is added to the difference between turbine speed TNH and station line frequency SFL, plus the 0.3% offset multiplied by gain, FSKRSYN2. Two (2) maximum select calculations are used for the lower limit of FSRSYN based on the minimum speed setpoint, FSKRSYN4, and minimum fuel stroke reference FSRMIN. The final summer calculation is used after synchronization or any time L83AS is not satisfied to ramp FSRSYN up at a rate determined by constant FSKRSYN3, until the speed control fuel stroke reference, FSRN, overrides FSRSYN. Maximum and minimum limits are applied to FSRSYN by the median select calculation.

4. SYNCHRONIZATION CALCULATION

Algorithm AUTO_SYNCH_INT monitors the generator to line phase differential, SSDIFF, and commands the breaker to close at the proper time. Reference Figure VII-G-3. SSDIFF is read as the clock count between two (2) voltage crossovers, and used in conjunction with the breaker closing time constant, SSKDIFF1, i.e., in cycles, to determine the predicted phase and slip at synchronization, i.e., SSDIFF1 and SFDIFF1 respectively. The equations are: SSDIFF1 = SFDIFF + D (SFDIFF) * SSKDIFF1 + D (SFDIFF) * (SSKDIFF **2)/2 SFDIFF1 = SFDIFF + D (SFDIFF) * SSKDIFF1

If the auto-synchronize permissive, L83AS, is satisfied and the line phase leads the generator, a time delay is initiated to give the synchronization calculation time to settle, i.e., [ABS (SSDIFF1) + ABS (SFDIFF1 - 38) > 58]. When the predicted phase and slip are on target, a command is given to close the breaker, L25 which is sealed for a pre-determined time. An algorithm sampling rate of 60HZ or 50HZ must be selected based on the station line frequency.

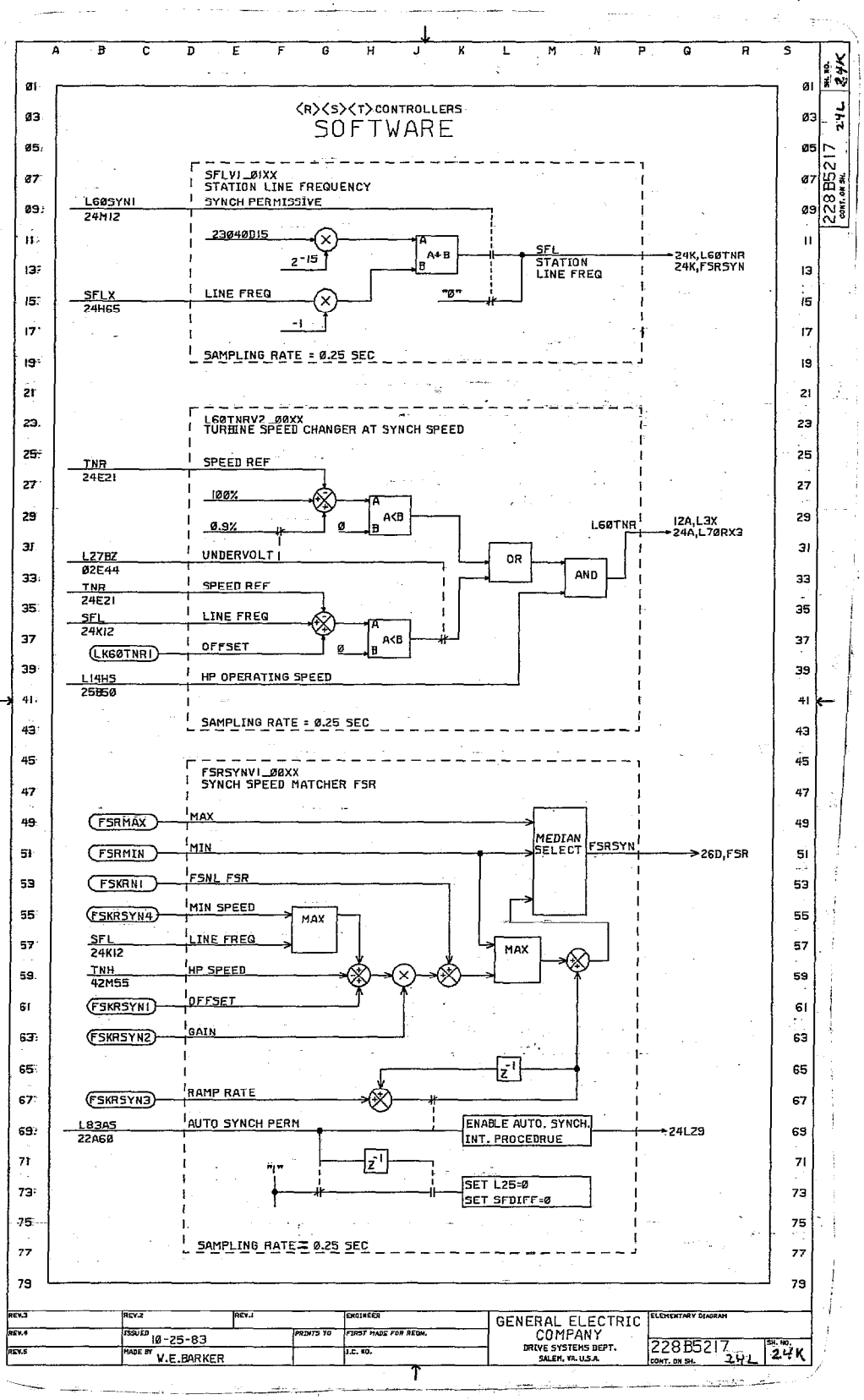


Figure VII-G-2: SYNCHRONIZING FUEL STROKE REFERENCE

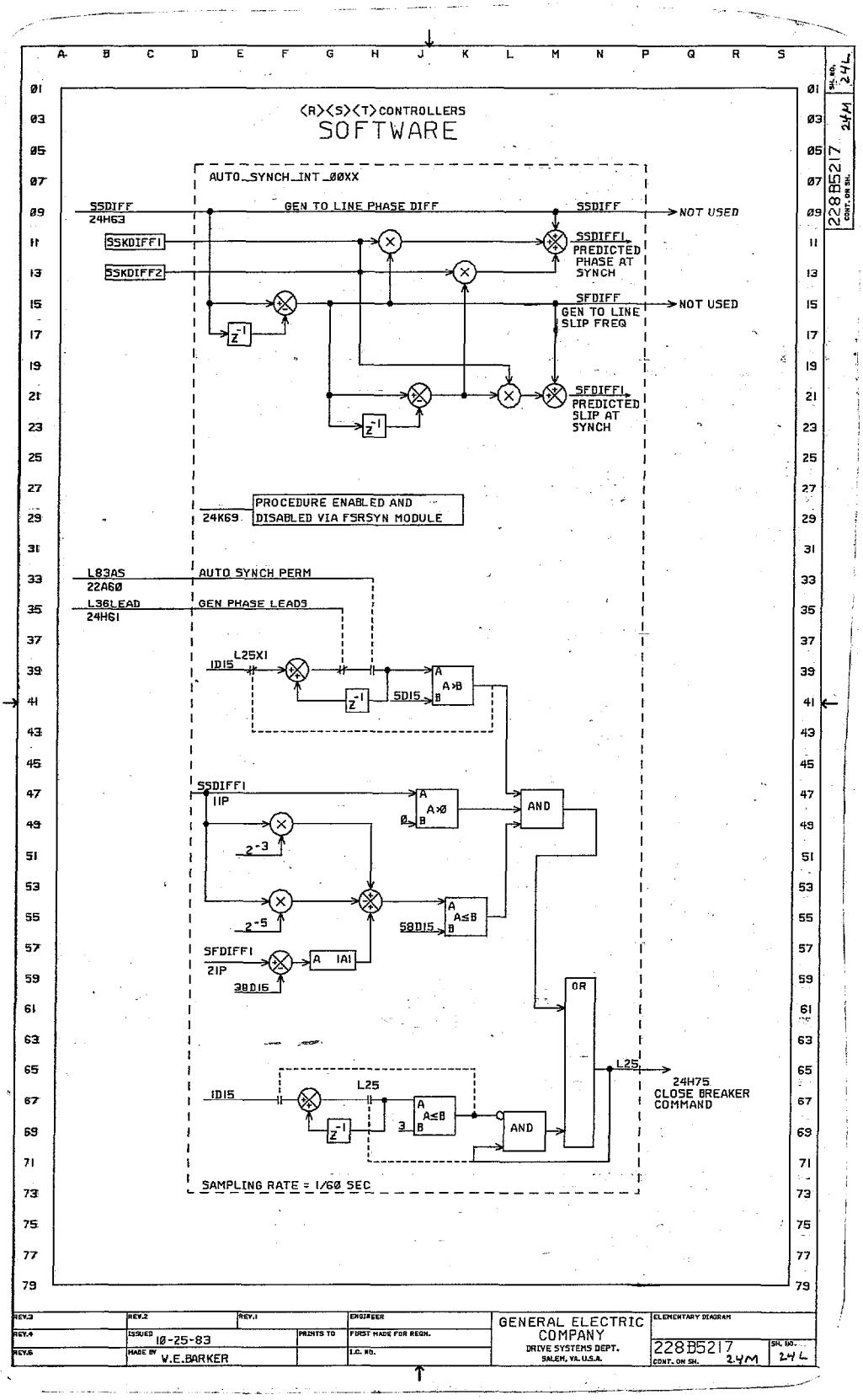


Figure VII-G-3: SYNCHRONIZATION CALCULATION

H. SERVO VALVE DRIVER SYSTEM

The servo valve driver system is the interface between the turbine panel control system and the hydraulic actuators that position the valves and other devices. Figure VII-H-1 shows a typical system such as:

Universal Liquid Fuel System	Modulated Inlet Guide Vane Position Control
Gas Fuel Control System	Second-stage Nozzle Position Control
Gas Fuel Pressure Control	Steam Turbine Control Valve Position Control

HSAA cards located in Controllers [R], [S] and [T], receive a digital reference signal which is converted to an analog voltage and compared to the position feedback from the LVDT, i.e., Linear Variable Differential Transformer. Usually, two (2) LVDT's measure the linear position of the control valve stem and send a 0.7VRMS to 3.5VRMS signal back to the HSAA card corresponding to the minimum and maximum stroke positions. Each LVDT feeds one (1) controller and both feed the third controller which uses a diode high-select circuit to pick the highest feedback signal. If one of the LVDT's fails low, two (2) of the controllers will continue to monitor the other LVDT. The two (2) controllers which are not performing the diode high-select function provide 7.0V, 3000HZ excitation to the LVDT's. Actually, one (1) oscillator is set at 2800HZ and the other is set at 3200HZ to eliminate the effect of the beat frequency.

The error between the reference and the position feedback is amplified by a DC gain selected with berg jumpers mounted on the card and is then fed to the servo driver amplifier. This amplifier is an integrator with an inner current feedback loop from the torque motor coil. The addition of the inner loop quickens the response and provides a more stable position loop. In addition, a separate integrator is available on the card for other types of feedback. For example, some turbines use the integrator for a fuel flow feedback in addition to, or in place of, an LVDT feedback. The integrator is not shown in figure VII-H-1 since it is not always used, and its implementation varies with the application. More detailed application information is given in the section of this manual pertaining to the subject function. The called for reference, position feedback and current feedback are multiplexed and monitored by control algorithms to determine the health of the system. Each algorithm compares the values of these parameters with values defined by control constants. These comparisons are monitored at particular points in the turbine sequencing and alarmed if their results are not correct.

If the control calls for the valve to open, the integrator output ramps up causing more current to flow in the torque motor coil. The current induces a magnetic field which acts in conjunction with a permanent magnet to move the armature and open the valve. Eventually, the feedback equals the reference causing the valve to maintain its called for position. Three (3) separate coils in the torque motor are driven by Controllers [R], [S] and [T], to extend the voting concept to the servo valve and avoid performing a median selection in the electronics which would be subject to a single point failure. A relay in the HSAA card shorts the integrator output if the CPU stalls. This keeps the stalled controller from contributing to the servo valve response but does not drive the stalled controller's coil in any particular direction. A mechanical bias is used to drive the valve closed if power is removed from all of the controllers.

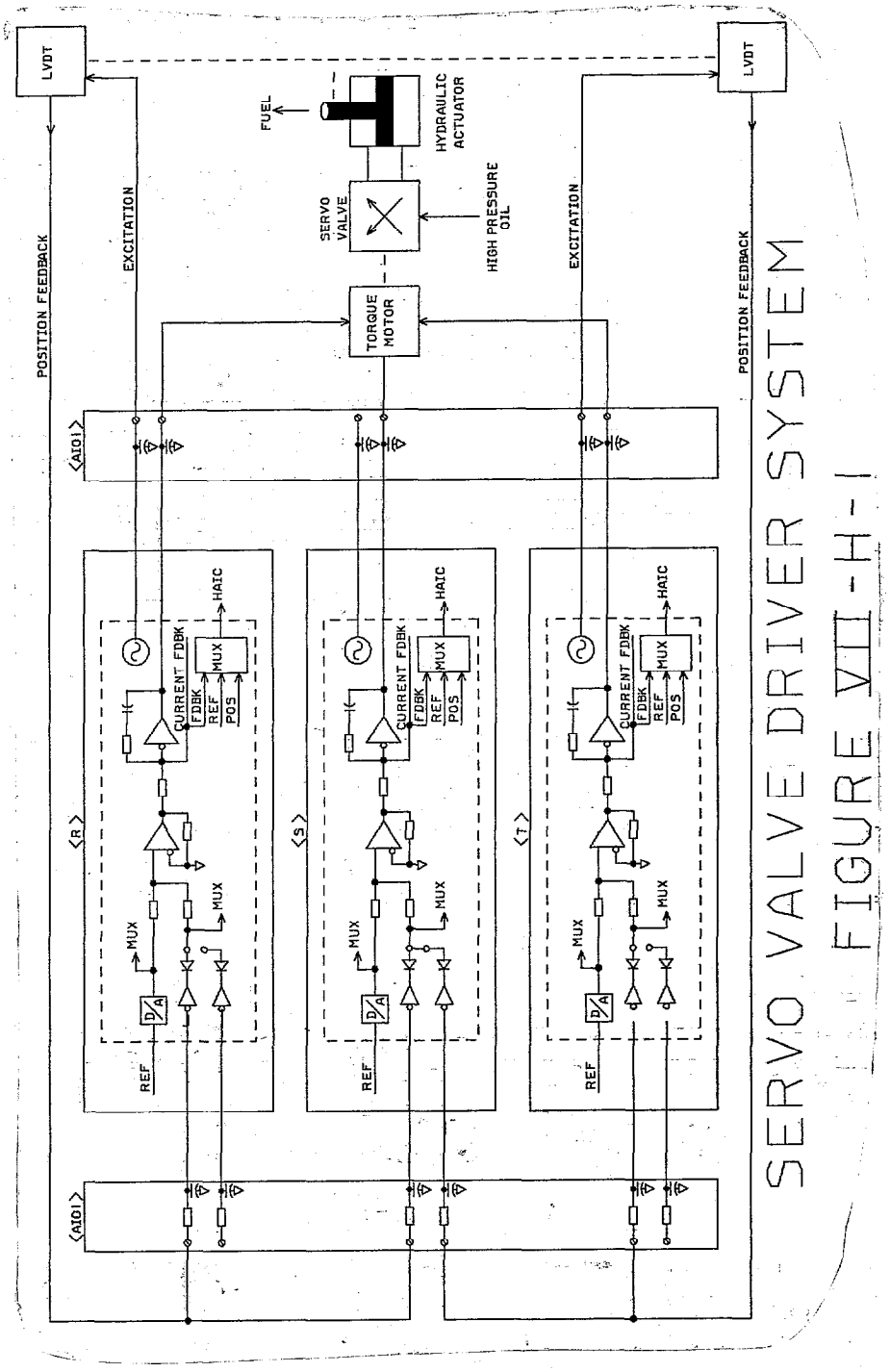


Figure VII-H-1: SERVO VALVE DRIVER SYSTEM

I. FUEL SYSTEM

1. APPLICATION

The turbine fuel system usually consists of a universal liquid fuel system which is designed for all types of liquid fuels and/or a gas fuel system. Figure VII-A-2 shows a typical block diagram of an MS6001 dual fuel system. In liquid fuel control systems, the fuel stroke reference, FSR, establishes the called for stroke of the bypass valve. The resultant fuel flow is proportional to the speed which can be represented by the equation, Fuel Flow = TNH X FSR. This relationship does not apply to a gas control system consisting of a gas control valve only because the fuel flow is a function of pressure rather than speed. This can be represented by the equation, Fuel Flow = P X FSR. Therefore, a stop/speed ratio valve is added which is electrically programmed by speed. Pressure is now a function of speed, and the liquid fuel control system and the gas fuel control System have the same characteristic.

2. LIQUID FUEL SYSTEM

Algorithm FQROUT calculates the liquid fuel flow command in Controllers [R], [S] and [T]. It is shown in Figure VII-I-1. The fuel stroke reference which feeds the algorithm is FSR1 instead of FSR, because FSR1 is the called for liquid fuel component of the master FSR. If the machine is a dual fuel application the FSR = FSR1 (liquid) + FSR2 (gas) as calculated by the fuel splitter algorithm. If it is a liquid fuel only application, then FSR = FSR1. FSR1 is multiplied by the high pressure set speed, TNH, and a gain constant, FQKOUTNG, prior to adding offset FQKOUTO. A fuel flow feedback is provided by the flow divider pick-up to the HPRB card which converts the pulse rate input to a DC voltage. This signal is wired to the HSAA servo amplifier card for control loop feedback and monitoring [R], [S] and [T] software. A correction is made for the error between the median fuel flow, FQ2C, calculated in [C], and the fuel flow calculated in the individual controllers, FQ2, at a rate determined by FQKOUTCC, provided that [C] is healthy, i.e., L3COMF. Logic L4 and L20FLX clamp the fuel flow command slightly negative until the master protective "4" relays are picked up and the liquid fuel bypass valve is commanded to close. In order to gradually increase the servo reference from the clamped negative value to the zero setpoint fuel flow FQ2 is used as the reference from the time the master protective "4" relays pick-up until stop valve 20FL opens. Constants FQKOUTO and FQKOUTG provide offset and gain conditioning for FQ2. GSADJ is used to calibrate the servo for the minimum and maximum stroke as calculated by the ANALOG CONTROL DISPLAY, List 15, as described in Section V.B.6. JADJ selects algorithm FQROUT to be calibrated when its corresponding ANALOG CONTROL DISPLAY - LIQUID FUEL CONTROL appears on the CRT which sets JADJ to "11" and presets the called for stroke to the calibrated limits.

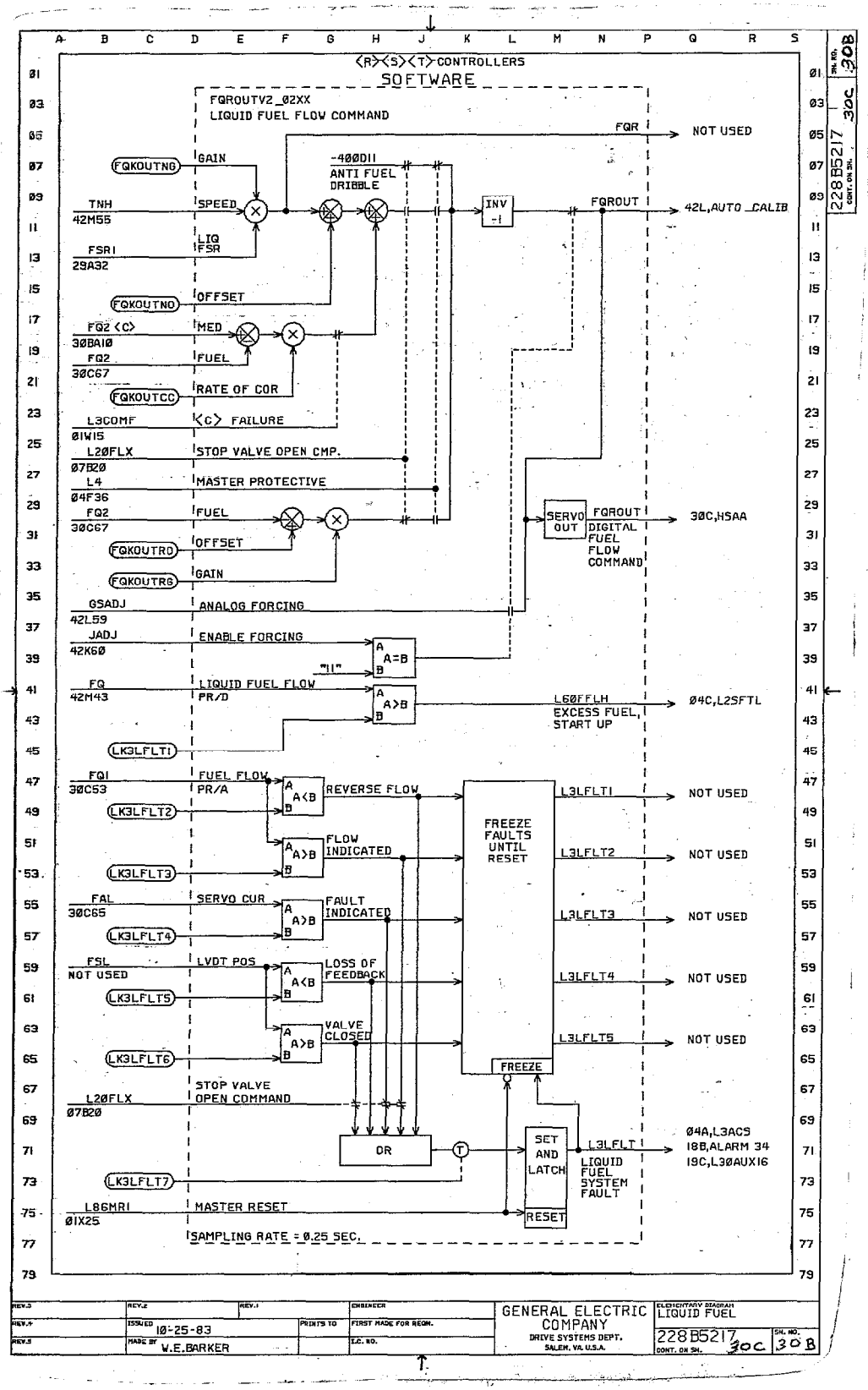


Figure VII-I-1: LIQUID FUEL CONTROL REFERENCE

I. FUEL SYSTEM (continued)

2. LIQUID FUEL SYSTEM (continued)

The algorithm calculates the called for reference which is converted to an analog voltage on the HSAA card for regulation with the feedback as described in section VII.H. This feedback may consist of position and fuel flow feedbacks, i.e., MS7001, or just a fuel flow feedback, i.e., MS6001, or just a position feedback on some turbines. An MS6001 fuel system diverts some of the discharge from the fuel pump through the bypass valve, and the remaining fuel feeds the flow divider. The flow divider is used to equally divide the fuel flow between the fuel nozzles. Three (3) magnetic pickups are mounted over 60 tooth wheels on the flow divider and are used to provide the pulse rate fuel flow feedback to Controllers [R], [S] and [T]. Then an HPRB card converts the pulse rate input to an analog voltage which is wired on the controller back-plane to the HSAA servo amplifier card. This manipulation results in four (4) separate signals:

FQ - is the result of the pulse rate to digital conversion on the HPRB card. It is scaled in algorithm PR_MEAS and displayed in GPM.

FQ1 - is the result of the pulse rate to analog conversion on the HPRB card. The timing of the PR/A chip is controlled by algorithm PR_MEAS.

FQ2 - is the output of the integrator amplifier on the HSAA card. This signal is multiplexed on the HSAA card and then wired to the HAIC card for the analog to digital conversion.

FQ_PR - is the frequency of the pulse rate inputs as calculated by algorithm PR_MEAS and displayed in Hertz.

Algorithm FQROUT monitors the various feedback signals and compares them against pre-assigned setpoints to determine the health of the fuel system. If FQ exceeds setpoint LK3LFLT1, then logic L60FFLH will indicate that the liquid fuel flow level is high and trip the machine. Five (5) additional comparator blocks provide fault detection logic which is combined, i.e., OR'ed, and time delayed. The resultant output, L3LFLT, generates an alarm, "Liquid Fuel Control Fault", and freezes the individual logic outputs to identify which one caused the alarm. The five (5) outputs are:

L3LFLT1, Reverse fuel flow is detected by the magnetic pickup.
L3LFLT2, Fuel flow is detected when the stop valve is closed.
L3LFLT3, Servo current is detected when the stop valve is closed.
L3LFLT4, Loss of LVDT position feedback
L3LFLT5, Bypass valve is not fully open when the stop valve is closed

Once L3LFLT latches, it must be manually reset from the Operator Interface Module, [OPM], by calling the RESET display.

I. FUEL SYSTEM (continued)

3. GAS FUEL SYSTEM

The gas fuel system consists of the stop/speed ratio valve and the gas control valve which are servo controlled and actuated by single-acting hydraulic cylinders moving against spring loaded plugs. As explained in Section VII.I.1, the gas control valve determines the fuel flow and the stop/speed ratio valve maintains a preprogrammed inter-valve pressure as a function of speed. The algorithms for these servo valves are shown in Figure VII-I-2.

Algorithm FSROUT calls for the gas fuel component, FSR2, of the overall fuel stroke reference, FSR, and conditions the reference with fuel system gain FSKOUTG, offset FSKOUTO, and mechanical bias offset FSKOUTB. Logic inputs L4 and L3GCV enable the fuel stroke reference to the gas control valve when the master protective "4" relays are energized and the gas fuel stop valve opens. GSADJ is used to calibrate the servo for the minimum and maximum stroke as calculated by the ANALOG CONTROL DISPLAY, List 15, as described in Section V.B.6. JADJ selects algorithm FSROUT to be calibrated when its analog control display, i.e., GAS VALVE CONTROL, appears on the CRT which sets JADJ to "12" and presets the called for stroke to the calibrated limits.

The gas ratio valve algorithm, FPRG, controls the inter-valve pressure as a function of speed TNH, with gain FPKGNG, and offset FPKGNO adjustment. Gas fuel pressure transducer 96FG is monitored by the NVCB card in each controller. The signal is wired to the HSAA servo amplifier card's integrator to bias the control loop and provide the FSRSR software feedback. Each controller calculates the error between its local FSRSR and the median FSRSRC and corrects the adjusted speed reference at a rate determined by FPKGSCC, provided that [C] is healthy L3COMF. A bias of -400D11 is used to keep the servo valve closed when the turbine is stopped. When the master protective "4" relays are picked up, FSRSR increases the reference from the -400D11 trip setting to the level where the speed reference can take over when the stop valve opens. The servo auto-calibrate function is similar to the gas control valve calibration described previously except that the ANALOG CONTROL DISPLAY for the gas ratio valve sets JADJ to "13".

Algorithm L3GFLT, not shown, monitors the reference output from the HSAA D/A converter, the LVDT position feedback, the integrator fuel pressure output, the servo current feedback for the gas control valve and the stop/speed ratio valve systems to detect a fault. The procedure for calculating the fault conditions is identical to the liquid fuel system described in Section VII.I.2.

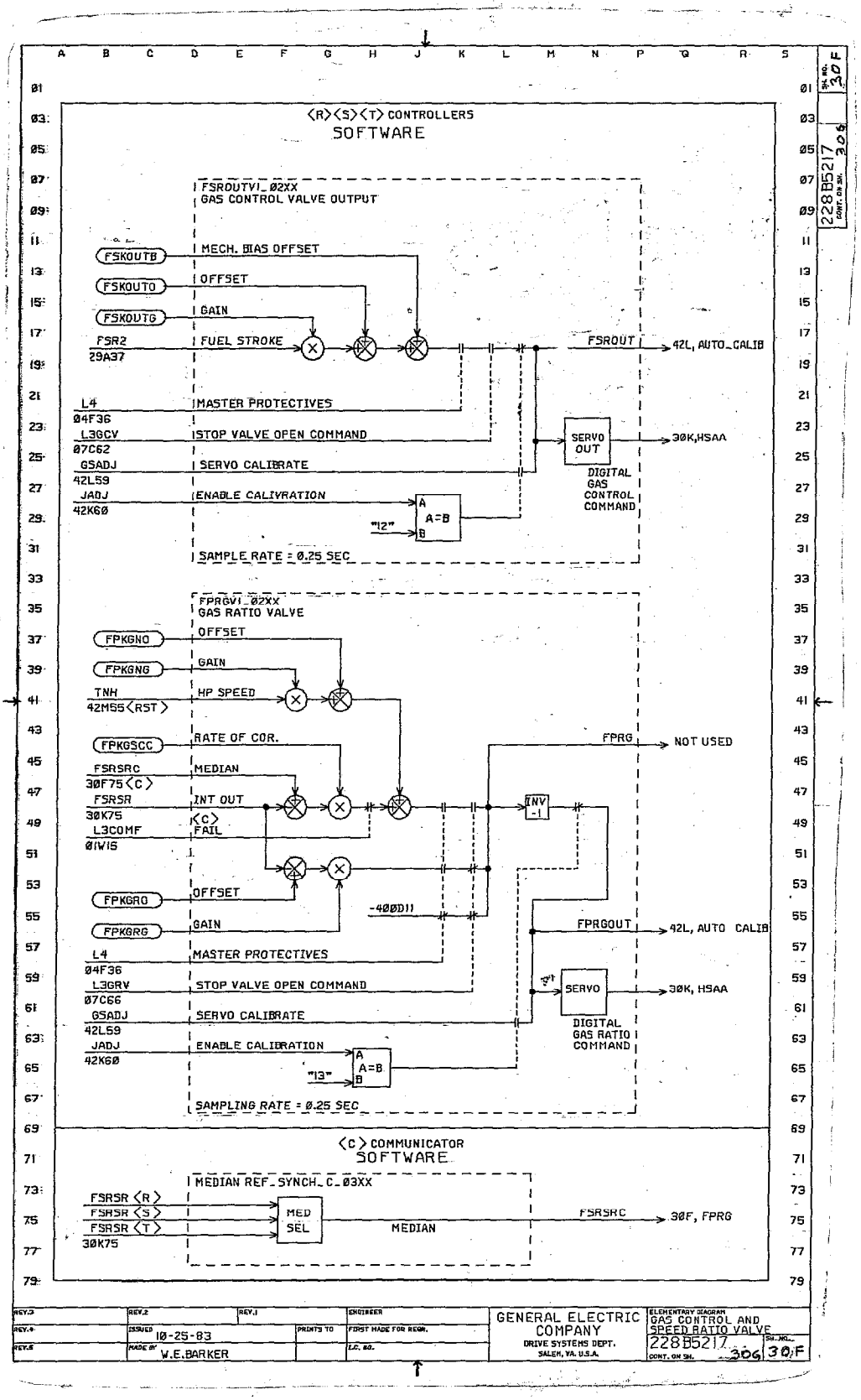


Figure VII-I-2: GAS FUEL CONTROL REFERENCE

I. FUEL SYSTEM (continued)

4. FUEL SPLITTER

Fuel splitter algorithm FSR1 receives the calculated fuel stroke reference, FSR, and divides it into liquid fuel component FSR1, and gas fuel component FSR2, as shown in Figure VII-I-3. The fraction of liquid fuel FX1 is calculated in Controllers [R], [S] and [T], and fed to a median select algorithm in Communicator [C]. The resultant median, FXC, is fed back to [R], [S] and [T] to correct the individual controllers. This occurs when a controller is re-initialized, i.e., recovers, as defined by logic L83REC in rung L83FXSET. It occurs once every minute as defined by logic L83MIN, as long as no simultaneous command is being received to increase liquid fuel L83FL, or increase gas fuel L83FG. A Communicator [C] failure, L3COMF, inhibits the algorithm from presetting to the median value. The called for fuel is determined to be totally liquid, L84TL, when the fraction of liquid fuel, FX1, equals the maximum limit, 7FFFH, i.e., 100%. The called for fuel is determined to be totally gas if the fraction of liquid fuel, FX1, equals the minimum limit, 0000H, i.e., 0%. If the fraction of liquid fuel is between the maximum and minimum limits, then logic L84MIXT tells the control that the governor is operating on a fuel mixture.

L83FZ tells the algorithm that all conditions have been satisfied to change the fuel mixture. L83FL adds the value of constant FXK1 to the fraction of liquid fuel, FX1, calculated during the last sample, and L83FG subtracts the value of FXK1. Changing the fuel mixture can be initiated before the turbine reaches minimum speed and the turbine vent timer times out, or after the turbine reaches operating speed. A time delay in sequencing of approximately 30 seconds, enables the fuel lines to be purged prior to changing the fuel mixture in either direction. If the turbine is required to run on a particular fuel mixture, the fuel transfer can be stopped at any mixture by pressing the MIX membrane switch on Operator Interface Module [OPM], which disables L83FL and L83FG until the operator initiates another command to change the fuel mixture. Other features are available in sequencing such as the ability to sense a loss of gas fuel pressure and automatically transfer to liquid fuel.

Four (4) versions of the fuel splitter algorithm exist:

Version 1 is used for dual fuel applications where the two (2) fuels are liquid and gas. It splits FSR into FSR1 and FSR2.
Version 2 is for liquid fuel only applications. It renames FSR to FSR1 to satisfy the standard liquid fuel algorithms which are fed by the splitter output.
Version 3 is for gas fuel only applications. It renames FSR to FSR2 to satisfy the standard gas fuel algorithms which are fed by the splitter output.
Version 4 is used for dual fuel applications where the two (2) fuels are undefined. It splits FSR into FSR1 and FSR2, but uses generic type I/O names which can be applied to various types of fuel systems.

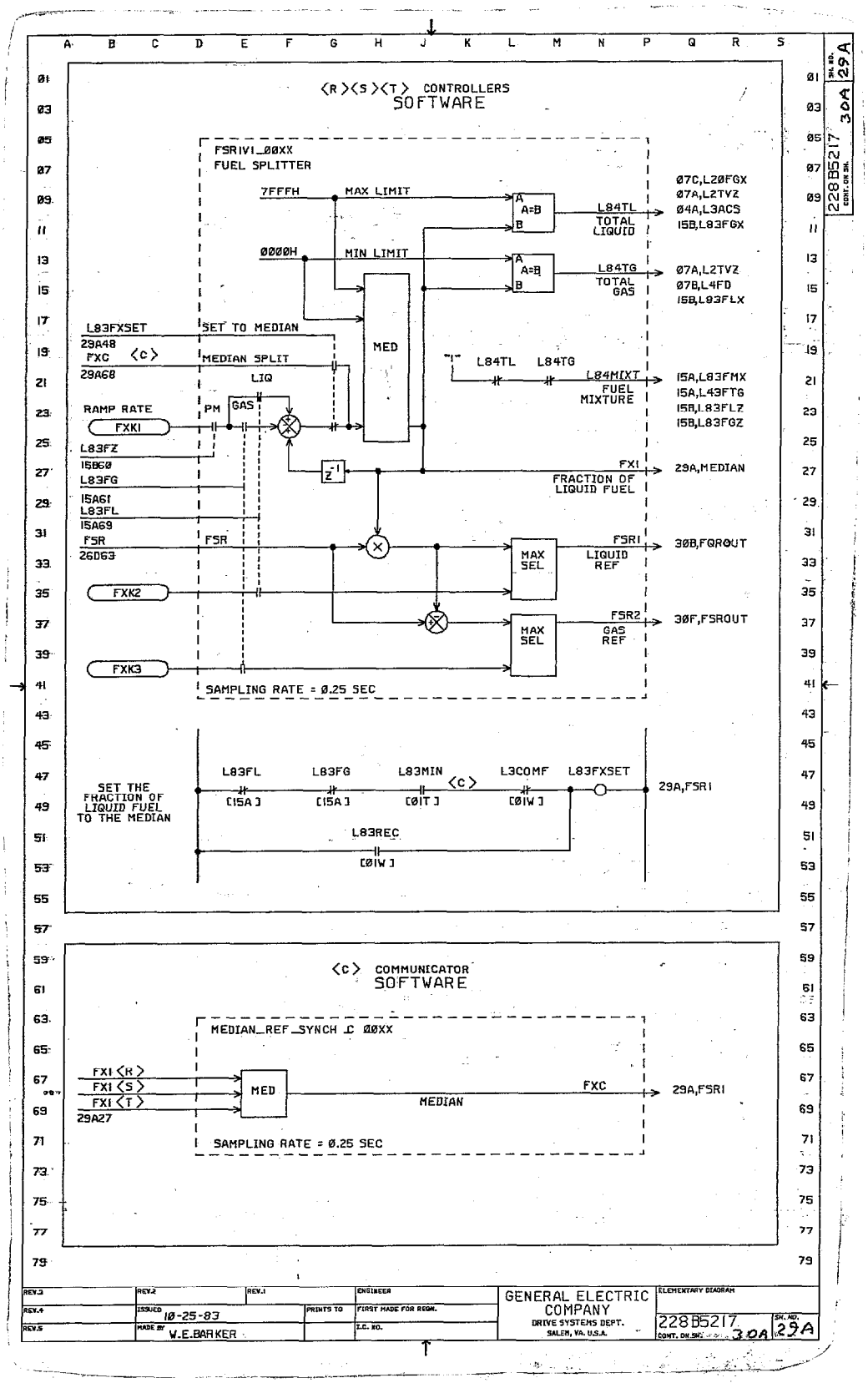


Figure VII-I-3: FUEL SPLITTER

J. MANUAL FSR CONTROL

The fuel stroke reference can be manually suppressed from Operator Interface Module [OPM] by pressing the MANUAL membrane switch and operating the FUEL STROKE REFERENCE MANUAL CONTROL DISPLAY. All relevant parameters are shown on this display including three (3) "Soft switches" which identify membrane switches to the right of the CRT as FSR GAG RAISE, FSR GAG LOWER and FSR GAG PRESET. The logic outputs from the display provide commands for algorithm FSRMAN in Controllers [R], [S] and [T], as shown in Figure VII-J-1.

FSRMAN is an open loop fuel stroke reference which is used in the minimum select algorithm, FSR. The reference is initially preset to FSRMAX when the power-up pulse, L83PUP, is received from the processor. This sets FSRMAN above the other control loop fuel stroke references where it stays until it is manually lowered to override the fuel stroke reference calling for the least fuel. Manually GAGGING the reference is calculated by comparing FSRMAX with FSRMAN, and the resultant logic, L60FSRG, generates an alarm.

L43FSRR raises the reference and L43FSRL lowers the reference at a ramp rate determined by constants FSKRMAN1 and FSKRMAN2. When a raise or lower command is given, FSKRMAN1 initially defines the ramp rate, and if the command is maintained, the ramp rate is increased to FSKRMAN2 to expedite the calibration. A raise command adds the value of the constant to FSRMAN calculated during the last sample, and a lower command subtracts the value of the constant. This process is continued as long as the calculated reference remains within the constraints of the FSRMAX and FSRMIN limits applied in the median select calculation.

Communicator [C] monitors FSRMAN calculated by Controllers [R], [S] and [T], and selects the median, FSRMANC, which is fed back to the controllers to preset their individual references to the median value. This occurs:

1. On a computer recovery, L83REC.
2. Every minute, L83MIN.

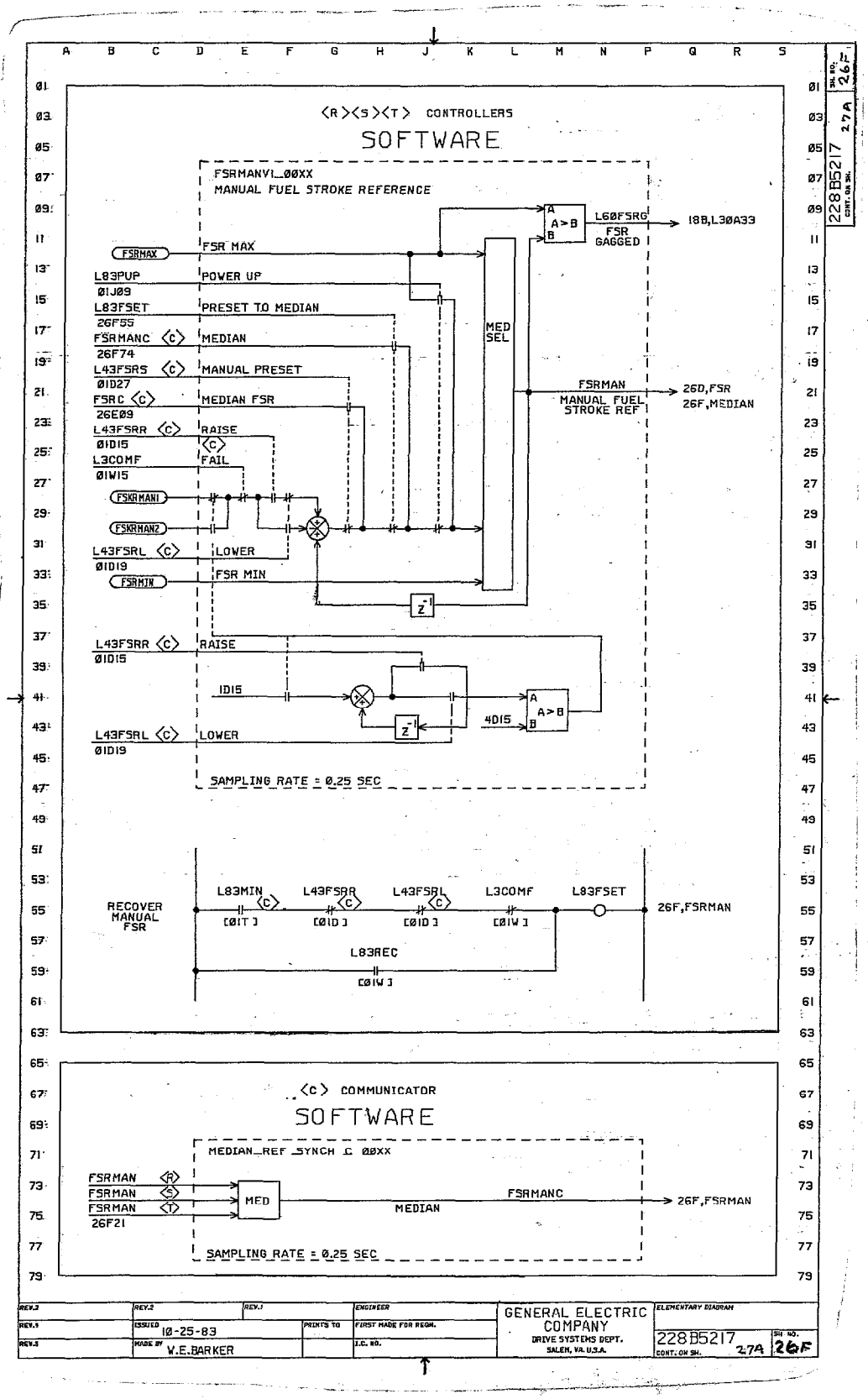


Figure VII-J-1: MANUAL FUEL STROKE REFERENCE

K. LOAD CONTROL

Algorithm, L90L, compares the called for load setpoint with the watts transducer feedback, DW, and raises or lowers speed setpoint TNR to keep the load within the defined limits. Figure VII-K-1 shows the algorithm located in Controllers [R], [S] and [T]. One (1) of the three (3) control constants are selected by sequencing logic to establish the called for load. Spinning reserve is selected by default if no other sequencing commands are received. Selection of a FAST LOAD START or PRESELECTED LOAD by the membrane switches on Operator Interface Module [OPM] forces logic L83PS to select the pre-selected load setpoint defined by LK90PSEL. Selection of BASE, PEAK or a manual load change causes logic L83RMAX to raise the maximum load limit to the level established by constant LK90MAX.

If the called for load setpoint exceeds the actual load, DW, by an amount greater than deadband limit LK90DB2, a raise command, L90LR is given to the speed/load reference, TNR, and sealed until the differential is reduced to less than LK90DB1. Conversely, if the actual load is greater than the called for load by an amount greater than LK90DB2, lower command L90LL is given.

A comparison of the fuel stroke references called for by the speed control loop, FSRN, and the temperature control loop, FSRT, calculates when the governor is on temperature control, i.e., FSRT $<$ FSRN, and it inhibits raising the speed reference if the differential exceeds the deadband, LK90DB3. A lower command is given if the deadband exceeds LK90DB4.

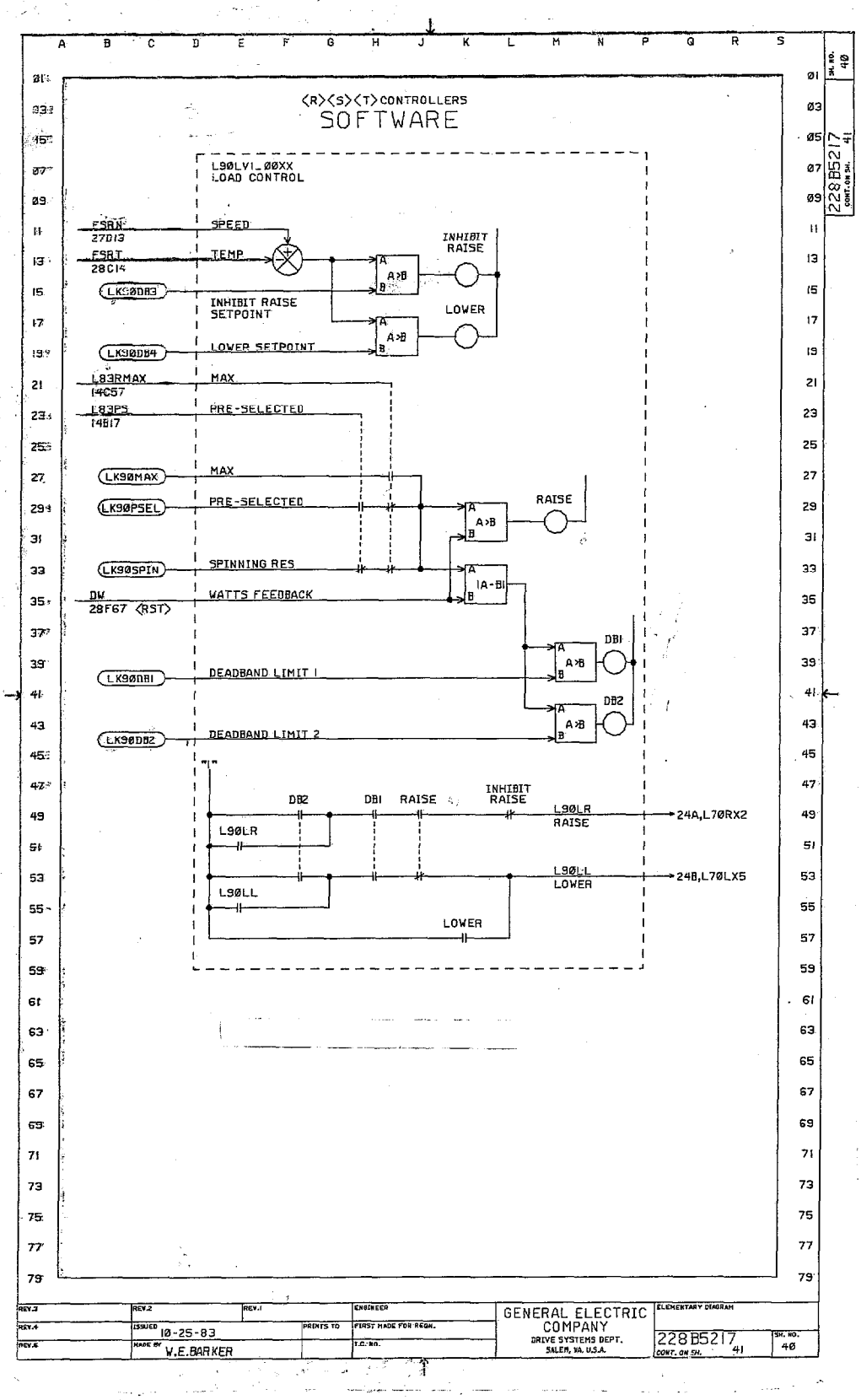


Figure VII-K-1: LOAD CONTROL

L. MODULATED INLET GUIDE VANES

1. GENERAL

Inlet guide vanes are normally controlled by an open or close command to a solenoid valve, but some applications require electrical modulation of the guide vanes for part speed or part load operation. In this case, the called for guide vane angle is calculated in software, and used as the reference for an HSAA servo amplifier. The amplifier drives a three (3) coil servo on the torque motor and regulates the loop by comparing the reference to the guide vane angle feedback via redundant LVDT's.

Figure VII-J-1 shows a typical control algorithm arrangement for an MS6001 inlet guide vane system. Algorithm CSRGV selects the control signal that calls for the smallest inlet guide vane angle and feeds it to the servo output algorithm, CSROUT. All algorithms in the inlet guide vane system are located in Controllers [R], [S] and [T], except for the median select algorithm which is located in Communicator [C].

2. MANUAL INLET GUIDE VANE CONTROL

The manual inlet guide vane control algorithm, CSRGVMAN, is similar to the manual fuel stroke reference control algorithm, FSRMAN, discussed and shown on pages VII-41 and VII-42. The inlet guide vane angle can be manually suppressed from Operator Interface Module [OPM] by pressing the MANUAL membrane switch and incrementing the display to the MODULATED INLET GUIDE VANE DISPLAY. All relevant parameters are shown. Two (2) softswitches are identified as IGV OPEN and IGV CLOSE. Outputs from the IGV OPEN and IGV softswitches are read by the CSRGVMAN algorithm which increases and decreases its reference angle at a rate determined by a control constant. Anytime the reference angle is forced below its maximum setpoint, logic output L60IGVG initiates an alarm message stating that the inlet guide vane angle reference is being manually gagged. The reference is preset to the median value once every minute and when a controller is initialized.

3. PART SPEED INLET GUIDE VANE CONTROL

During turbine start-up and shutdown sequences, the part speed algorithm, CSRGVPS, controls the guide vane angle as a function of corrected speed. The angle is increased as speed TNH increases. Ambient temperature CTIM biases the speed signal to avoid compressor pulsation and negative fifth stage extraction pressure. Consequently, an increasing ambient temperature causes the guide vanes to close and cause a more positive fifth stage pressure.

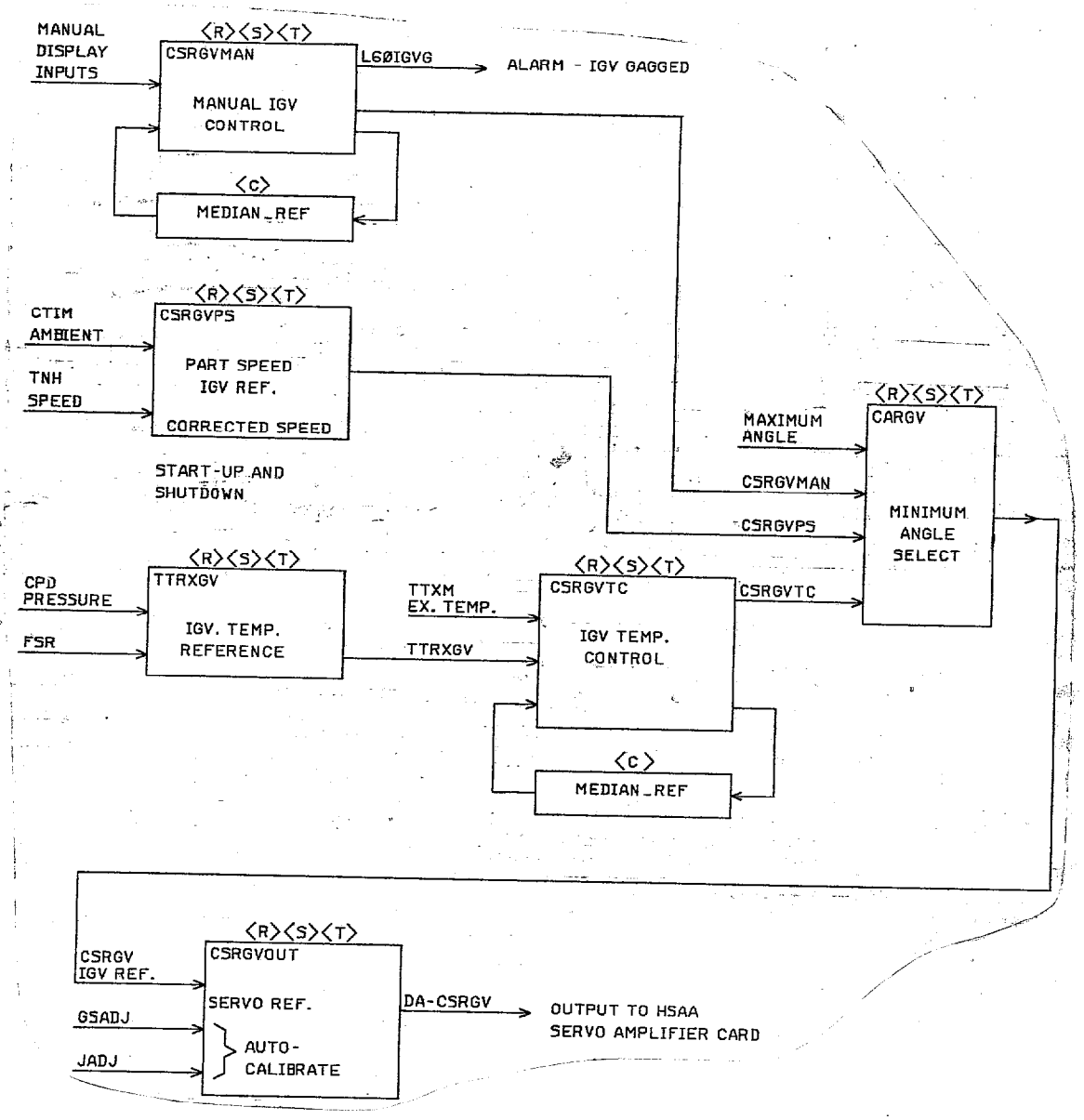


Figure VII-L-1: MODULATED INLET GUIDE VANE CONTROL

L. MODULATED INLET GUIDE VANES (continued)

4. TEMPERATURE CONTROL OF INLET GUIDE VANES

Selection of IGV temperature control is performed by the operator by pressing the IGV TEMP CONTROL ON or OFF membrane switches on the front of Operator Interface Module [OPM]. IGV temperature reference algorithm TTRXGV calculates the called for temperature similar to FSR temperature reference algorithm TTRX, described and shown in Section VII.F.2. Increasing ambient temperature causes reduced mass flow in the turbine and reduced compressor discharge pressure, CPD. Algorithm TTRXGV then calls for a larger inlet guide vane angle to increase the mass flow and compensate for the higher ambient temperature.

Inlet guide vane temperature control algorithm CSRGVTC is used during part load operation to maintain a high firing temperature and to increase efficiency. Reference figure VII-J-2 on the next page. IGV temperature reference TTRXGV is compared with median exhaust temperature feedback TTXM times a gain defined by CSKGVTPG. It is then corrected for the error between each controller's called for reference, CSRGVTCX, and the median value, CSRGVTCC, at a rate determined by CSKGVTCC. The rate of change of CSRGVTCX is calculated by the DPMA subroutine, and the result is then added to the CSRGVTCX calculation. An increase in the temperature reference causes the called for guide vane angle to increase on a ramp as long as it remains between the maximum and minimum limits established by constants CSRGVMAX and CSRGVMIN. A controller recovery, i.e., L83REC, presets the DPMA output to the median value, CSRGVTCC.

The final called for IGV angle due to temperature control CSRGVTC is a function of the temperature calculation, CSRGVTCX, and an optional called for IGV angle, CSRGVAQ, as a result of the air flow, AQRDSP. The parameter which calls for the maximum IGV angle will govern CSRGVTC, and logic L60GVTC will indicate a "1" if the temperature control reference is larger than the air flow reference.

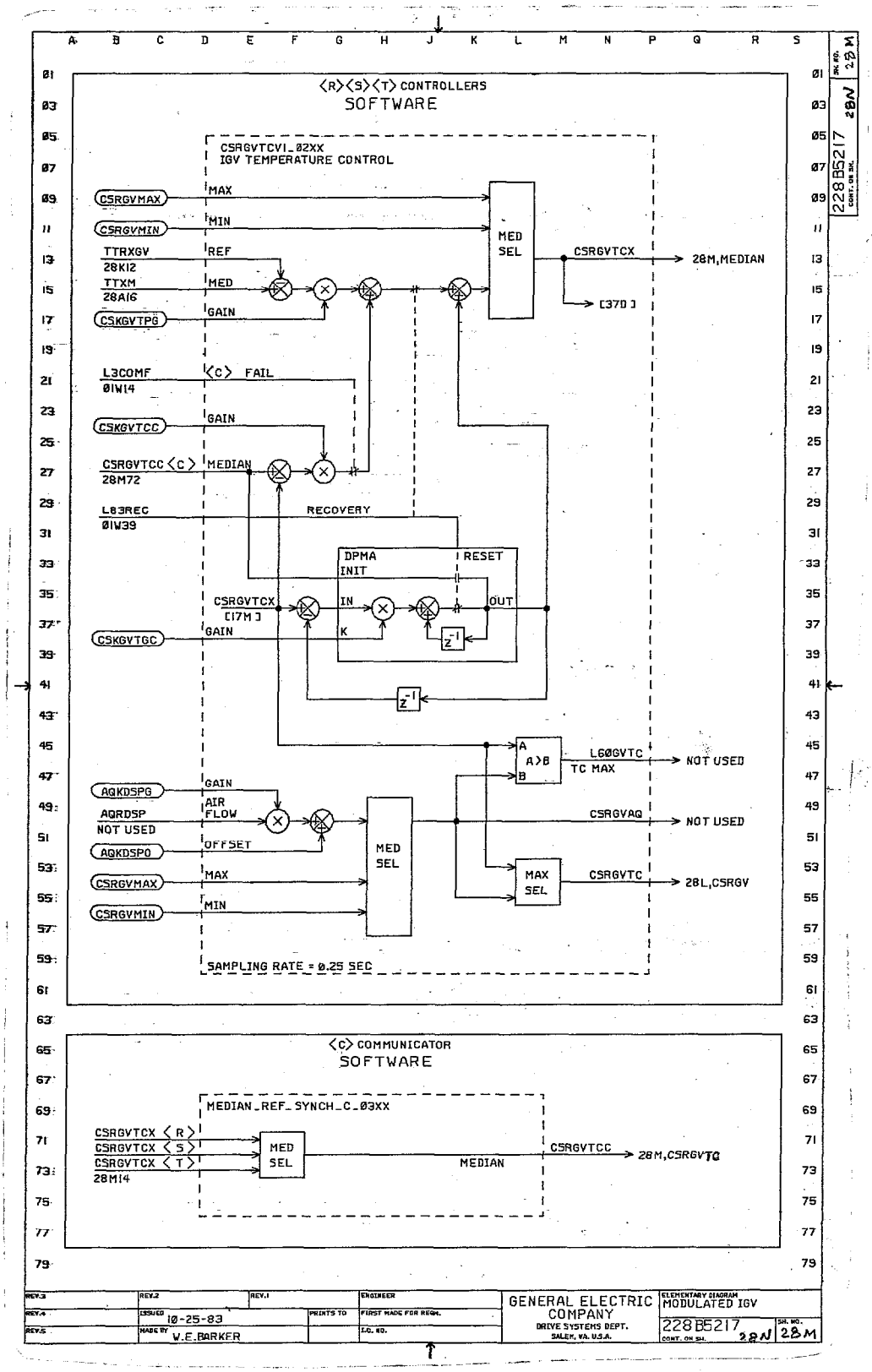


Figure VII-L-2: INLET GUIDE VANE TEMPERATURE CONTROL

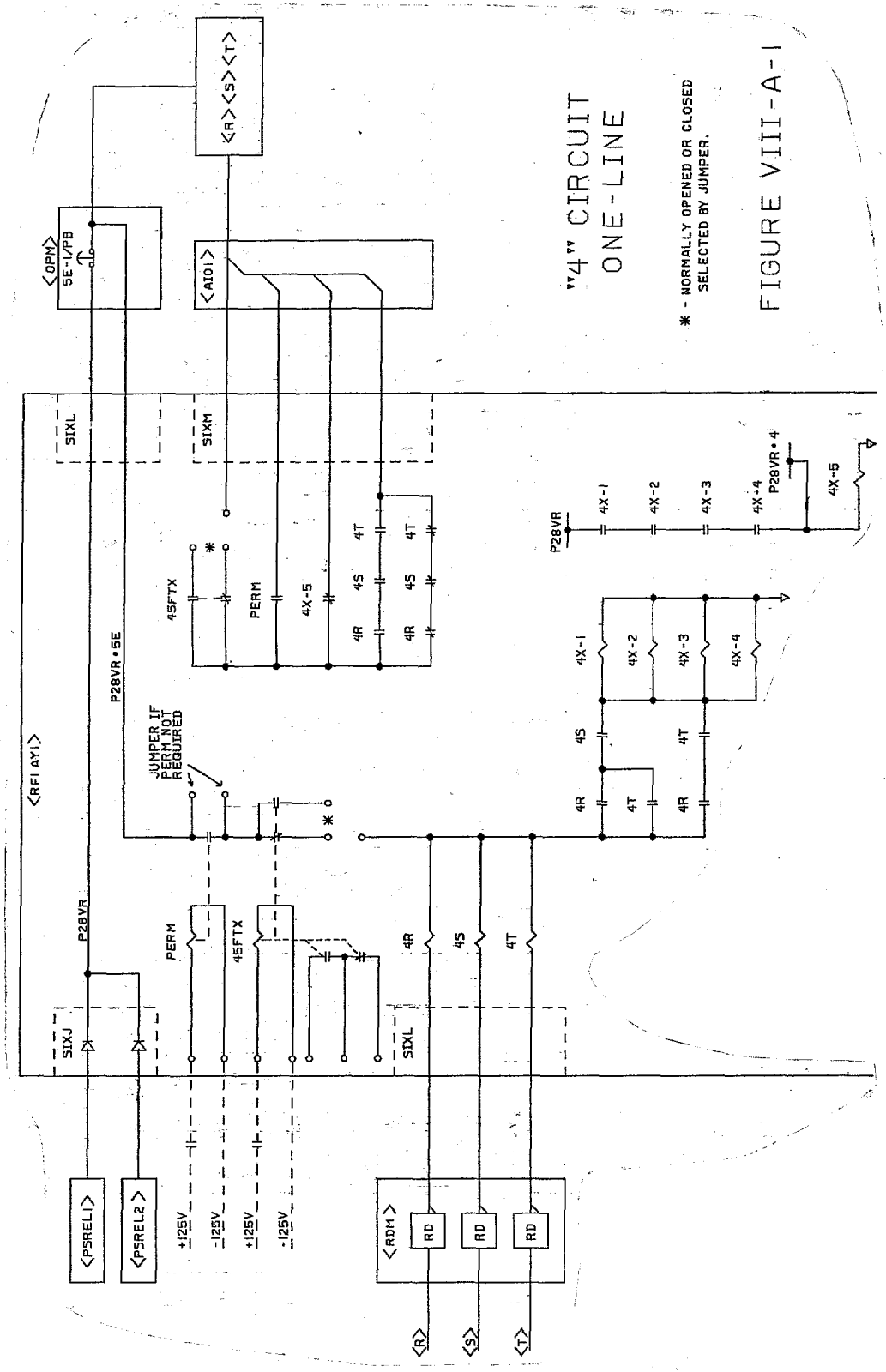


Figure VIII-A-1: MASTER PROTECTIVE CIRCUIT "4" BUS

VIII. PROTECTION SYSTEM

A. MASTER PROTECTION CIRCUIT ("4" BUS)

The master protection circuit is called the "4" bus, and it is shown in Figure VIII-A-1. Any trip command from Controllers [R], [S] and [T], de-energizes a separate relay for each controller, i.e., 4R, 4S and 4T, in relay module #1, [RELAY1]. A 2/3 vote is performed by contacts from these relays to decide whether the master 4X relays should be energized or de-energized. This voting of the run/trip command is performed in hardware rather than software to ensure that no single program or device contains a single point failure and to ensure that the final component in the circuit, the relay, has a predictable failure mode, i.e., de-energized. Magnetic relays are still used instead of solid-state relays because they have the predictable failure mode, their contacts are independent of the voltage level or type, i.e., AC or DC, their contacts are form "C", etc. Two (2) redundant 28VDC power supplies are used to power the bus, and each supply is monitored by Communicator [C] to alarm any undervoltage condition. A contact from the emergency stop pushbutton is wired in series with the bus to directly de-energize the "4" relays independent of the computers. In addition, the status of the pushbutton is monitored by Controllers [R], [S] and [T], to trip the software sequencing and prevent any restart while the pushbutton is pressed in. The red, mushroom head, shrouded pushbutton is located on the front of the Operator Interface Module, [OPM], and it will remain pushed in until the mushroom head is turned and pulled out.

Two (2) other relays are wired to directly interrupt the "4" bus: the 45FTX relay, i.e., fire protection, and the PERM relay, i.e., permissive to run. 45FTX can be made to energize to trip or to de-energize to trip by jumpering the normally closed or normally open contact in series with the "4" bus. The controller's HAFA cards monitor both relays and the status of the "4" relays. Any disagreement between the "4" relays is alarmed. Any discrepancy between the output command to the "4" relay drivers and the resultant state of the "4" relays is alarmed. All relays which must be de-energized to trip the turbine have the "4" relay contacts wired in series with the 28VDC bus feeding their coils to ensure that the relay will de-energize if the relay driver fails turned-on. Since a relay's application varies from installation to installation, its coil can be jumpered to the "4" bus or the continuous 28VDC bus.

B. FLAME DETECTION

Flame detectors monitor ultraviolet scanners on the turbine and generate logic signals to controllers [R], [S] and [T]. The panel mounted detectors receive 28VDC from the redundant 28VDC power supplies and produce 350VDC, 120HZ outputs to the scanners. Flame in the turbine's combustion chambers ionizes the gas in the scanners which causes a 120HZ current feedback to the panel detector. Voltage follower amplifiers in the detectors turn on switching transistors when a flame is present, thus generating a logic "0" output. The outputs are filtered in the Analog I/O Modules and fed to the HAFA cards in [R], [S] and [T], for monitoring.

B. FLAME DETECTION (continued)

There are two (2) standard types of flame detection systems called the redundant and non-redundant systems. The redundant system has four (4) channels which require flame detection on only two (2) channels to run and loss of flame on any three (3) channels to trip. Non-redundant systems have two (2) channels. They will run if flame is detected on either channel and trip if flame is lost on both channels. Both systems have a one (1) second time delay on loss of flame before the turbine reaches its operating speed, i.e., flame flicker protection. Both alarm and trip on a loss of flame. In addition, the NORMAL DISPLAY shows which channels flame has been detected on during a start-up sequence, and the current logic of any flame input can be displayed on the screen at any time by calling the desired signal name, i.e., L28FD1, L28FD2, L28FD3 or L28FD4.

C. OVERSPEED PROTECTION

The overspeed protection system uses the same three (3) magnetic pick-ups per shaft that are used by the speed control system. Each magnetic pick-up is terminated at analog I/O module #3, [AIO3], which filters the signals and connects them to the HPRB cards in Controllers [R], [S] and [T], for the pulse rate to digital conversion. Algorithms L12H and L12L monitor the high pressure set speed, TNH, and the low pressure set speed, TNL, and compare them with a setpoint to establish the speed limit. If the speed exceeds the setpoint, a latched logic output signal is generated to trip the turbine. The trip signal will remain latched until the operator manually resets the latch via the MANUAL RESET DISPLAY. Provision is made to raise the electrical overspeed setpoint by a pre-defined offset to facilitate testing the mechanical overspeed bolt. This is done by selecting the MANUAL CONTROL DISPLAY for the MECHANICAL OVERSPEED TEST and pressing the softswitch labeled OVERSPEED TEST.

The overspeed bolt assembly consists of a plunger whose mass is located eccentric to the turbine shaft centerline, and it is held together by a spring during normal turbine operating speeds. When the speed of the turbine reaches the trip setpoint, the centrifugal force overcomes the compression of the spring, and the plunger protrudes from the shaft. The plunger strikes the "trip finger" which activates a valve and dumps the high pressure trip oil. Simultaneously, a contact from a limit switch is fed back to the control panel to alarm the trip. Typical overspeed mechanisms are specialized two-way valves with a latch. The trip mechanism may be activated automatically by the overspeed bolt assembly or manually tripped at the device. In either case, it must be manually reset at the device.

D. STARTING DEVICE BOG DOWN DETECTION

Algorithm L60BOG detects when the starting device, such as a diesel, is bogging down. The algorithm is enabled from the time the "4's" pick-up, i.e., L4 = 1, until the turbine reaches its operating speed, i.e., L14HSX = 1. High pressure set speed from the current sample is compared with the highest speed ever reached during the start-up sequence, and if the differential exceeds the limit established by control constant

D. STARTING DEVICE BOG DOWN DETECTION (continued)

LK60BOG1, trip logic L60BOG will be generated. A time delay defined by constant LK60BOG2 insures that a transient speed droop will not cause an erroneous trip.

E. EXHAUST OVERTEMPERATURE PROTECTION

Algorithm L86TXT calculates the exhaust overtemperature alarm and trip points in Controllers [R], [S] and [T]. It is shown in Figure VIII-E-1. The algorithm compares median exhaust temperature TTXM with called for temperature reference TTRXB plus offset TTKOT3 and generates alarm logic L30TXA if the differential exceeds the offset. A similar calculation is used to determine the the trip setpoint. In this case, control constant TTKOT2 contains the maximum allowable spread between the median feedback and the reference. The resultant trip logic, L86TXT, latches until manually reset by logic L86MR1. Note that the alarm and trip characteristics are a function of compressor discharge pressure, CPD which biases the temperature reference in algorithm TTRX. An isothermal trip will occur if the median temperature exceeds the isothermal limit established by constant TTKOT1.

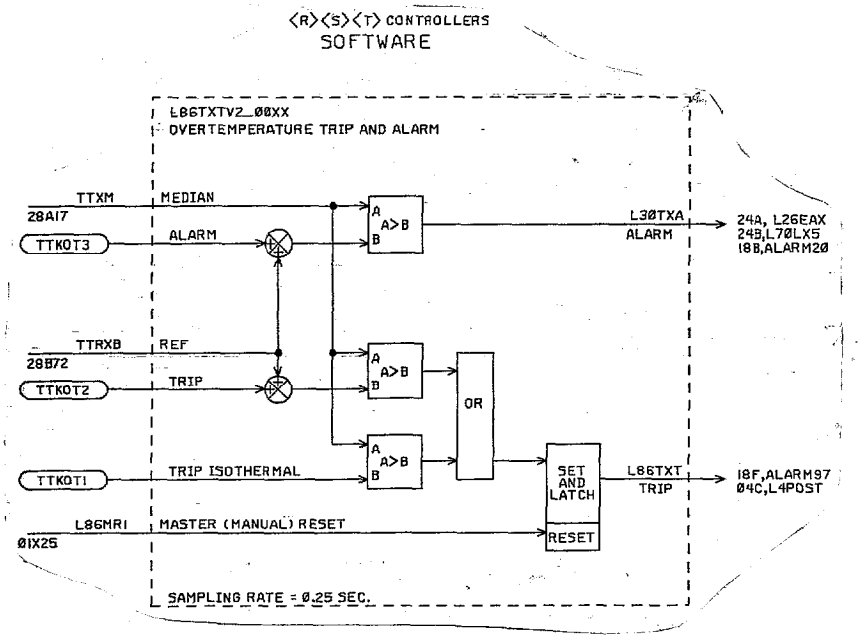


Figure VIII-E-1: "TX OVERTEMPERATURE"

F. COMBUSTION MONITOR

Algorithm TTXSP calculates the allowable temperature spread between the exhaust thermocouples and generates logic outputs to alarm and trip if the spread exceeds the established limits. It is located in Communicator [C] and is shown in Figures VIII-F-1 and VIII-F-2. The algorithm selects the value of the hottest compressor discharge thermocouple input, CTDAn, and determines if it is between the maximum and minimum limits defined by constants TTKSPL1 and 2 respectively. This median temperature is multiplied by constant TTKSPL3 and then used to inversely bias the scaled average exhaust temperature, TTXC. For a given average exhaust temperature, an increase in the compressor temperature will reduce the allowable exhaust temperature spread, TTXSPL. An offset, TTKSPL5, is added to this calculation, and the maximum and minimum limits are applied by constants TTKSPL6 and 7, respectively.

During rapid FSR changes, such as start-up, fuel transfers, etc., the allowable spread is increased by the temperature bias contained in constant TTKSPL10. Sequencing logic L83SPMB is a "1" during these periods. The resultant bias is the difference between TTKSPL10 and the output of the minimum select calculation which is zero (0). When L83SPMB resets to "0", the output of the minimum select calculation ramps up at a rate determined by TTKSPL8 and 9 until it equals TTKSPL10 which nullifies the bias.

Non-adjustable constants TTKXM2R, S and T, define the number of thermocouple inputs connected to each controller, and their sum defines the total number of inputs to be monitored for the speed calculation. Figure VIII-F-2 shows the comparison of the maximum allowable temperature spread with the actual spreads. TTXD2n is an array of thermocouple temperatures from the hottest (0) to the coldest (n). Three (3) different spread calculations determine the differential temperatures between various groups of inputs, and if the spread, i.e., TTXSP1-3, exceeds the allowable margins determined by constants TTKSP1-4, then their respective logic outputs, L60SP1-4, will become a "1". These outputs are enabled when L83SPM is a "1" which generally occurs when the turbine is above its operating speed. In addition, failure of any controller, L3COMFO-2, will change the spread calculation for TTXSP1 and inhibit the spread calculation for TTXSP2 to prevent an erroneous spread indication because fewer thermocouples are being monitored.

JXDn is an array of the thermocouple device numbers sorted from the hottest (0) to the coldest (n-1). For example, if the coldest thermocouple is #9 and the second coldest is #10 which is physically adjacent to #9 on the turbine, then the difference calculation is #9 - #10 = 1 which sets logic L60SP5 to a "1". L60SP5 and L60SP6 are used as trip permissives to enable a turbine trip due to an excessive temperature spread only if the coldest reading is an accurate measurement.

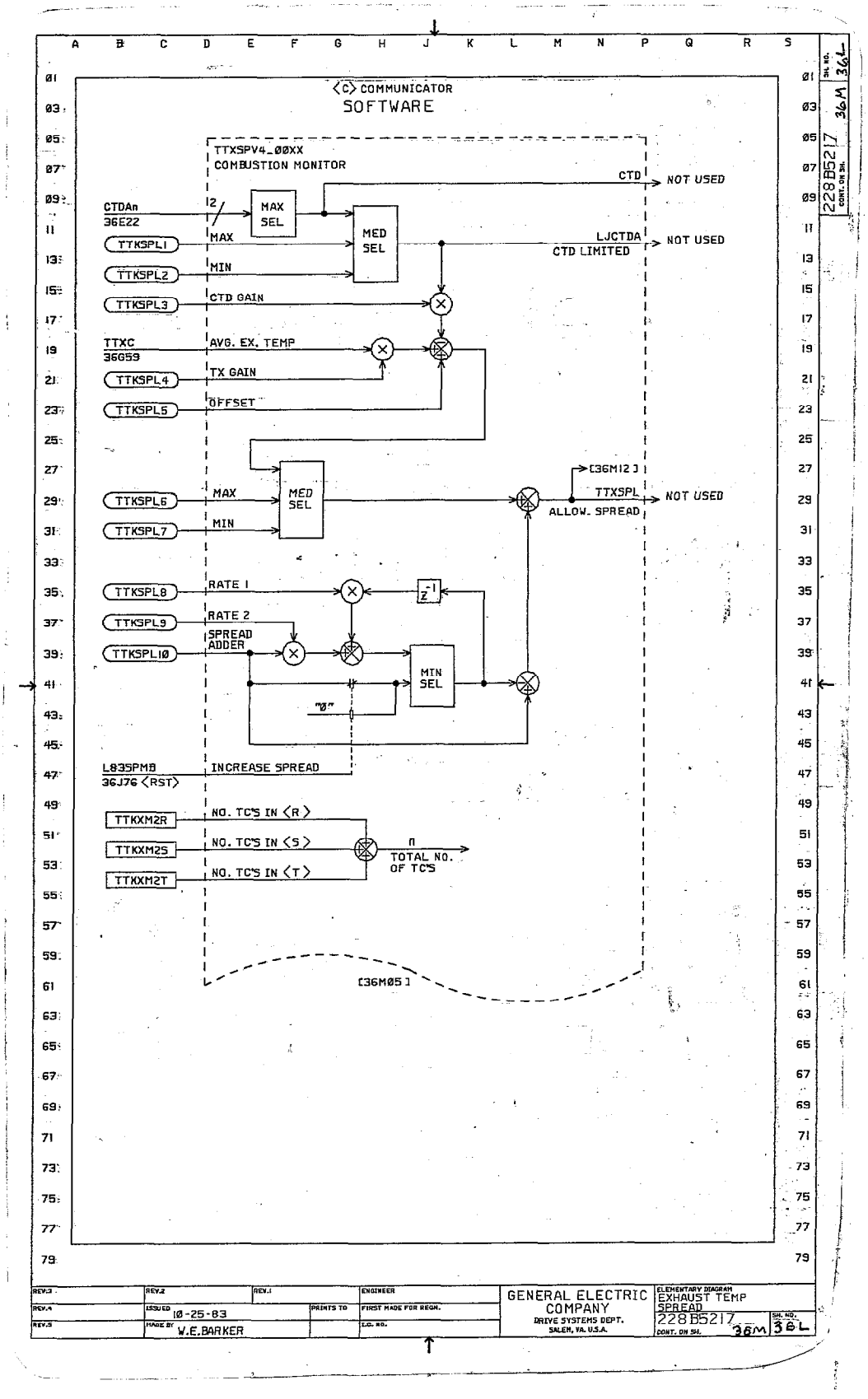


Figure VIII-F-1: COMBUSTION MONITOR

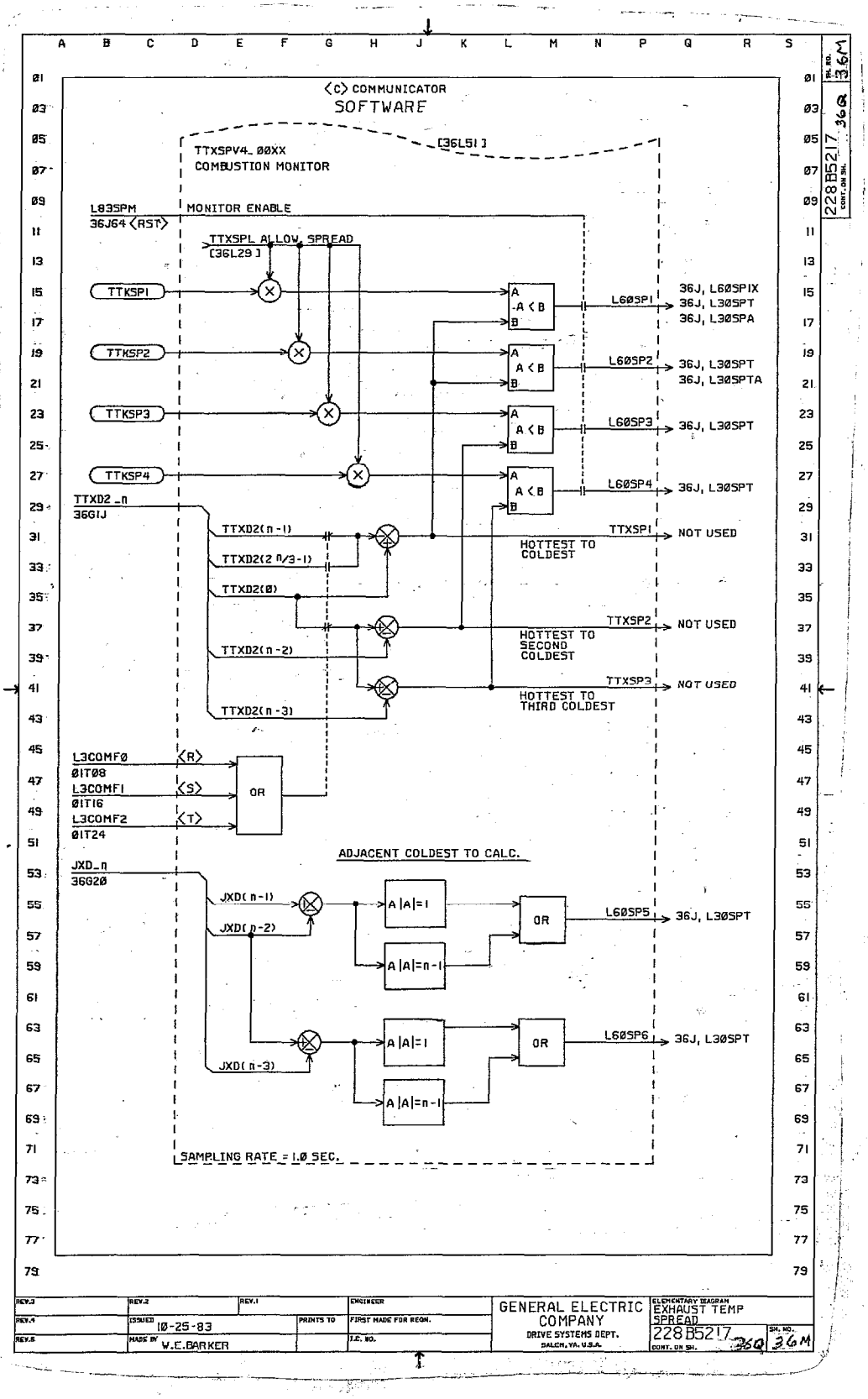


Figure VIII-F-2: COMBUSTION MONITOR

G. WHEELSPACE TEMPERATURE MONITOR

The wheelspace thermocouple monitor algorithm, L30WSA, alarms an overtemperature condition or an excessive differential between a pair of thermocouples. An overtemperature condition is determined by comparing the average of a thermocouple pair with a temperature limit contained in a control constant. In addition, the temperature limit is lowered a specified time after the turbine reaches complete sequence. Separate control constants establish the temperature limit for each thermocouple pair but a single control constant defines the differential limit for all thermocouple pairs. The algorithm is located in Communicator [C] because the overtemperature logic, L30WSA2, and the differential logic, L30WSA1, alarm only.

H. VIBRATION PROTECTION

There are two (2) types of vibration protection systems called the redundant sensor system and the non-redundant sensor system. We will examine the redundant system in this chapter since it is more frequently used. The non-redundant system is a sub-set of the redundant system. Algorithm L39V is located in Controllers [R], [S] and [T], and is shown in Figures VIII-H-1 through VIII-H-4. This algorithm is designed for a STAG, i.e., Steam and Gas Turbine, system and receives five (5) velocity type vibration sensor inputs from the gas turbine, three (3) from the generator and three (3) from the steam turbine. The electrical interface to the NVCB cards in the controllers is rated 0.0 - 1.0V-Peak, 8-275HZ, 200-1,200 ohms, for non-grounded transducers.

Reference Figure VIII-H-1. Each input is assigned a signal name, BBn, where "n" specifies one (1) of the eleven (11) inputs. Comparators monitor the inputs to detect if the vibration level falls below the fault level LK39VFn, rises above the alarm level LK39VAn, or rises above the trip level LK39VTn. If any of the sensors fail high or low the erroneous input can be rejected by selecting the particular enable/disable control constant JK39n and setting it to zero. This clears the fault, alarm or trip output but retains the disabled logic, L39VD1, which is recorded as an alarm message. Constant JK39Un is a non-adjustable constant which is contained in PROM and used to determine which of the eleven (11) input channels are being used. For example, if your application is not a STAG system, inputs #9, 10 and 11 would be disabled to prevent the algorithm from reading those inputs and incorrectly alarming a sensor fault condition.

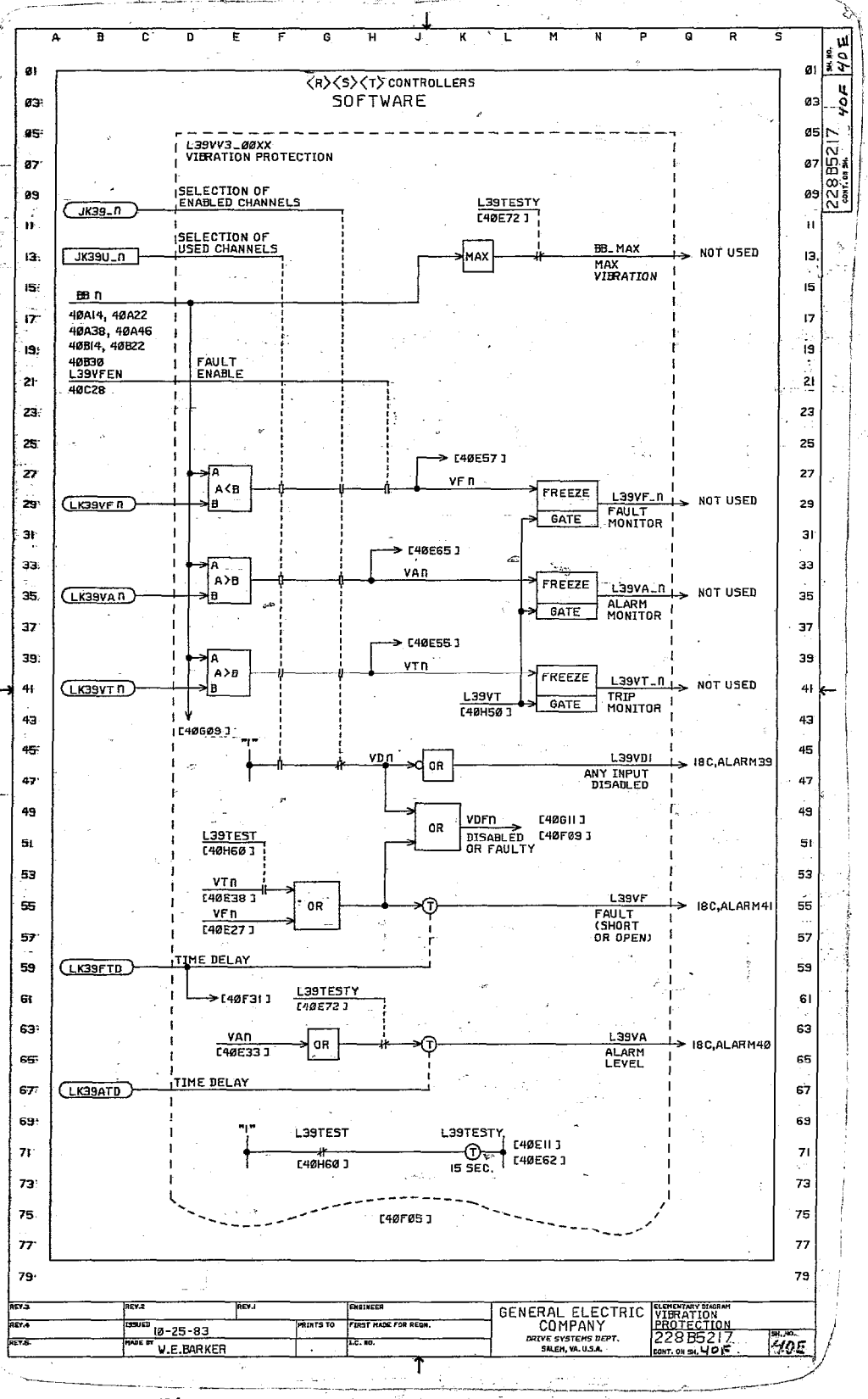


Figure VIII-H-1: VIBRATION PROTECTION

H. VIBRATION PROTECTION (continued)

The algorithm commands the NVCB vibration card to generate a test signal for all sensors when the gas turbine is stopped, L39TEST, and all permissives are satisfied, L39TPERM, as shown in Figure VIII-H-4. The resultant feedback will be greater than the fault setpoint, LK39VFn, and less than the trip setpoint, LK39VTn, if the sensors are functioning properly. Alarm logic L39VA and trip logic L39VT are inhibited during this test. In addition, the maximum select function which selects the highest vibration level, BBMAX, and displays it on the NORMAL DISPLAY is inhibited during a shutdown vibration test. Two (2) vibration fault calculations are used to determine if an actual sensor fault condition has occurred. L39VF becomes a logic "1" if any vibration level remains below setpoint LK39VFn, or above setpoint LK39VTn, for longer than the time allowed by constant LK39FTD. Logic input L39VFEN enables the "low level" fault calculation when the sensor is being tested during a turbine shutdown and typically when it is running above its operating speed. During a turbine shutdown, logic L39TEST enables the "high level" fault calculation. A high vibration level above the allowable alarm level is indicated by logic L39VA. This calculation compares each input with the setpoint, LK39VAn, and sets L39VA to a logic "1" if the high level persists for longer than the time contained in constant LK39ATD, and if the vibration system is not in the shutdown test mode as determined by logic L39TEST.

A turbine trip due to excessive vibration will result in an alarm message documenting the reason for the trip. The "freeze" blocks will retain the vibration logic of the individual channels which caused the trip. These freeze blocks are activated by trip logic L39VT. Their resultant logic outputs can be displayed on the CRT by calling their logic names.

Reference Figure VIII-H-2. If all of the vibration sensors from a given machine are manually disabled for longer than the time defined by constant LK39FTD, then logic L39VD2 will become a logic "1" initiating an alarm message and usually initiating an automatic shutdown sequence. If three (3) of the five (5) gas turbine sensors are manually disabled or calculated to be faulty for longer than the time in constant LK39FTD, then logic L39VD3 will become a logic "1" causing an alarm message and usually inhibiting a start-up sequence. Similarly, this will occur if two (2) of the three (3) generator or steam turbine sensors have been manually disabled or calculated to be faulty. The generator and steam turbine calculations are inhibited if constant JK39Un defines these sensor inputs to be not used.

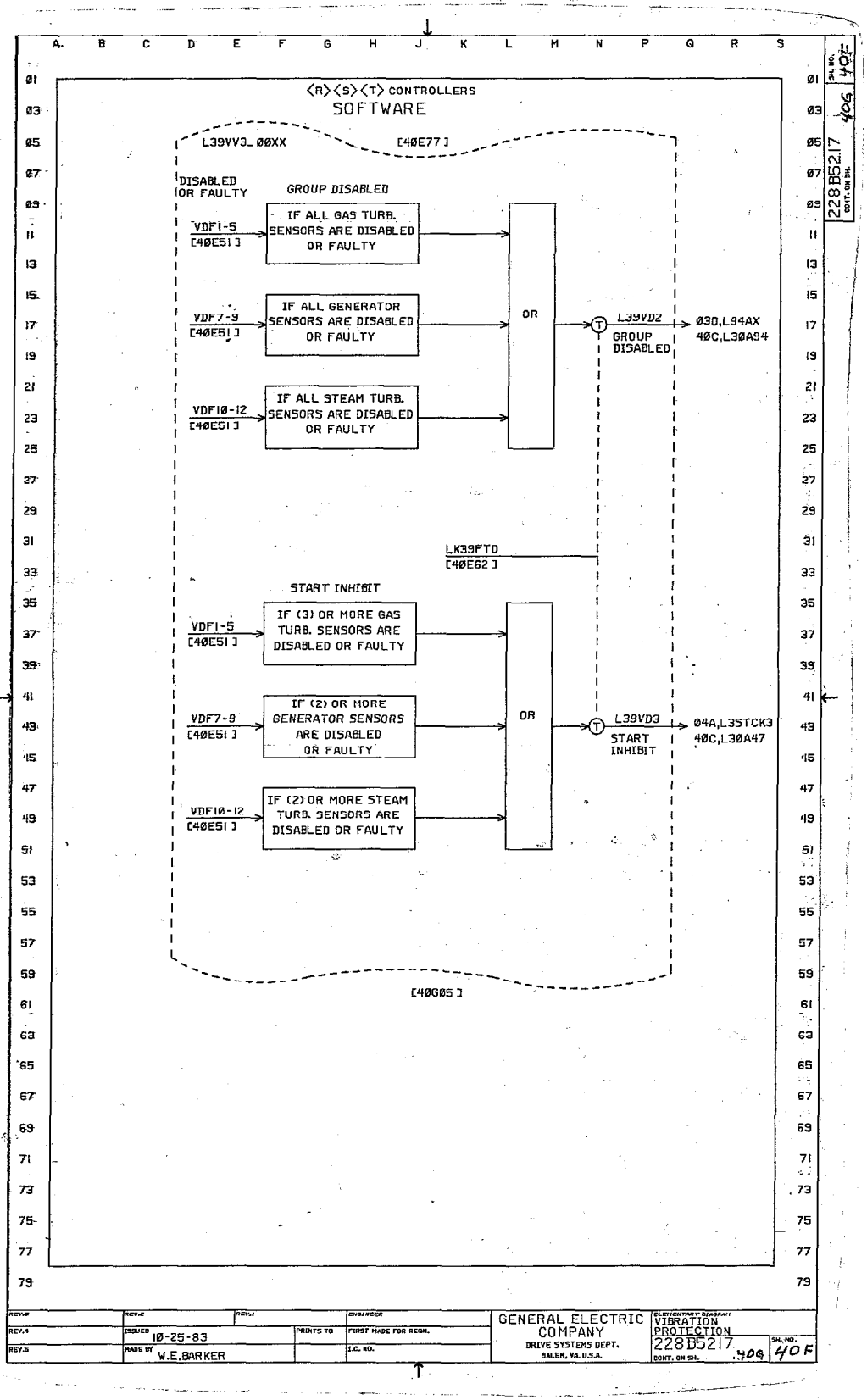


Figure VIII-H-2: VIBRATION PROTECTION

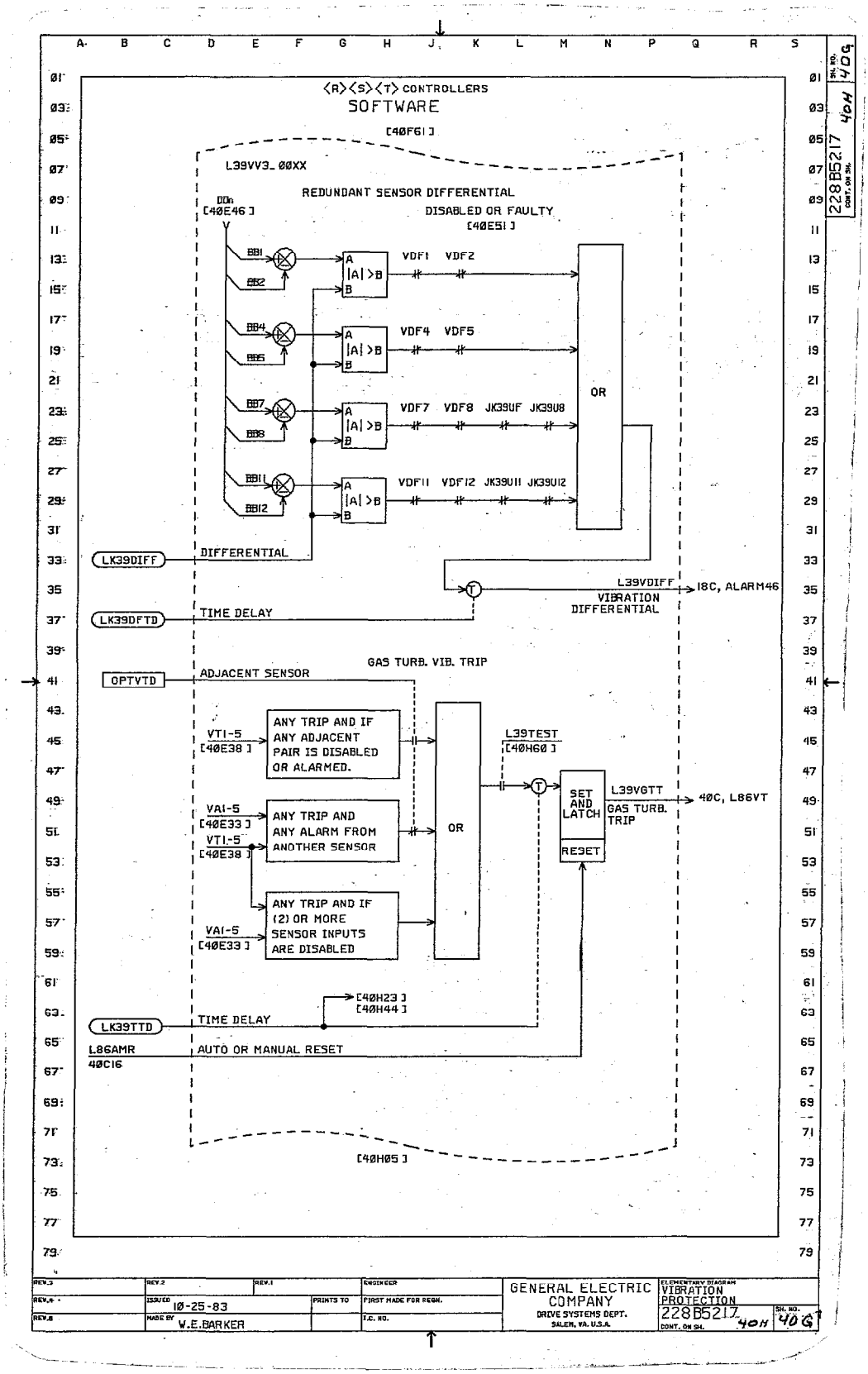


Figure VIII-H-3: VIBRATION PROTECTION

H. VIBRATION PROTECTION (continued)

Reference Figure VIII-H-3. The difference between pairs of vibration inputs which monitor particular bearings are compared with an established differential setpoint, LK39DIFF. An alarm output, L39VDIFF, is generated if the differential persists for a time defined by LK39DFTD. Since manually rejecting one (1) of these inputs will automatically cause an excessive differential, the alarm is inhibited if either input is rejected from a given pair. A fault condition will cause a similar inhibit of the alarm. The term VDFn is defined as "vibration disabled or fault".

A gas turbine trip will occur if the vibration exceeds the trip level on any input and any adjacent pair of inputs is disabled, or if the vibration has reached the trip level on one (1) input and it has reached the alarm level on another channel. Note that these two (2) trip calculations are mutually exclusive as defined by the non-adjustable constant, OPTVTD, which is a function of the mechanical sensor arrangement for a given gas turbine. In addition, a turbine trip will occur if the vibration reaches the trip level on any input and two or more of the gas turbine sensors are disabled or faulty. Time delay LK39TTD ensures that a trip will not occur due to a system transient, and once a trip does occur it will latch until reset by logic L86AMR. This logic usually automatically resets the latch for applications which require a multiple start feature and manually resets the latch for applications that do not require a multiple start.

Reference Figure VIII-H-4. The generator and steam turbine vibration trip calculations are performed similar to those of the gas turbine except for the quantity of sensors and the ability to inhibit the function via constant JK39n if no sensors are monitored from one (1) or both of the machines. A trip will occur if the vibration level reaches the trip level on one (1) channel and the alarm level on any other channel or if two (2) sensor inputs are disabled or faulty from a given machine. In either case, the trip is latched until manually reset by logic L86MR1. The individual machine trip logic is available for monitoring, and a trip from any machine's sensors will result in logic L39VT becoming a "1" if the turbine is not in the shutdown mode, i.e., L39TEST = "1".

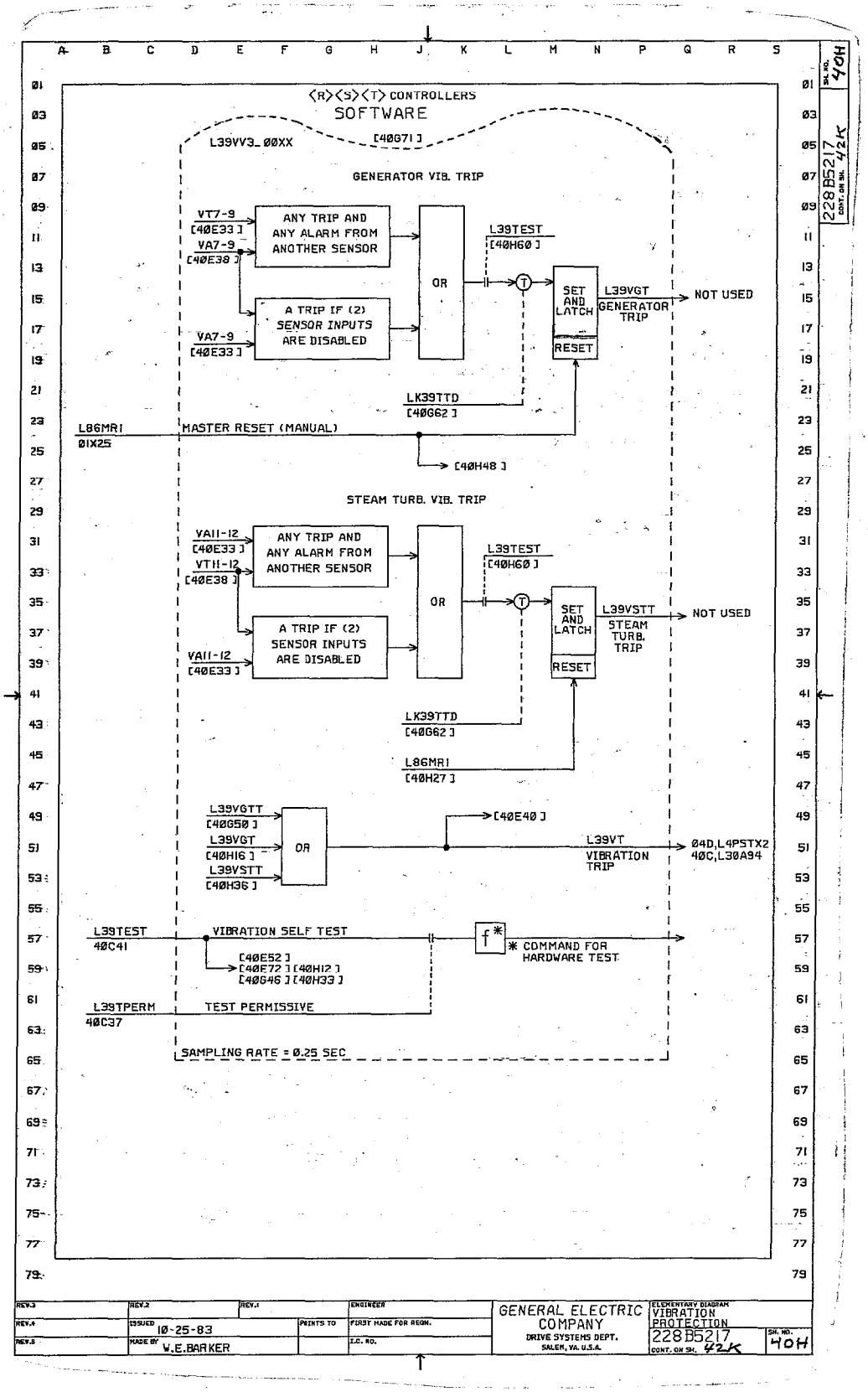


Figure VIII-H-4: VIBRATION PROTECTION

IX. GLOSSARY

The following are definitions of the terminology used in the control system.

Actuator — A self-contained device designed to deliver a controlled or regulated force in order to activate some other device.

Address Bus — A bus used to transmit the identification code that distinguishes one memory location or I/O port from another.

Algorithm — In this control system, the term algorithm usually refers to a PL/M-86 application program dedicated to a specific function like speed control or vibration protection.

Analog — A continuous signal or representation of a quantity that can take any value. Note that this is a hardware term.

Application Software — Application oriented programming such as sequencing and control algorithms which are customized to the needs of each installation.

ASCII — Abbreviation for American Standard Code for Information Interchange. Each character is assigned a number between 32 and 127.

AS/M-86 — A low level programming language designed for the Intel 8086 microprocessor. It is used in some parts of the computer operating system which require a fast execution time.

Asynchronous — Operation of a switching network by a free running signal. Completion of one instruction triggers the next instruction.

Back-plane — The internal wiring of a module between pins of the cards located in the module. Some modules have a "flow-soldered" back-plane which is a printed circuit card that the other cards plug into.

Base Load — The maximum temperature control setpoint at which the turbine can be operated without sacrificing the machine life expectancy.

Baud Rate — A measure of the rate of data flow. The number of signal elements per second based on the duration of the shortest element.

Binary — A system of numbers using 2 as the base in contrast to the decimal system which uses 10 as the base.

IX. GLOSSARY (continued)

Bit	A single binary digit which can be in either of two (2) states, i.e., 0 or 1.
Black Start	The means of starting a turbine without AC power.
BRAM	Battery backed Random Access Memory is used to retain field editable software during power outages for a given length of time.
Bypass Valve	A device which regulates the flow of fuel or air in a fuel bypass system on a liquid fuel system using a positive displacement pump or in an air control valve used for pulsation protection systems.
Byte	A data element consisting of 8 bits.
Clock	The pulse generator which controls the timing of switching circuits in the microprocessor. Hardware architecture and programming are other factors in determining the rate of data manipulation.
Cold Junction	The junction between the thermocouple wire and the screw terminals on the thermocouple modules generates an EMF at the point of termination called the "cold junction".
Control Bus	The bus carries commands from and to the CPU for control of the operating system, i.e., read, write, etc.
Control Compart.	The compartment, Control Cab, which contains the gas turbine electrical controls and protection equipment.
Control Constant	An input to a control algorithm used to define a gain, time constant, offset, etc. Each control constant can be called by its signal name as shown in the control elementary.
CPU	The Central Processing Unit is the heart of the computer system consisting of storage elements called registers, computation circuits, the control block and I/O.
Cranking	The slow turning of the turbine rotor during start-up or shutdown.
Data Bus	A bus used to transfer coded information to and from the CPU, memory storage and peripheral devices.
Differential Input	An analog input which has a high impedance to ground on both input wires.

IX. GLOSSARY (continued)

Download	A dump of memory from one area to another. For example, if power to a controller is turned off and then turned on, the current sequencing program is dumped or downloaded from the Communicator, [C], to the controller.
EPROM	Erasable Programmable Read Only Memory. It can be erased by exposure to ultraviolet light.
EEPROM	Electrically Erasable Programmable Read Only Memory is used to store some application software which will not be lost during a power outage.
Firmware	A method of system control program design in which all control instructions are contained in ROM or PROM.
Flame Detectors	Sensors, usually ultraviolet, used to detect flame.
Flow Divider	A device which distributes fuel flow equally to the fuel nozzles.
Fuel, Gas	Either natural gas with a high heat content or manufactured gas.
Fuel Light Distillate	Also known as No. 2 fuel. A volatile distillate fuel having good combustion properties, clean burning and readily atomized. Preheating is usually not necessary.
Fuel Oil Stop Valve	A spring loaded, closed, hydraulically opened device used as a positive shutoff of liquid fuel.
GAC	Abbreviation for the generator auxiliary compartment which contains high voltage switch gear and excitation.
Governor	The Mark IV is the gas turbine process control system or governor.
Handshaking	A method of controlling data transfers in which the transmitting device generates a data ready signal. This signal directs the receiving device to accept the data. The receiving device then generates a data accepted signal to inform the transmitting device that it may remove the data and proceed.
Hardware	Physical equipment as opposed to software programming.
Hexidecimal	A system of numbers using 16 as the base in contrast to the decimal system which uses 10 as the base.

IX. GLOSSARY (continued)

High Pressure Turbine	The compressor or gas generator turbine.
Hydraulic Ratchet	A form of turning gear which turns the rotor slightly at periodic intervals.
IDATA	A file dedicated to storing application integer variable signal names such as speed, i.e., TNH, or fuel stroke reference, i.e. FSR.
Inlet Guide Vane	The guide vanes are at the inlet to the compressor. They are used to direct and control the air flow to the first stage of the axial flow compressor.
Interrupt	The suspension of the normal programming routine of a microprocessor in order to handle a request for service.
IVAR	A file dedicated to storing application integer data.
LDATA	A file dedicated to storing application logic variable signal names such as complete sequence, i.e., L3, or flame detected, i.e., L28FD.
Load Turbine Nozzle	The variable angle nozzle between the high pressure and low pressure turbine wheels on two (2) shaft turbines to proportion energy between the turbines.
LVAR	A file dedicated to storing application logic data.
LVDT	Abbreviation for Linear Variable Differential Transformer.
MDATA	A file dedicated to storing predefined miscellaneous application logic variable signal names.
MEM	A Medium Electronics Module can contain up to 24 circuit cards. The Communicator, [C], is a MEM.
Membrane Switch	One of the pushbuttons which are beneath the overlay on the Operator Interface Module.
Memory	The part of the computer which stores information in binary form.
Microprocessor	A central processing unit generally consisting of an arithmetic and logic unit, control block and register array, and a memory storage system.

IX. GLOSSARY (continued)

Modem	A device that adds or removes a carrier frequency to an existing signal which allows data to be transmitted or received on a high frequency channel.
Multiplexing	A process of transmitting more than one signal at a time over a single link via time-sharing, i.e., serial, or frequency-sharing, i.e., parallel.
MVAR	A file dedicated to storing predefined miscellaneous application logic data.
Off-line	A function performed with the turbine stopped and/or the control disconnected from the process.
On-line	A function performed regardless of the operating status of the turbine or whether the control is connected to the process.
Operating System (Computer)	The Operating System of the computer performs all tasks needed to run the computer independent of the variable application software.
Optical Isolation	A semiconductor device consisting of an LED and a photodiode or phototransistor in close proximity. Current through the LED causes an internal light emission that forces current to flow in the phototransistor. Voltage differences have no effect because the devices are electrically separated.
Party Line	A large number of devices connected to a single line originating in a CPU.
Peak Load	The exhaust temperature control setpoint above the base load setpoint which produces more power but reduces the life expectancy of the turbine parts.
PL/M-86	A high level language designed for systems and application programming of the Intel 8086 microprocessor. Control program algorithms are written in PL/M-86.
Port	The point where the I/O is in contact with the outside world.
Pre-selected Load	The load setpoint between spinning reserve and base load.

IX. GLOSSARY (continued)

Pressure Ratio	The ratio of the compressor discharge pressure to the inlet pressure.
PROM	Programmable Read Only Memory used to store software which is not field adjustable and will not be lost during a power outage.
Protocol	A procedure for data communication.
RAM	Random Access Memory can be read and written to during normal operation and must be backed up to retain its contents during a power outage.
Real Time Operation	A data processing technique which allows the machine to use information as it becomes available, as opposed to batch processing at a time unrelated to the time the information is generated.
Register	A temporary small scale memory used by the CPU for logic, arithmetic or transfer operations.
RS232	An IEEE communication standard used for communication between [C], [R], [S], [T] and the panel mounted printer. It is available for remote communication to customer's computers.
RS422	An IEEE communication standard used for remote communication to General Electric supplied computers.
RTD	Abbreviation for a resistance temperature detector which has a resistance directly proportional to the measured temperature.
Rung	A series of software sequencing commands, i.e., contacts, which result in a store statement, i.e., coil.
Sampling Rate	The frequency at which a given program is run. For example, some control algorithms are run four (4) times every second.
SEM	A Small Electronics Module containing up to 12 circuit cards. Controllers [R], [S] and [T], are SEM's.
Serial Link	An interface between two (2) computer systems that permits one (1) of the systems to transmit data to, and receive data from the other system.

IX. GLOSSARY (continued)

Servo Valve — A hydraulically powered valve with provisions for direct control, i.e., positioning, in direct relation with a primary control of a comparatively low level of force. It is used for proportional control.

Single Ended Input — An input which has a high impedance to ground on one of the two incoming wires only.

Single Shaft Turbine — A gas turbine whose rotating components, compressor blading and the turbine blading are arranged on one (1) shaft.

Softswitch — One of the six (6) pushbuttons located at the right of the video display which is assigned a function by the display which currently appears on the screen.

Software — Programming in the computer.

Solenoid Valve — A valve specifically designed to control the flow of fluid by means of the magnetic action of an electric coil on a movable core or plunger which actuates the valve stem or pilot needle. It is used for on/off control.

Spinning Reserve — The minimum load control point which is based on the generator output.

Stack — A data structure used for temporary storage which receives data on the top of the stack and pushes existing data further down in the stack. Data is removed from the top of the stack creating a last in first out arrangement.

Starting Clutch — The hydraulically positioned jaw clutch which connects the torque converter or turning gear output to the accessory gear box and disengages when the turbine reaches a self-sustaining speed.

Starting Device — The machine part used to produce adequate torque for the starting system. Some types of starting devices are:

1. Diesel Engine
2. Electric Motor
3. Steam Turbine
4. Natural Gas Expansion Turbine
5. Turbine Impingement
6. Air Motor

Stall — A cessation of processor operation.

IX. GLOSSARY (continued)

String A sequence of character codes stored sequentially in memory. For example, the ladder diagram sequencing rungs consist of one (1) or more strings.

Thermocouple A pair of dissimilar metal connectors joined in series to form a closed circuit which will generate a thermoelectric current when heated.

Torque Converter A hydraulic device coupled to the turbine starting means which transfers and amplifies torque, causing the turbine compressor shaft rotation during start-up.

TTL Transistor-transistor logic; the most widely used bipolar technology for digital integrated circuits.

Turning Gear The machine part which is used to break the turbine away when starting and rotate the shaft during cooldown and inspection.

Two-shaft Turbine A turbine arrangement where the high pressure and low pressure turbine stages are coupled aerodynamically and run at different speeds.

Voting Voting means that the control will respond to the majority logic from the three (3) controllers.

Wheelspace (Temp) The temperature of the air in close proximity to the surface of the turbine wheel below the platform surface of the turbine buckets.

Word 16 consecutive bits which are treated as an entity.

Wire Wrap Wiring within modules is terminated on the card or relay socket pins by wrapping the wire around the pins.